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Tracking Europe's AI transition: A new STOXX 600 earnings-call framework

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How quickly is AI moving from experimentation into everyday business, and where are companies reporting tangible gains? The answers matter for institutional investors and policymakers assessing productivity, operating efficiency and labour demand. Yet comparable evidence across Europe remains limited, making it difficult to track the pace and scale of the transition. [1]

We address this gap with a proprietary framework that turns what is said on STOXX 600 earnings call into three structured indicators: AI adoption intensity, reported productivity benefits, and the impact on workforces. Large language models analyse what management teams say about AI's operational role, applying a common scoring framework across companies, sectors and countries. Our new approach provides a timely pan-European comparable tracker that captures not just whether AI is being mentioned, but its real-world impact as the adoption of the new technology unfolds.

Key Takeaways:

- **Adoption is broadening, with signs of faster progress.** AI-use scores rose gradually from a low base before strengthening over the past 12 months. The increase extends across sectors, although the pace and level of adoption remain uneven.
- **Workforce impacts remain limited in corporate disclosures.** The framework does not show a broad negative workforce signal attributable to AI. The aggregate score remains close to the level of no impact on the labour force across all of our sample. However, this does not rule out AI displacing workers at a future date.
- **Geographic differences persist within the listed-company sample.** Since 2022, German companies in the STOXX 600 have generally recorded above-average AI-use and reported productivity scores. Dutch and Spanish companies in the STOXX 600 have tended to score lower on AI use, although Spain has recently gone through a rapid catch up.
- **The technology J-Curve:** The findings are consistent with an early stage of the AI transition. Companies report increasing adoption of the new technology, but transformational productivity gains and extensive workforce restructuring remain uncommon.

[1] 1 Our US team regularly publish on data from the USCB showing the impact of AI on businesses, and our German team have recently begun their own chartbook. In the UK, the Office for National Statistics publishes on the topic too.

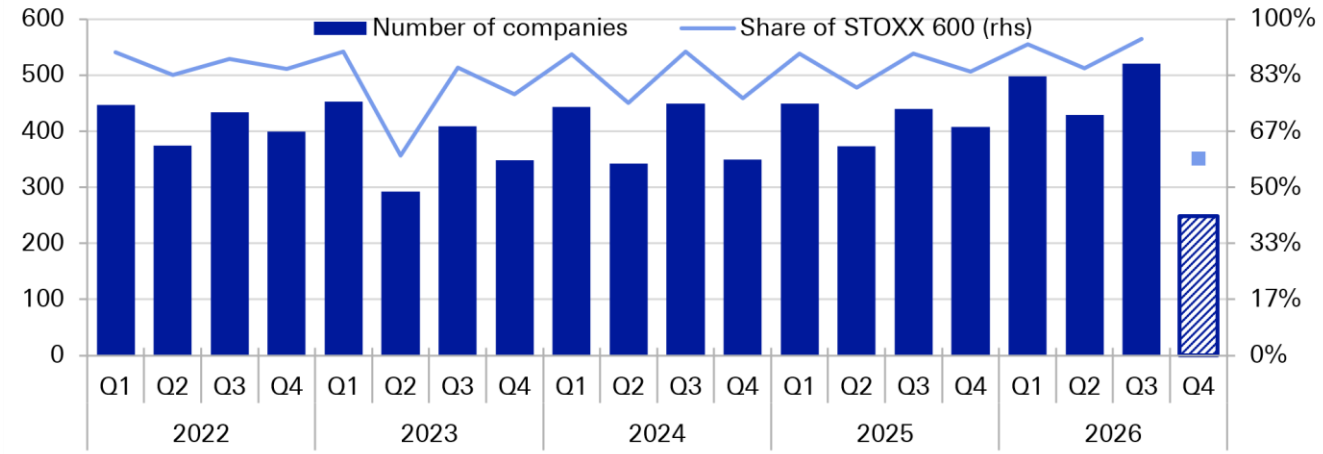
Methodology

Sample period: Our sample begins in January 2022, providing a baseline before ChatGPT's launch on 30 November 2022. AI in this framework covers both established and emerging technologies, so the launch of ChatGPT is a reference point rather than the start of all corporate AI adoption.

Timing of observations: STOXX 600 companies report on different financial calendars and at quarterly or semi-annual intervals. Their earnings calls also mix discussion of completed activity with forward-looking guidance.

We assign each transcript to the calendar quarter in which the earnings call takes place, regardless of the financial reporting period discussed. The resulting indicators therefore track when information about AI use and its operational impacts becomes public rather than when implementation or the reported effect occurred. This provides a common dating convention across the sample while retaining differences in reporting frequency and disclosure timing.

Figure 1: Coverage of the STOXX 600 earnings calls



Source: Deutsche Bank Research. Notes: Share of the STOXX 600 is based on the current company weight in the index. Data collection for the current quarter (Q4 2026) is incomplete as some companies are still to report

Constructing the STOXX 600 AI corpus: We use a specialised AI transcription engine to process earnings calls from STOXX 600 companies, restricting the analysis to calls for which we obtain complete transcripts of sufficient accuracy. In Q3 2026, the latest full quarter, the dataset covered more than 500 companies, representing approximately 90% of the index's current market capitalisation weights (Figure 1). In the current quarter (Q4) earnings calls are still taking place. Nearly 250 companies with an equivalent weight to around c.60% of the STOXX index having already reported.

Historical coverage typically ranges from 300 to 500 companies and represents more than 80% of the index's current market-capitalisation weights. Coverage peaks twice a year because some European companies report semi-annually.



Algorithmic Scoring

Each transcript is analysed by an LLM using a structured prompt to classify disclosures across three dimensions: AI use, reported productivity impact and workforce impact. The LLM approach bypasses simple keyword counts to evaluate semantic context, distinguishing between casual mentions and concrete operational disclosures. A common scoring framework enables comparisons across the covered companies.

- **AI Use Score** summarises the intensity of AI adoption described in a company's earnings call on a scale from 0 to 10. It distinguishes limited experimentation from operational implementation and AI-centred business models:
    - **0:** No mention of AI, machine learning, or automated algorithmic systems.
    - **1–3 (Low):** Minor mention of AI (e.g., pilot programs, exploring ideas, client-side adoption only).
    - **4–7 (Medium):** Active implementation (e.g., AI is integrated into specific operational workflows, products, or core services).
    - **8–10 (High):** Core business driver (e.g., business model is centred on AI, massive R&D spend dedicated to AI, transformational integration).
- **AI Productivity Impact** score summarises company disclosures about realised productivity and profit benefits attributed to AI on a scale from 0 to 10:
    - **0:** No productivity impact mentioned, or AI is not used.
    - **1–3 (Low):** Minor efficiencies (e.g., slightly faster administrative tasks).
    - **4–7 (Medium):** Notable operational improvements (e.g., significant cost savings or volume increases in specific departments).
    - **8–10 (High):** Transformational productivity gains (e.g., doubling output, rendering legacy processes obsolete).
- **AI Labour Force Impact** score summarises disclosures about AI-related changes in labour demand and workforce development on a scale from −5 to +5. Negative scores reflect workforce reductions or hiring restraint; positive scores include hiring and the creation of AI-focused divisions:
    - **-5:** Significant workforce reduction or hiring freezes directly attributed to AI automation.
    - **0:** No impact on headcount or hiring decisions, or AI labour impact is not mentioned.
    - **+5:** Aggressive hiring or creation of new divisions dedicated to AI.

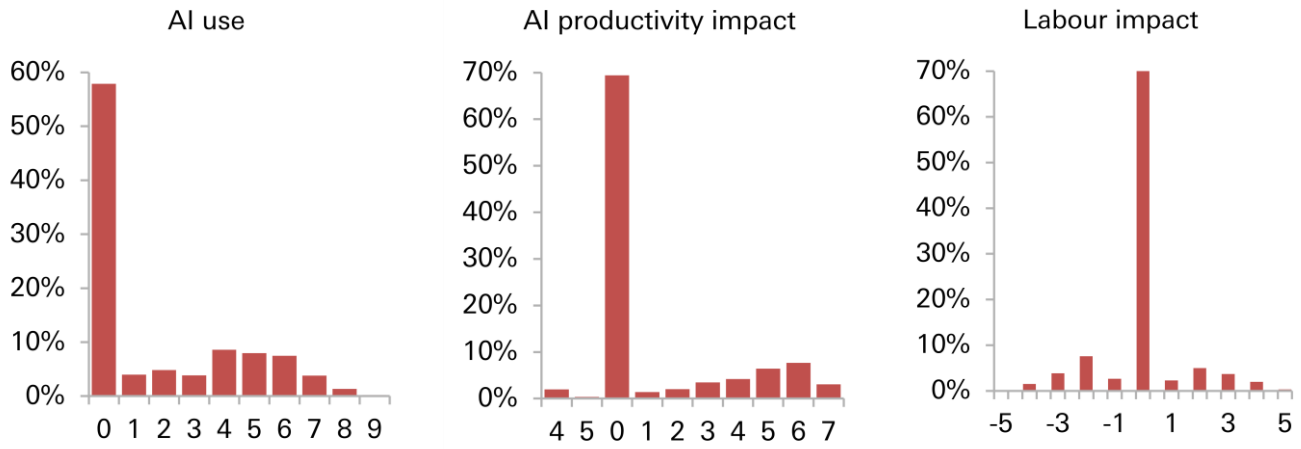
For examples of the AI's scoring and justifications, see the tables presented in the Annex at the end of this report.

Summary statistics

Figure 2 shows the distribution of scores across all earnings-call transcripts in the historical sample, without weighting for company size or reporting frequency. These are transcript-level distributions rather than shares of companies, and they combine early and recent observations. They provide context for the quarterly indicators:

- **AI use score:** Approximately 60% of earnings call transcripts do not discuss AI use. Among the minority that do discuss it, most fall into the "Active implementation" range (a score of 4-6 out of 10).
- **AI's productivity impact:** Nearly 70% of earnings calls do not mention a tangible AI-related productivity benefit. Among calls that do, scores are generally in the medium-impact band, indicating notable operational improvements. Transformational gains, scored at 8 or above out of 10, are rarely reported.
- **AI's labour impact:** Only around 10% of earnings calls discuss AI-related workforce adjustments. Reported changes typically receive scores of 1–2 in magnitude on the –5 to +5 scale, with a slight tilt towards hiring and upskilling rather than redundancies.

Figure 2: Raw distribution of scores across the entire sample



Source: Deutsche Bank Research. Note: Results are based on the raw number of transcripts, and are independent of firm size, and biased towards firms reporting more often.

Taken together, the distributions are consistent with an early adoption phase: companies discuss AI implementation more often than transformational productivity gains or extensive workforce restructuring. **A technology J-curve offers one interpretation of this pattern,** with investment and organisational adaptation preceding broader benefits.

This J-curve pattern is strongly corroborated by external micro-data, such as the Banco de España's business activity survey (EBAE) of Spanish firms, which

similarly indicates that AI has a more immediate impact on upfront investment and productivity than on headcount.

AI use: A broad-based rise, with signs of faster recent adoption

The new AI-use indicator points to a broadening transition across STOXX 600 companies, with early signs that adoption has accelerated recently. Scores rose only gradually from 2022 and 2024, but the latest observations suggest a stronger pace across industries and country groups.

The rise is broad-based rather than confined to one sector. Technology has led much of the sample, while financials show tentative recent outperformance and utilities have caught up from a low starting point. Among the large country groups, German companies have persistently recorded above-average scores.

AI use across European industries

To trace sectoral dispersion, we group companies by STOXX industry classification and calculate a weighted average of their scores for each calendar quarter, using their latest STOXX 600 index weights.[2]

We focus on four sectors spanning different business models: financials, the largest services sector; industrials, the largest physical-assets sector; technology, a likely early adopter; and utilities, an infrastructure-heavy sector where adoption may proceed more slowly.

Figure 3 presents the AI use score for these four key industries alongside the range for all industry scores (represented by the shaded area).

- The AI use score is generally low over time but is gradually rising from around 1 point in 2022 to around 3 in 2026.
- AI-use scores are rising across industries, but differences between sectors are widening. The common direction therefore points to broadening adoption rather than uniform progress: sectors can participate in the transition while remaining at different levels of use.

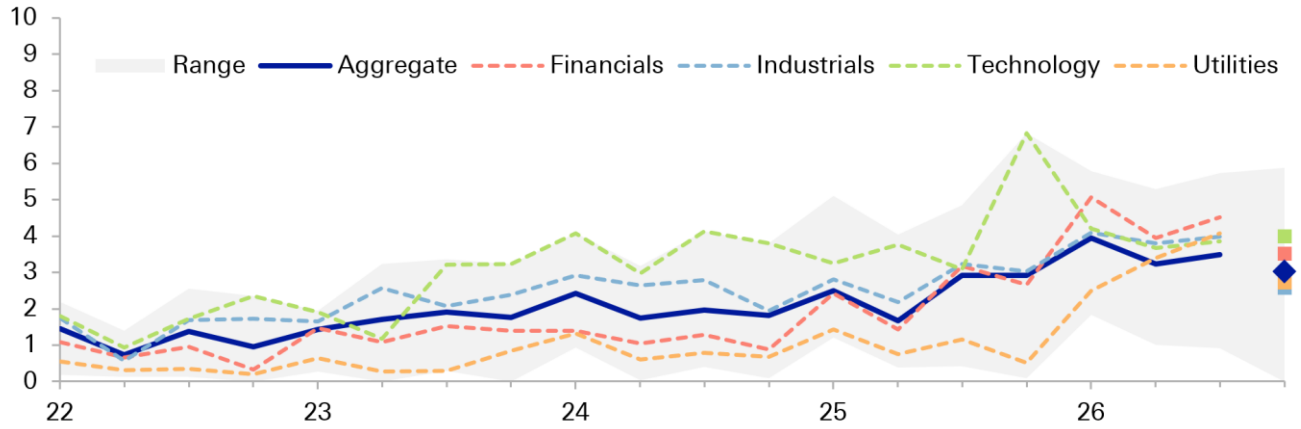
In terms of the industries of interest:

- **Technology's early lead has narrowed.** The sector's AI-use score was near the top of the industry range through to 2024 but has recently plateaued as the broader STOXX 600 average caught up during 2025–26.[3]
- **Financials and industrials broadly track the index average.** Both sectors have remained close to the aggregate AI-use score over history, with financials generally a little below average, and industrials a little above. But financials show tentative signs of moving above average in recent quarters.

[2] The industries are: (1) Basic Materials, (2), Consumer Discretionary, (3), Consumer Staples, (4) Energy, (5) Financials, (6) Health Care, (7) Industrials, (9) Real Estate, (10) Technology, (11) Telecommunications, (12) Utilities.
[3] Q4 2025 saw a large increase in the AI use score for the "Technology" industry, jumping up to around 6. This one-off spike was driven by ASML's announcement of a significant strategic partnership with and investment in Mistral AI" to 'embed AI across our entire holistic portfolio in order to increase the performance and productivity of our systems and the yield of our customers' processes'. Because of ASML's large market cap (around 49% of our Technology category), it distorts the weighted average significantly.

- **Utilities have caught up from a low base.** After recording lower AI-use scores than other sectors for much of the sample, utilities converged towards the STOXX 600 average during 2026. The result shows that the rise in reported adoption extends beyond the sectors most closely associated with AI.

Figure 3: AI use score (0-10) by industry



Source: Deutsche Bank Research. Notes: The value for each industry is computed as a weighted average of all firms within the industry where the weights are derived from the current STOXX 600 share of each company. The markers denote the readings for the current incomplete quarter

As a cross-check, our findings align with the European Commission's October Eurobarometer Flash survey of MSEs and Mid-Caps and the Eurostat annual ICT surveys, which show low overall intensity but a similar expansion in service- and tech-related sectors.

AI use across countries

Companies in the STOXX 600 tend to be large multi-nationals. That makes it difficult to pin down where physically the AI transformation is taking place within their organisation. Instead, our country comparison is done using two classifications of a company's country:

- **Country of domicile:** AI is a transformation technology, and this type of change is likely to be driven in organisation from the top down. We therefore consider a company's country as that in which it is domiciled (which is likely to be where the headquarters is also).
- **Country of primary listing:** While companies can generally list in other countries, there tends to be a home bias in their choice of exchange.

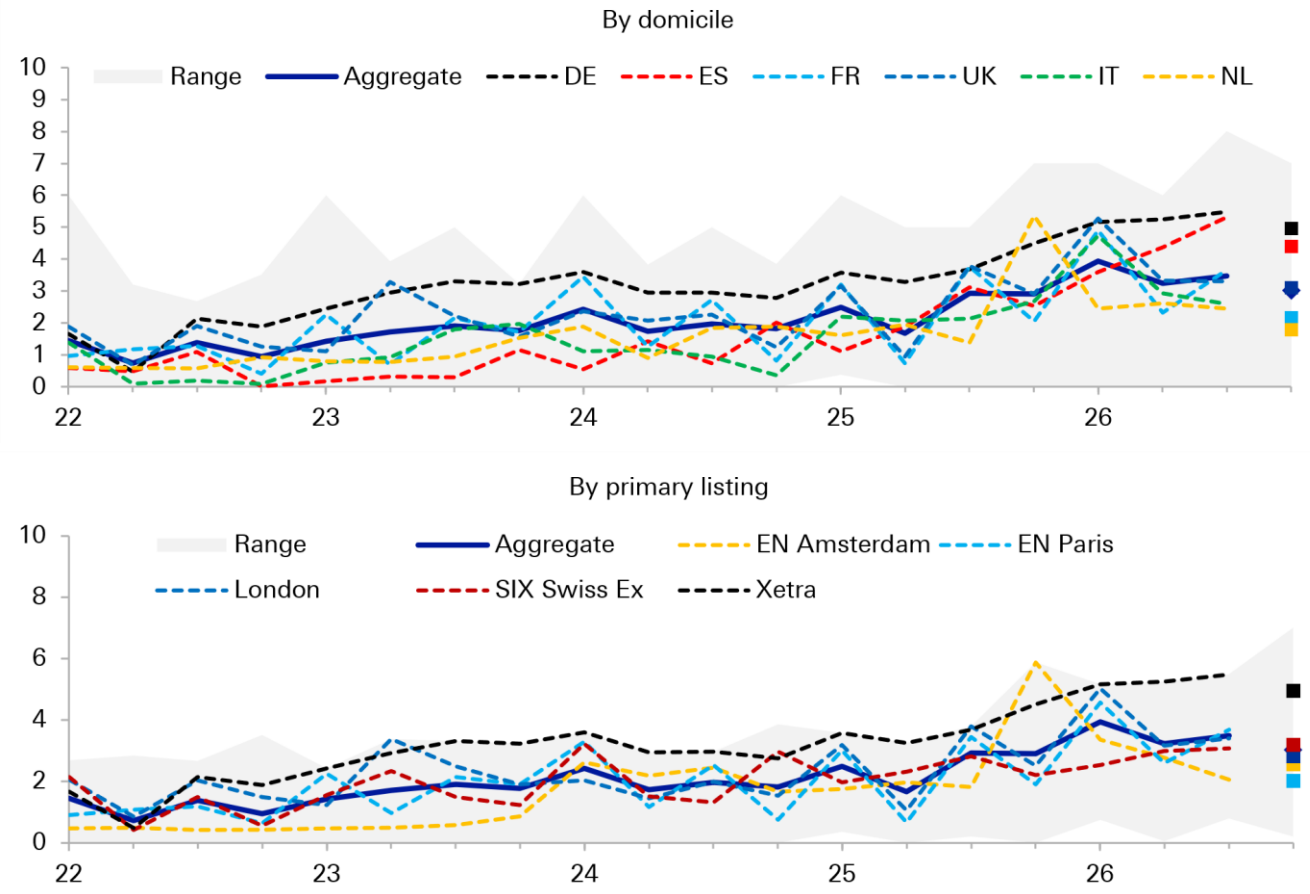
Figure 4 presents the AI use intensity metric by both of these measures of country (the company's domicile and primary exchange listing). Again, the shaded area shows the range across all classifications, with the largest countries/exchanges singled out.

German-domiciled companies and the Xetra-listed (Frankfurt stock exchange) group record above-average AI-use scores in most quarters since 2022.

Spain's AI-use score has risen noticeably over recent quarters and now exceeds the aggregate average, reversing its earlier below-average position.

AI-use scores rise across country groups, with indications of faster growth from 2025. The shared direction suggests a broad European listed-company trend, even though differences in score levels and timing persist between countries.

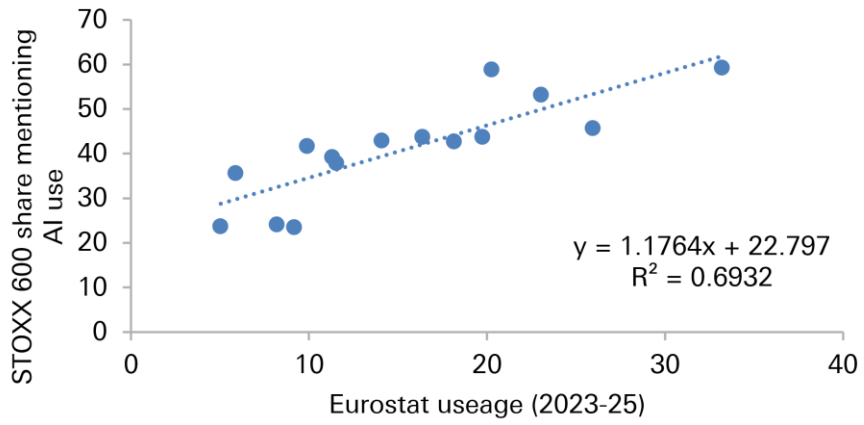
Figure 4: AI use intensity (0-10) by country



Source: Deutsche Bank Research. Notes: The markers denote the readings for the current incomplete quarter

As a geographical cross-check, our index matches the broad trends of the Eurostat ICT annual survey, which places Germany as a clear frontrunner among its large peers. But some differences do exist. For instance, Eurostat shows the Netherlands with a very high AI adoption. As Figure 5 demonstrates, there is a positive correlation between the AI use score metric derived here, and that from Eurostat, even though the Eurostat data measures if firms are adopting AI, and our new use measure also incorporates the concept of how intensively firms are using AI.

Figure 5: Relationship between Eurostat usage and Weighted share of STOXX 600 mentioning AI usage



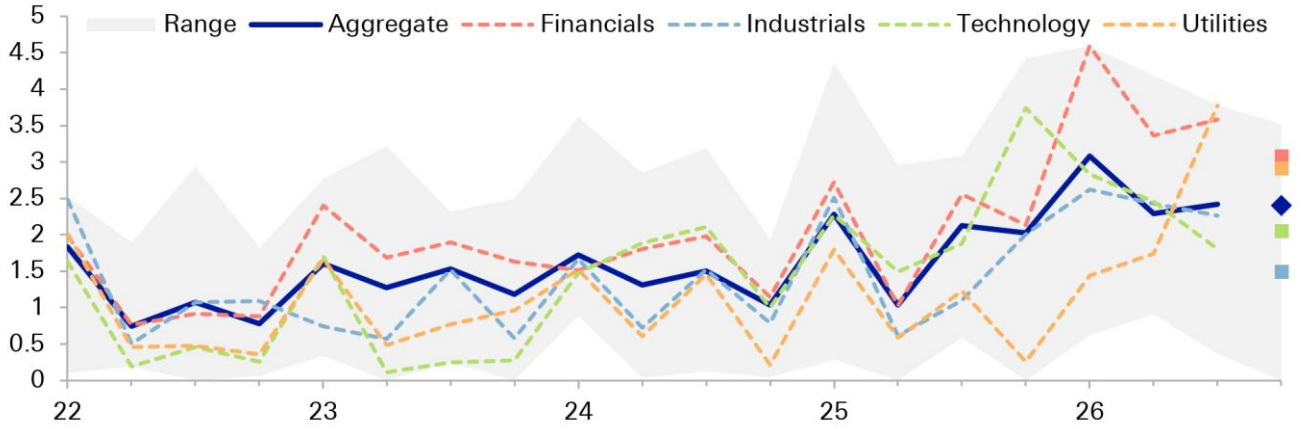
Source: Deutsche Bank Research, Eurostat. Notes: each dot represents a country-year pair of AI usage>0

AI productivity impact: Reported benefits are rising, but transformational gains remain uncommon

Higher intensity of use does not necessarily mean firms are benefiting more from AI. But our AI productivity impact metric does show a similar pattern to our AI usage metric, although it tends to be slightly lower, indicating that companies may not yet be reaping the full rewards for their AI investment.

Financials stand out more clearly on reported productivity benefits than on AI-use intensity, while German companies again record relatively strong scores, but only in more recent history.

Figure 6: AI's impact on productivity (0-10) by industry



Source: Deutsche Bank Research. Notes: The markers denote the readings for the current incomplete quarter

AI productivity impact across European industries

Reported AI-productivity benefits have become more prominent across industries. Figure 6 shows scores rising gradually during 2022–25, followed by

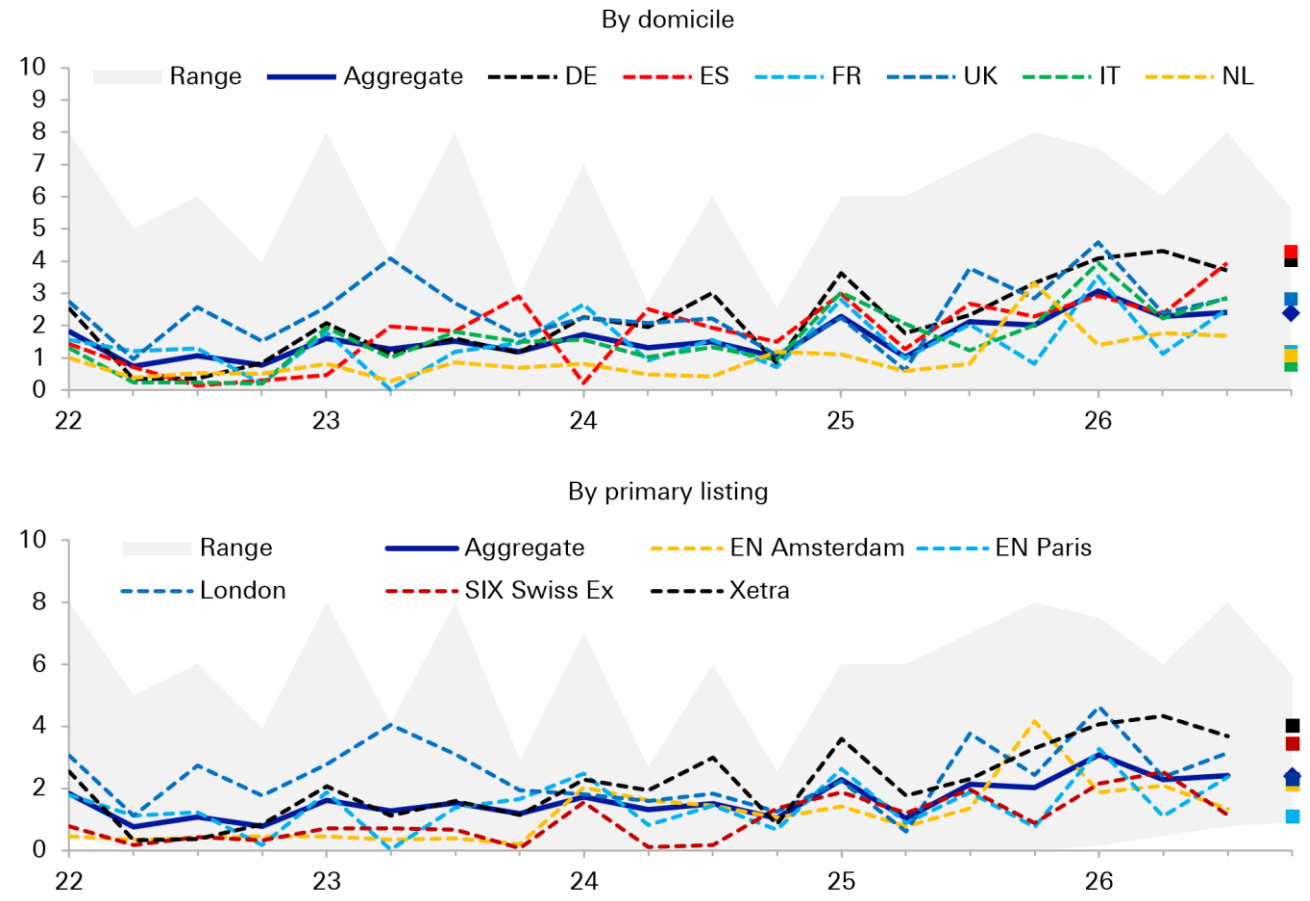
indications of a faster increase from late 2025. The profile resembles the AI-use trend in Figure 4, suggesting that the stronger adoption narrative is accompanied by more discussion of operational gains.

The aggregate trajectories of the AI use and AI productivity impact scores are similar, but sector rankings differ. Technology's early relative strength in AI use is not matched by its productivity score, while financials report above-average productivity benefits despite more modest use scores.

AI productivity impact across countries

Figure 7 plots the self-report impact of AI on company productivity by country. Between 2022-23, German companies recorded average productivity impact, even though their AI use score was relatively high. However more recently, their AI impact score has persistently been above average. The UK grouping shows persistent early outperformance in AI impact during 2022–23, followed by convergence towards the aggregate average.

Figure 7: AI's impact on productivity (0-10) by country

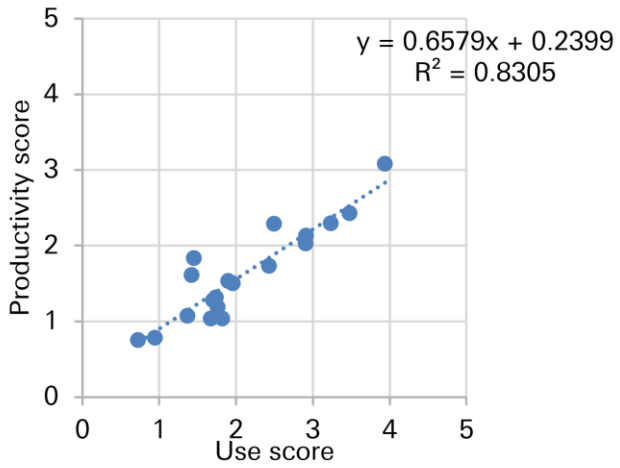


Source: Deutsche Bank Research. Notes: The markers denote the readings for the current incomplete quarter

Figure 8 shows that there is a strong positive correlation between the AI use score and the AI productivity score across the STOXX 600 (r-squared = 0.83). The relationship between the scores is approximately linear, with a slope of around

0.66: a one-point higher use score is associated with roughly a 0.66-point higher productivity score for each country-year.

Figure 8: Productivity and use scores are correlated, but less than 1:1

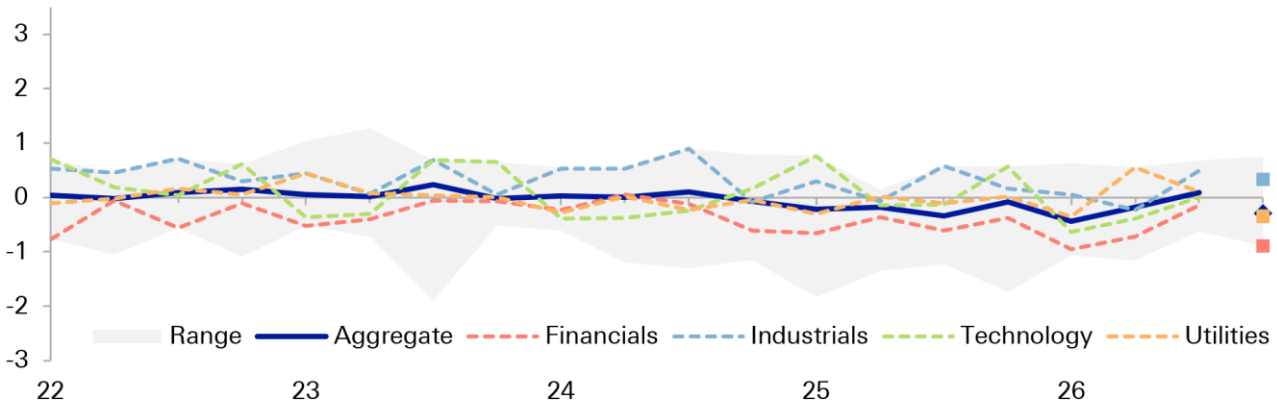


Source: Deutsche Bank Research

AI labour force impact: On net, AI may be benefiting employment, for now...

Across industries and countries, the recorded score for AI's impact on the labour force remains relatively small. On average, the reported impact on companies' employment has been fairly neutral, with a marginal negative bias over the last year or so.

Figure 9: AI's impact on the labour force (-5 to +5) by industry



Source: Deutsche Bank Research. Notes: The markers denote the readings for the current incomplete quarter. Negative (positive) scores are associated with job losses (gains) associated with a company's AI use.

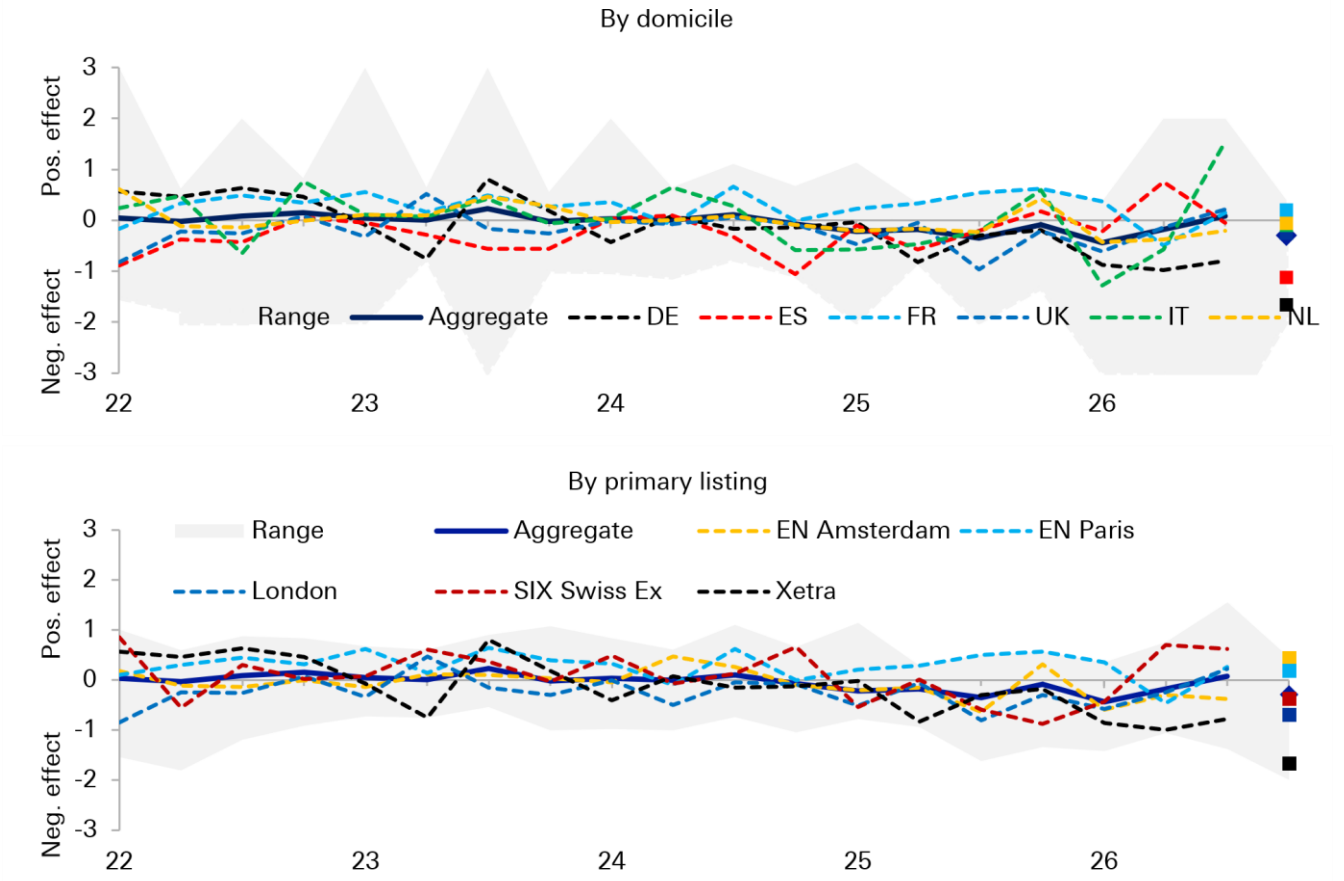
AI labour force impact across European industries

Despite rising productivity, AI has not triggered the negative employment shock feared by the public. Aggregate labour impact scores across nearly all sectors hover near zero, with a slightly positive balance over history, and a slightly negative balance more recently in 2025-26 (Figure 9). But overall the magnitude of AI's impact on labour remains low. For the STOXX 600 aggregate average, the score's maximum deviation from 0 (no impact) is 0.43 on our 0-5 scale.

AI labour force impact across countries

Workforce-impact scores are also small across the main country groupings, with most countries slowing a slight negative tilt in recent years (Figure 10).

Figure 10: AI impact on the labour force (-5 to +5) by country



Source: Deutsche Bank Research. Notes: The markers denote the readings for the current incomplete quarter. Negative (positive) scores are associated with job losses (gains) associated with a company's AI use.

The lack of significant impact so far from AI on companies' labour forces so far is consistent with results from our previous work looking at the other side of the labour market. Data from our dbDataInsights household survey showed that workers across Europe reported a fairly low level of fear about losing their job to AI over the next year. The analysis in that report found results that were more consistent with workers adapting to the new AI world by upskilling and



demanding higher wages. Those results would not be consistent with a world in which AI was actively replacing labour.

These findings of a low impact on labour are highly consistent with recent academic research (e.g., Yotzov et al., 2026), which found that nearly half of surveyed UK and German firms reported absolutely no impact on headcount from AI over the past three years,

Conclusion

The newly developed framework provides a recurring way to monitor company disclosures about AI's impact on productivity and labour. **The central finding of this report is a broadening AI transition rather than clear evidence of an economy-wide productivity breakthrough.** The new metrics indicate that AI use remains limited but is rising across the European countries and industries represented in the sample. Companies report low but growing productivity gains and little aggregate workforce impact so far. This pattern is consistent with a technology J-curve, with firms still integrating the new technology, rather than proof that this trajectory will unfold. Limited reported workforce effects today do not rule out subsequent layoffs or wider organisational restructuring.

Quarterly updates to this framework will track Europe's AI transition and provide timely signals of whether companies are reporting more substantial productivity and employment effects.

Beyond AI, the framework could be adapted to extract signals about employment decisions, investment decisions and the general economic outlook from earnings calls. These potential extensions could form the basis of alternative nowcasting models for important macroeconomic data; they are applications that will be explored in future work.

Annex

The tables below provide examples of the LLM's scoring of earnings call transcripts for the various AI metrics developed in this report. Some examples have been shortened for brevity in the tables, and names of companies, people, etc have been redacted.

Figure 11: AI adoption intensity

Scoring matrix	Examples
0: No mention of AI, machine learning, or automated algorithmic systems.	There is no mention of Artificial Intelligence (AI), machine learning, or any related automated systems in the provided transcript.
1–3 (Low): Minor mention of AI (e.g., pilot programs, exploring ideas, client-side adoption only).	**Score = 2:** The company's discussion of AI is minimal and primarily future-oriented, focused on investment rather than current internal adoption. The main mention of AI comes when CEO [NAME] discusses a new EUR 2 billion venture capital fund, stating, 'The areas we wanna invest into is network security, IoT, GenAI, and some opportunities for beyond growth activities in the core business.' However, he clarifies that the fund is still being set up and has not made any investments yet. This indicates an exploratory or planning stage.
4–7 (Medium): Active implementation (e.g., AI is integrated into specific operational workflows, products, or core services).	**Score=5:** The primary example is the recent acquisition of [COMPANY], a company with IoT solutions and 'software that interprets all of those signals' from vibration sensors to 'predict from that data what's gonna go wrong'. This technology is being integrated into the core Process Automation division to accelerate aftermarket growth. Supporting this, the company also highlights developing 'intelligent' products like the [PRODUCT], and using data analytics in its CRM to 'find patterns' for commercial decisions. This indicates an active implementation of AI-related technologies into specific products and services, warranting a medium score.
8–10 (High): Core business driver (e.g., business model is centred on AI, massive R&D spend dedicated to AI, transformational integration).	**Score=8:** ... The CEO, [NAME], dedicates a significant portion of the call to AI, calling it a "once-in-a-generation opportunity" where the firm intends to "lead the charge." The adoption is not theoretical; the company reports that an exceptionally high "96% of the firm already using AI tools on a daily basis." This is supported by concrete examples of active, value-additive implementation, such as a discretionary portfolio manager using a "proprietary AI toolkit" to double their coverage universe. The company is also building its own "cutting-edge AI infrastructure" and connecting advanced models to its proprietary data. The strategy is to "reshape the entire firm," applying AI as a "multiplier" across both investment functions (quant research) and operations (client onboarding). This goes far beyond pilot programs and represents a deep, strategic, and transformational integration of AI into the core business.

Source: Deutsche Bank Research

Figure 12: AI's impact on company productivity

Scoring matrix	Examples
0: No productivity impact mentioned, or AI is not used.	The transcript does not contain any mention of AI or its impact on the company's internal productivity or efficiency.
1–3 (Low): Minor efficiencies (e.g., slightly faster administrative tasks)	**Score=2**: The company's Vice President, [NAME], made a very brief mention of AI in his closing remarks. He stated, "Just to put an end, talk just a second on the prudent deployment of AI and digitalization that is going on in the company. The company is improving a lot... we are expanding it over the company." While this confirms the internal use of AI and claims positive impact ("improving a lot"), the statement is extremely high-level and lacks any specific examples, metrics, or details about productivity gains or cost savings. The description of a "prudent deployment" suggests a cautious, incremental approach. The brevity and lack of substance indicate that the impact to date is likely minor.
4–7 (Medium): Notable operational improvements (e.g., significant cost savings or volume increases in specific departments).	**Score=5:** The company explicitly states it is using AI to improve internal productivity, particularly within its large technology division. In response to a question about productivity, CFO [NAME] noted, 'we've got something like 11,000 employees involved in technology. And we're becoming more productive as we're able to use better tooling, including AI tooling, in the product development lifecycle.' This productivity gain is cited as a component of the company's 'strong year of organic margin progression.' While many AI mentions relate to customer-facing products, this specific point confirms a notable operational improvement within a core internal function, justifying a score in the medium range.
8–10 (High): Transformational productivity gains (e.g., doubling output, rendering legacy processes obsolete).	**Score=8:** The company explicitly details a significant productivity impact from its use of artificial intelligence. CEO [NAME] stated that through AI-powered value stream mapping, they were able to dramatically reduce production throughput time at their Malta site from 50 days down to 25 days. This 50% reduction in a core operational process represents a transformational gain, effectively doubling the speed and potential output of that production line. The CEO described this as 'top class' and noted it also helps release cash and improves responsiveness to customer demand. The success has prompted a plan to implement this AI-driven approach across the entire [COMPANY] network, indicating the high value and impact of this initiative.

Source: Deutsche Bank Research

Figure 13: AI's impact on the labour force

Scoring matrix	Examples
Negative scores (-5 to -1)	**Score=-4:** The company's CEO, [NAME], explicitly links a workforce reduction to the company's transformation strategy involving automation. When discussing the UK market, he states, "we are actually lowering the amount of people that we have. We started the year with 20,000 people. We're going to end up with 18,000. So simplification and automation is well underway." This represents a 2,000-person (10%) reduction in that geography directly attributed to automation... **Score=-2:** CEO [NAME] stated, "We've also launched a new machine learning backed version of contact form that now enables customers to resolve the problems within seconds using bots in over 40% of cases." Automating 40% of this workflow directly reduces the demand for human labor in that specific area. This aligns with the company's broader [INITIATIVE NAME] initiative and the CFO's comment that hiring is slowing down: "We've only added 4.6% of new staff during the last 6 months and it continues to slow." While no layoffs were announced, the implementation of labor-saving AI at this scale points to a clear negative impact on the company's demand for workers.
Positive scores (+1 to +5)	**Score=3:** CEO [NAME] explicitly stated that the company's recruitment efforts, which are 'even slightly ahead of the plan,' are 'instrumental... for the acquisition of new competencies to support our ambitions in digitalization, automation, and decarbonization.' This directly links an increase in labor demand (hiring) to the company's adoption of AI-type technologies. There was no mention of AI or automation leading to workforce reductions; the entire focus was on hiring to build capabilities in these areas. **Score=5:** CEO [NAME] announced a EUR 2 billion investment in generative AI over the next three years, a significant portion of which is allocated to personnel. He stated, "we intend to double our data and AI teams to 60,000 within the next three years." This represents a plan to hire 30,000 new employees for their data and AI teams. While the company's overall headcount has slightly decreased, this was attributed to market softness and optimizing operational buffers, not AI-driven automation. The clear and substantial plan to double the AI-specific workforce is a strong indicator of AI increasing the company's demand for labor.

Source: Deutsche Bank Research

Appendix 1

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Why hasn't there been a major financial crisis in recent years?

November 3, 2026

— The economist Hyman Minsky warned, in essence, that "stability is destabilising". He argued that a lengthy period of financial stability could be problematic, as it induced risk-taking and complacency amongst investors, laying the foundation for the next crisis. For example, lenders relax their terms and borrowers are more willing to extend themselves, as nobody wants to miss out. This dynamic makes economies inherently prone to financial panics and boom-bust cycles.

— But since the 2008 crisis, and the subsequent sovereign crisis in the Eurozone, we have yet to see another financial crisis happen on a scale that had a lasting market and macroeconomic impact. There have been clear moments of concern, as in early 2023 with the regional bank turmoil, but nothing that has caused a lasting impact. So this led us to ask why.

— In this paper, we examine why the post-GFC era has seen fewer financial crises. Explanations include the reforms made after 2008 to avoid a repeat, the lower proportion of financial assets held by the banking system, as well as the speedier reaction of policymakers when wobbles have occurred. We then consider a few candidates that could cause the next crisis, including AI vulnerabilities, private credit, and stretched fiscal positions.

— Ultimately, financial crises haven't been abolished, but the reforms of recent years have made it harder for shocks to propagate through the traditional banking system. Moreover, policymakers have become far more inclined to intervene when initial signs of stress have emerged.

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Important research disclosures located in Appendix 1



Section 1: What is a financial crisis, and what causes them?

There isn't a single technical definition of a "financial crisis", and they come in many different types. Commonly, we think of banking crises like 2008, but they can also encompass currency/balance of payments crises, debt crises and other forms too. Given currency and sovereign crises have been more common in emerging markets (the UK's ERM crisis in 1992 and the euro crisis being the two most notable DM occurrences), historical analysis in developed economies has tended to focus heavily on banking crises. Non-bank funding or market-liquidity crises are often not categorised separately in the literature, but they have become more common in recent years. This includes the "dash-for-cash" stress at the onset of the Covid pandemic in March 2020 and the 2022 LDI crisis in the UK.

But for all the different types of financial crisis, there are common themes between them. Often, they are accompanied by asset bubbles that burst in a non-linear fashion. We generally see a severe tightening in credit conditions, which in turn flows through into a broader economic contraction. Frequently, there are contagion-like dynamics that can be difficult to stop. And in the modern world, they usually require policymakers to step in, stabilising the situation with interventions.

At risk of oversimplification, we note four factors that determine the emergence and the severity of a financial crisis. The first three are well-established in the financial literature, sometimes pooled together into a build-up and a crisis phase (Brunnermeier & Oehmke 2013), while the fourth has become increasingly important over the past 15-20 years:

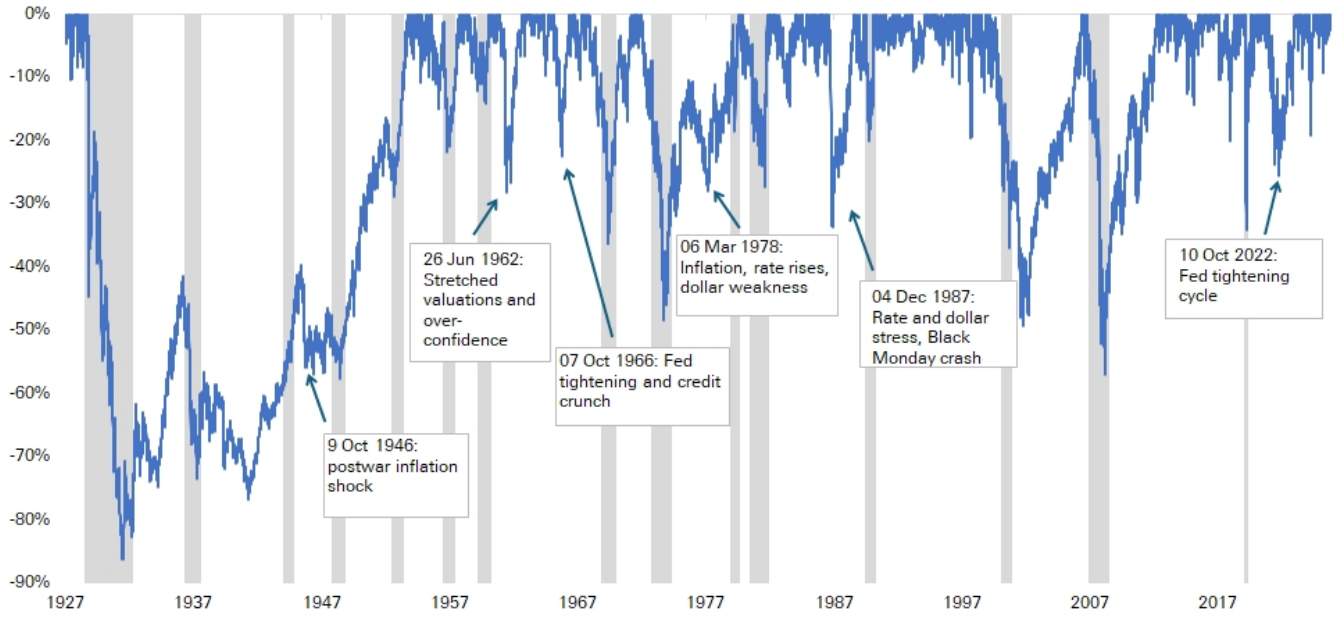
1. **Underlying imbalances.** Bubbles and financial imbalances build during the run-up phase, leaving accumulated risks that materialise when the crisis erupts. This is the build-up typically described by the early phases of Kindleberger's model of financial crises.
2. **Trigger.** This is the shock that turns an existing vulnerability into clear financial stress. These are often very difficult to predict and we mostly leave these outside the focus of this report.
3. **Amplification and financial system fragility.** Various factors determine whether a shock propagates through the financial system, and the intensity of that shock. These include leverage, liquidity and maturity mismatches, interconnectedness, and the concentration of exposures. Other channels in the financial system can then amplify the shock, including bank runs, fire sales and margin calls, and negative net worth effects, sovereign–bank loops etc.
4. **Intervention/policy response.** This has been increasingly important in recent decades as the size and scope of policy response to crises has increased. The outright size can vary, and intervention can occur across several dimensions, ranging from direct intervention in financial markets, to counter-cyclical macroeconomic policy that mitigates the impact on the real economy.

Clearly, these factors are not independent and the interaction between them is what determines the severity of a financial crisis.

One important aspect is the joint role that underlying imbalances and amplification play. While theoretical models such as Diamond & Dybvig highlight the potential for a bank run that triggers a broad crisis, empirical research shows that in practice, it is exceedingly rare for a bank run to lead to bank failure in the absence of an underlying solvency issue (Correia, Luck and Verner 2026). In other words, the presence of amplification channels alone does not lead to a crisis.

Conversely, in the absence of amplification, it is possible to get significant turbulence in asset prices, but without a financial crisis. For instance, a bear market, wider credit spreads, or even a recession do not in and of themselves imply there is a financial crisis. This was seen clearly with the bursting of the dot com bubble, a severe equity decline but not a financial crisis, though it did cause a recession. At the same time, in the chart below, we can see several US equity bear markets that took place outside of recessions. On other occasions, however, a "Minsky Moment" can then happen, where even a seemingly minor event can trigger a high-profile incident that cascades across the financial system.

Figure 1: S&P 500 drawdowns over time, with shaded recessions showing select episodes when the index fell over 20% without coinciding with a recession



Source :Bloomberg Finance LP, NBER, Deutsche Bank Research

Looking back through history, we can see how variation in these factors has led to different regimes over time in terms of frequency/severity of financial factors.

Section 2. Why have financial crises been more or less likely in certain periods of history?

1. Pre-WWII: More frequent financial instability, albeit under a very different system.

Financial instability was a recurrent feature of the pre-WWII era. During the 19th and early 20th centuries, there were frequent panics that often led to major slumps. For instance, the Panic of 1873 sparked a period often known as the "Long Depression", which extended into the 1890s for many countries.

In part, financial panics occurred more frequently because policy operated under very different settings. For instance, the Gold Standard as the monetary system constrained a countercyclical policy response and transmitted deflation internationally, while falling prices increased real debt burdens. Moreover, the Federal Reserve wasn't established until 1913, and the lack of an effective lender of last resort meant there was a private intervention during the US Panic of 1907. The first wave of globalisation also brought new vulnerabilities. Between 1870 and 1913, many emerging economies accumulated debts denominated in gold or foreign currency, leaving them exposed when capital inflows reversed or exchange rates weakened.

The biggest and most famous crisis occurred in 1929, when the Wall Street Crash of that year sparked the Great Depression. The length and depth of that crisis helped to stimulate the study of modern macroeconomics, with huge efforts to prevent a repeat over the years ahead. Indeed, the failures of the response to the Great Depression played a large role in the much swifter response to the 2008 financial crisis, helping to avoid a second Great Depression.

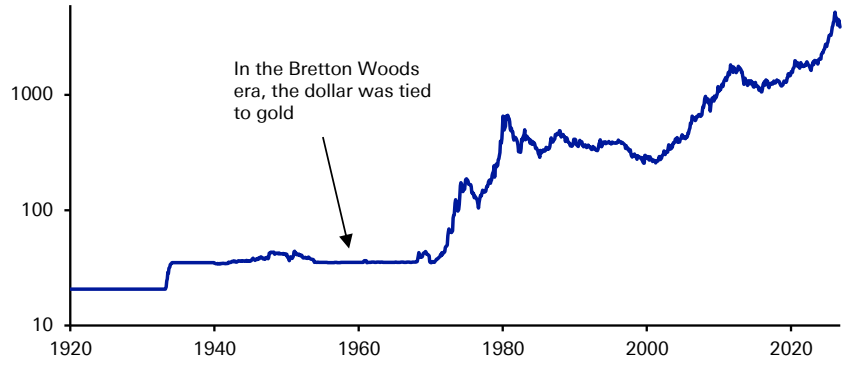
That experience established much of the modern financial safety net that exists to this day. For instance in the US, the Federal Deposit Insurance Corporation was established in 1933. The US and many other countries came off the Gold Standard as well in this period. There were also regulatory reforms, and the Glass-Steagall Act separated out commercial from investment banking.

2. 1944-1971: The Bretton Woods era coincided with fewer financial crises

Around the end of WWII, economic policymakers sought to create an international monetary system for the post-war era. The relative financial calm that followed reflected a combination of restrictions on international capital movement, tightly regulated domestic financial systems, and a system of fixed exchange rates that aimed to provide stability and prevent the competitive devaluations that occurred in the 1930s. This saw exchange rates pegged against the US Dollar, which in turn was convertible to gold at $35 per ounce.

At the same time, capital controls helped preserve monetary policy flexibility, while maintaining the system of fixed exchange rates. These controls had several downsides but they also insulated countries from "sudden stop" crises that can occur, when capital rapidly flees a country. Moreover, capital controls prevented the rapid buildup of cross-border systemic leverage. So it was more difficult to generate a global financial crisis in the same way.

Figure 2: Gold price (USD/oz, log scale)



Source : Finaeon, Deutsche Bank

Alongside capital controls, there were also domestic reforms in the aftermath of the Great Depression, which sought to avoid a repeat. As mentioned earlier, the US Glass-Steagall Act of 1933 separated commercial banking from investment banking. Another example of the era's regulated financial system in the US was Regulation Q, which imposed maximum interest rates that commercial banks could pay on deposits. That meant depositors could not search for yield and cause a run on deposits as easily. Another rationale was that with institutions not competing for deposits, it would avoid a compression in the margin between the rates banks paid on their lending and borrowing, which would disincentivise a push into riskier investments.

Finally, economic growth was also stronger in this period, with the period between WWII and the 1973 oil shock often described as the "Golden Age". This played a role in avoiding financial crises too, as faster growth helped keep debt default rates down, avoiding the kind of "trigger" events above that might spark a financial crisis. Coupled with financial repression, strong growth also helped to bring down the high post-war public debt burdens.

Over time however, the peg between the dollar and gold found itself under growing strain. The economist Robert Triffin identified the "Triffin dilemma". Supplying the world with dollar reserves resulted in outstanding dollar liabilities around the world that exceeded the gold reserves in the US, weakening confidence that dollars could ultimately be redeemed for gold. Conversely, restricting dollar supply to protect those reserves and preserve convertibility risked depriving the system of the liquidity needed to support trade and growth.

These tensions intensified in the late 1960s amid expansionary US policies, including spending on the Vietnam War and Great Society programmes. The continued drain in US gold reserves eventually led the US to suspend the convertibility of the US Dollar into gold, effectively ending the Bretton Woods system. Initially, there were attempts to bring some kind of fixed-exchange rate system back, but the world of fiat currencies (with no link to a precious metal) persists to this day.



3. 1971-2008: Financial crises become more frequent after the breakdown of Bretton Woods to the Global Financial Crisis

After the breakdown of the Bretton Woods system, financial crises became a more regular occurrence. While the end of the gold convertibility did not by itself trigger future crises, it opened the door for wider financial liberalisation, with the resulting rise in capital mobility and looser domestic financial regulation then helping to facilitate a global build-up of imbalances.

The 1980s and 1990s were a period of frequent crisis events. The transition away from capital controls meant that capital could flow more freely across borders. However, this could also cause a "sudden stop" if investors suddenly pulled their capital out of a country, contributing to bank runs and currency collapses. Moreover, many emerging markets borrowed in foreign currencies like the US Dollar, meaning that a depreciation in their own currency would see this USD-denominated debt surge in local currency terms. This meant it was a particular problem if US interest rates were increasing, as it became harder for many emerging markets to repay their debts. This was a contributing factor behind the Latin American debt crisis in the early 1980s. And the rate hikes also had domestic ramifications, being a contributing factor behind the Savings and Loan crisis in the United States.

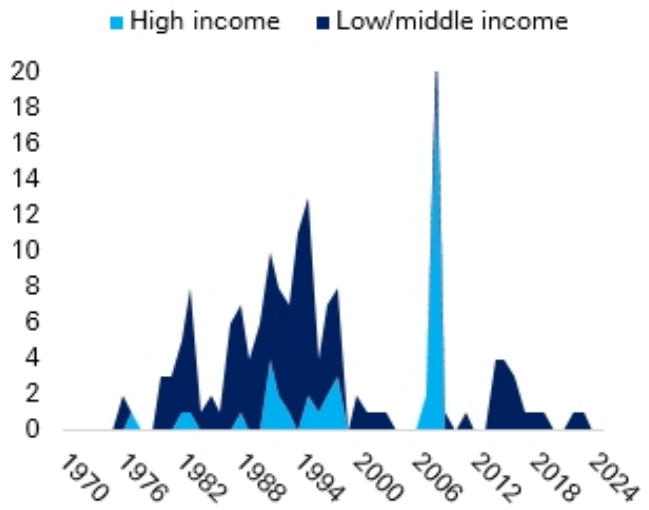
With capital flowing more freely, that also played a role in the 1994 Mexican Peso Crisis, and the 1997 Asian Financial Crisis. A sudden reversal in capital flows led to crisis and ultimately recession in many countries. The following year, the Russian financial crisis then took place, and fears grew in the United States about the wider ramifications, as the hedge fund Long-Term Capital Management (LTCM) came under pressure. Indeed, this period saw 75bps of pre-emptive rate cuts from the Federal Reserve.

This volatility also saw the Fed proactively seek to step in and smooth out the economic volatility. This was broadly successful, but in the process also helped engender complacency and a further buildup of leverage. Indeed, after the dot com bubble burst, the federal funds rate was cut to just 1% in mid-2003, where it remained until mid-2004.

This period saw clear optimism that another Great Depression was highly unlikely. For instance, Nobel Prize-winning economist Robert Lucas argued in his 2003 AEA Presidential Address that "macroeconomics in this original sense has succeeded: Its central problem of depression prevention has been solved, for all practical purposes, and has in fact been solved for many decades."

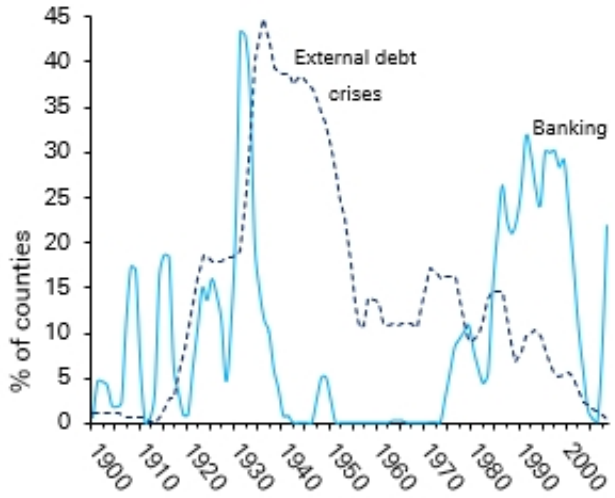
However, as we now know, the 2008 financial crisis occurred, driving the biggest global downturn since the Great Depression of the 1930s. It turned out that Minsky had been correct, and a lengthy period without a crisis had once again helped to induce complacency, laying the foundations for a repeat.

Figure 3: DM economies have avoided systemic financial crises since the post-GFC period



Source : Laeven and Valencia (2026).
Number of financial crises by year since 1970 (by start date)

Figure 4: Banking crises were very uncommon in the post-WWII period, but then proliferated after the 1970s



Source : Reinhart and Rogoff (2013)
Countries with banking and external debt crises, weighted by their share of world income, 1900-2008

Section 3: The Post-2008 world: Why have we not had another big financial crisis?

Since the Global Financial Crisis of 2008, we have thankfully avoided a major financial crisis. Despite clear moments of stress, most notably around the collapse of SVB in March 2023, we have not seen a similar wave of bank collapses on that scale. There are several components to this.

First, post-GFC deleveraging cleared out risks in the system, with countries and sectors that had excess indebtedness before the GFC experiencing pronounced declines in leverage.

Second, there have been reforms explicitly designed to avoid a repeat of 2008. After the crisis happened, there was a clear recognition that gaps in regulation had exacerbated the crisis, leading to effects that cascaded across the financial system.

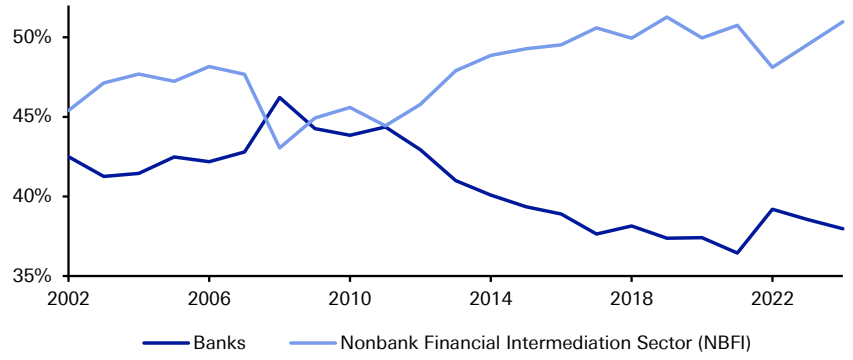
For instance, the Basel III reforms required banks to hold more equity, with a minimum requirement for Common Equity Tier 1 (CET1) capital. They also introduced the liquidity coverage ratio, which requires banks to hold enough high-quality liquid assets to ensure they can cover their net cash outflows under a 30-day stressed scenario. In addition, the net stable funding ratio was brought in, requiring banks to have sufficient stable funding that exceeded the required amount of stable funding for a one-year stressed scenario.

At the national level, further reforms were also introduced. In the US, the Dodd-Frank Act was introduced, with measures including the Volcker Rule that restricted banks from engaging in proprietary trading, and required the Federal Reserve to perform an annual stress test of large institutions. Meanwhile in the European Union, which also experienced a sovereign debt crisis, reforms included a "Single Supervisory Mechanism", whereby banking supervision was brought under the ECB, ensuring harmonised regulatory standards.

Third, the share of financial assets held by banks has fallen since the 2008 crisis. This hasn't eliminated risk, but commercial banks are funded by retail deposits that can quickly fly. By contrast, non-bank financial institutions (NBFIs), e.g. hedge funds, are not as prone to these bank run dynamics. For instance, a private credit closed-ended fund is not one where investors can simply redeem their funds on demand.

We can see this shift in the data. Looking at the FSB's group of 21 plus the Eurozone, we can see that the share of assets held by banks has fallen from a peak of 46% in 2008, to 38% in 2024. By contrast, the share held by NBFIs has risen from 43% to 51%.

Figure 5: Share of Assets of Financial Corporations (FSB 21 + EA Group)



Source :Haver Analytics, Deutsche Bank

Fourth, the lessons of 2008 have meant that when the financial system has had wobbles since, policymakers have acted with incredible speed to stop the turmoil.

This was most obvious when the Covid-19 pandemic began. Whilst a pandemic was very different to a financial crisis, there was initially huge turmoil in financial markets. Indeed, the S&P 500 fell by more than a third in just over a month, a decline at a speed and scale unseen since the Great Depression. Meanwhile, US HY credit spreads soared to levels last seen around the financial crisis, and at the height of the turmoil, there was a huge spike in real yields, despite the economy entering a recession.

Against this backdrop, central banks wasted no time in acting. For instance, the Fed delivered an emergency 50bp rate cut, followed by another 100bp rate cut. Moreover, they swiftly resumed Quantitative Easing, and announced the establishment of multiple new facilities to ensure the smooth functioning of credit markets. These included a Primary Market Corporate Credit Facility, to support new bond and loan issuance, along with a Secondary Market Corporate Credit facility, which offered liquidity for outstanding corporate bonds. Meanwhile, a third facility, called the Term Asset-Backed Securities Loan Facility, was set up to enable the issuance of asset-backed securities. Similarly in Europe, the ECB set up a new Pandemic Emergency Purchase Programme (PEPP), which ultimately had a total envelope of €1.85tn.

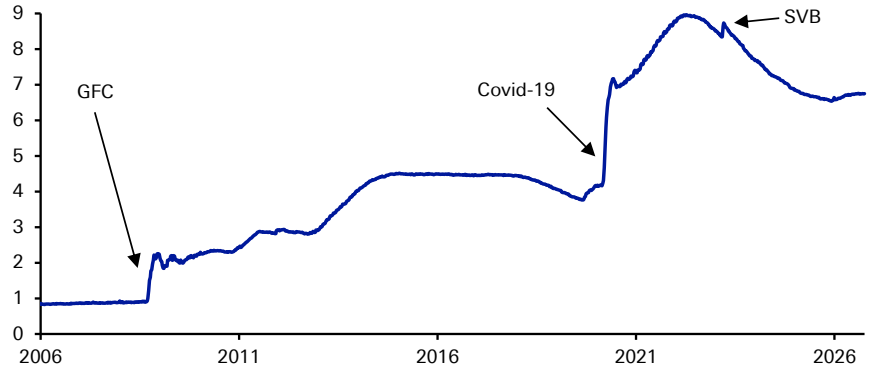
Whilst the pandemic caused a huge economic shock, the swift action meant it did not translate into a financial one, and the brief tightening in credit conditions was

unwound over the coming months.

A similar pattern was seen during the regional banking turmoil of March 2023. This was a financial panic, which initially began with the collapse of Silicon Valley Bank. However, this began to spread, and Signature Bank also failed. Then in Europe, Credit Suisse came under investor scrutiny and saw large deposit outflows, which culminated in a purchase by UBS that included guarantees from the Swiss government. This coincided with huge market moves, with the 2yr Treasury yield posting its biggest daily decline in over 40 years, with a 61bps decline on March 13.

However, policymakers swiftly moved to stop any contagion. The US Treasury, Federal Reserve and the FDIC released a joint statement stating that all deposits at SVB would be protected, and that the Federal Reserve would provide additional funding to ensure banks could meet the needs of depositors. Meanwhile, the Fed's balance sheet briefly expanded again, despite the fact it had begun the process of quantitative tightening. These actions ultimately meant the crisis proved contained and did not lead to a wave of further bank collapses in the manner of 2008.

Figure 6: Federal Reserve Total Assets (USD trillions)



Source : Bloomberg Finance LP, Deutsche Bank

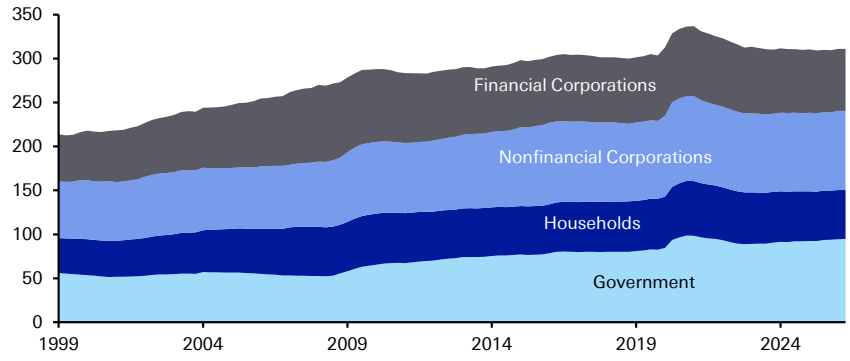
Fifth, policymakers have also responded to recent shocks with huge fiscal transfers, meaning that debt has migrated from private towards sovereign balance sheets.

Whilst global debt has risen slightly over the last decade, if you dig deeper into the numbers, that's entirely driven by a rise in government debt. If we compare the statistics for Q2 2026, relative to Q2 2016, there's actually been a decline in debt for households, financial corporations, and nonfinancial corporations. The only one of the four segments that's gone up in the last decade is government debt.

In relative terms, the transformation is even bigger if we look back to 2007, before the Global Financial Crisis kicked off. Back then, government debt was the smallest of the four categories, but because of the GFC and the pandemic, it's now become the biggest of the four categories.

That in turn has led to separate issues, with government bond yields at multi-year highs, alongside rising net interest spending. So the fiscal space has become limited. However, public balance sheets don't face the same kind of system-wide run that a bank would.

Figure 7: Global Debt (% of GDP) - Since 2007, government debt has gone from the smallest to the biggest category



Source : Haver Analytics, Deutsche Bank

Overall, the post-2008 experience offers three key forward-looking lessons. First, stronger bank balance sheets and the shift towards non-bank finance suggest future stress may originate outside banks. As such, historical analyses centred on banking crises may provide a less complete guide. Second, policymakers have become increasingly effective at containing liquidity pressures and amplification. And third, the shift in debt towards sovereign balance sheets has cushioned private-sector shocks but narrowed the capacity for future intervention.

Section 4: What are potential candidates for the next crisis?

This section considers some of the current risks in the financial system and whether they could be sowing the seeds for the next financial crisis. This section is not designed to be a comprehensive list of financial risks, but rather an assessment of three important areas where risks have risen in recent years.

1. Could AI spark broader vulnerabilities?

A huge driver of the current equity bull run has been the growth in AI-related stocks. This report won't explore the debate as to whether this is a bubble, but examine whether a correction could eventually propagate into a systemic financial crisis. As noted earlier, a reversal in the investment cycle need not cause a crisis in itself, and the bursting of the dot com bubble also did not lead to a broader financial crisis.

Nevertheless, there are issues that could spark vulnerabilities. First, the rapid rise in AI capex raises questions about whether the investment will pay off. The AI build-out is on course to exceed the scale of past technological investment cycles, which have often involved overinvestment even when the technologies ultimately proved transformative. Second, funding is shifting from predominantly operating cash flow towards external capital, and rising debt issuance in particular. Third, greater equity-market concentration increases exposure to a reassessment of technology valuations. Together, these developments make a correction a clear risk.

To assess the potential for systemic stress, we apply the framework in Greenwood, Hanson, Shleifer and Sørensen (GHSS). This associates heightened crisis risk with the combination of a strong run-up in asset prices and excessive credit growth, estimating a 45% probability of a crisis within 3 years if the two thresholds in Figure 11 are met. On a backward-looking basis as of 2025, an economy-wide assessment finds that the equity-price increase meets the framework's "red zone" test, but business credit growth has been considerably more subdued, given the slowdown



in credit following the 2022-23 tightening cycle. That is, going into 2026, the pattern was one of a displacement of credit from other interest-sensitive sectors towards AI and technology rather than any broad-based rise in leverage.

Nevertheless, US data does show a clear acceleration in credit trends so far in 2026. To make a more forward-looking assessment, we incorporate the rise in issuance in 2026 and indicatively extend it into 2027 using the estimated rise in AI capex plans, assuming credit growth elsewhere in the US economy remains stable (footnote: We see two-sided risks to this – we may understate the broader credit demand in tech adjacent areas; conversely, this may understate the drag of higher rates on credit demand in non-AI related sectors). We also extend the estimated 2027 trend into 2028, including an assumption that equity prices continue their trend over the past 12 months, to provide an illustrative scenario over the next two years. Under these assumptions, we would see a visible rise in aggregate US business sector leverage, leaving the sector just shy of economy-wide red-zone levels by 2028. As such, approaching that territory by 2028 is conceivable if the AI cycle continues to accelerate, both in terms of increased debt borrowing and a renewed rise in valuations.

While the GHSS framework gives a useful quantification of the potential risks, its indicators also warrant some qualification, which we judge to come in opposing directions:

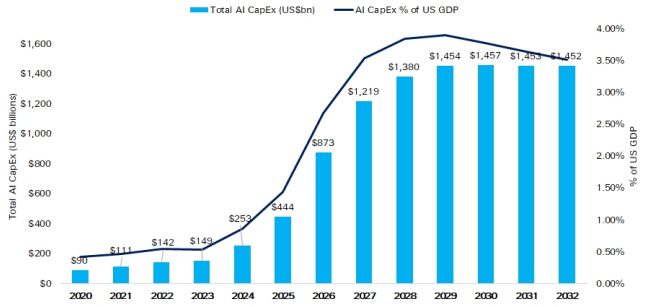
- On the one hand, there is a risk that aggregate credit measures may understate vulnerabilities when borrowing is highly concentrated. Circular financing, leases and off-balance-sheet structures complicate the picture, and while we incorporate the recent rise in payables in our leverage metric to capture some of this risk, it may well be underestimated. Moreover, realised interest costs remain below current market borrowing rates after the lengthy period of ultra-low rates, so leverage trends alone may miss the pressure from rising debt-service costs.
- On the other hand, the price-based metric used by GHSS to assess excess asset price growth (the 3-year real rise in equity prices) may overstate valuation excesses given the strong rise in earnings. Indeed, P/E multiples have already seen a sizeable correction since late 2025, which was most pronounced for software stocks, and currently there is little evidence of overly stretched tech valuations, especially when compared to the dot com boom.

The risk of stress amplification is also likely to depend on the nature of any AI correction. A downgrading of the AI capex cycle due to slower growth in AI demand, financing constraints or supply bottlenecks may result in weaker valuations, leading to negative wealth effects and slower capital expenditure. However, while this would create an economic headwind, it need not produce a systemic crisis. And while parallels to the dot com bubble may well be relevant, currently there are factors that suggest downside risks would be more moderate. For instance, operating cash flow still plays an important role in financing the AI buildout, whilst there has been a limited run-up in borrowing and much less stretched valuations relative to the late-1990s.

Our assessment therefore remains that an AI correction is a clearer risk than an AI-led systemic crisis at this point. For these risks to become more material, it would need leverage to keep building but AI returns to eventually disappoint, raising the likelihood of credit events and amplifying the investment slowdown through the real economy. We will focus on this interaction between debt-service pressures and

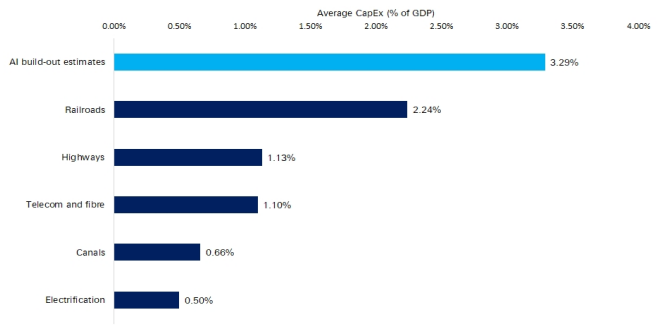
the role of credit events more in the next section.

Figure 8: AI CapEx Total Estimate Spend, Nominal and % of US GDP (2020–2032)



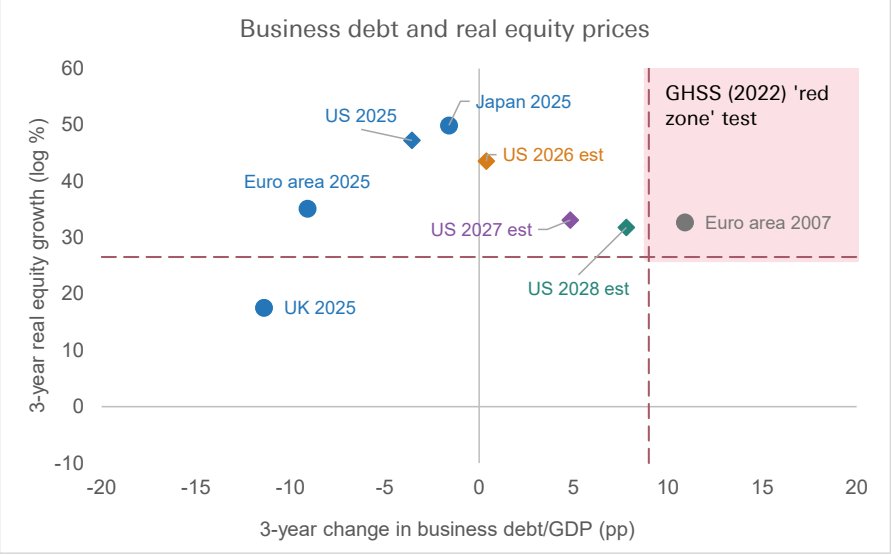
Source : Bloomberg Finance, IMF. CapEx totals include Oracle, Meta, Amazon (AWS), Microsoft, Google and Tier 2/Neoclouds Estimates from Bloomberg.

Figure 9: Annual AI capex spend likely to be bigger than prior US investment cycles



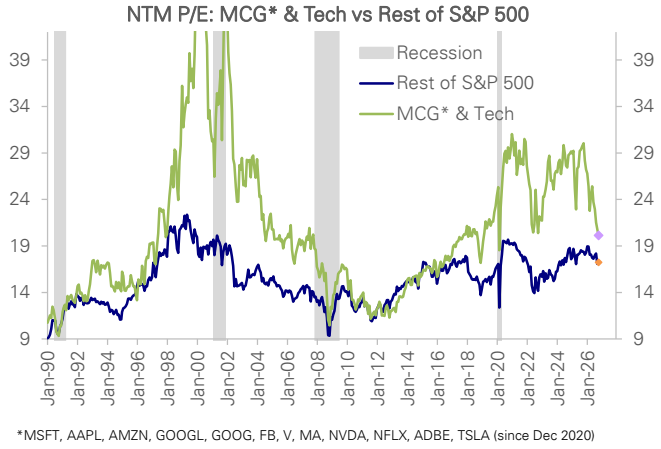
Source : Van Nieuwerburgh (2026), AI projections are based on the Bloomberg estimates in Figure 8.

Figure 10: The combined run up of asset prices and leverage of the US business sector is some way from risky levels, but could approach these by 2028



Source : Deutsche Bank Research, FRB, BIS, Bloomberg Finance LP, Greenwood, Robin, Samuel G. Hanson, Andrei Shleifer and Jakob Ahm Sørensen (2022). "Predictable Financial Crises".
"Red zone" criteria chosen from GHSS paper are a 3-year rise in business-debt of more than 9pp and 3-year rise in real equity prices of over 26%. US estimates for 2026-28 based on: DB macro projections for GDP and CPI; 2026-27 credit growth assumptions used in Figure 14, extended to 2028 at 2027 pace; annual US equity growth of +12.5%, as per the post-GFC trend channel estimated by DB equity strategists.

Figure 11: Unlike the dot com bubble, tech valuations are far from stretched



*MSFT, AAPL, AMZN, GOOGL, GOOG, FB, V, MA, NVDA, NFLX, ADBE, TSLA (since Dec 2020)

Source : Deutsche Bank Research, Bloomberg Finance LP
Latest data point: 27 October 2026.

Figure 12: Borrowing by US corporates has accelerated markedly in 2026



Source : Deutsche Bank Research, Bloomberg Finance LP, FRB Flow of Funds; end-26 and end-27 projections based on (i) indicative AI debt funding based on capex estimates in Figure 8 and (ii) assuming non-AI related credit growth remains unchanged at Q2-26 levels.

2. Risks from higher rates and private credit

In the previous section we noted that modest economy-wide debt growth doesn't necessarily imply modest credit risk when debt-service costs are rising. As existing debt reprices or is refinanced, borrowers can face greater pressure even without additional borrowing. This is a key feature of the debt cycle, whereby vulnerabilities accumulate during the expansion, and rising rates (often amid a monetary tightening due to an overheating economy) can then expose them. These risks are relevant in the context of the current rate hiking cycle, especially as much of the catch up in debt service costs is yet to play out (see chart below).

With banking-sector risks having moderated since the GFC, the rapid expansion of private credit and its opaque structure have understandably drawn attention as a possible source of the next crisis. Parallels with the subprime credit boom have been periodically made. However, we think these should not be overstated. Even though the dollar size of the US private credit market is similar to that of subprime around 2006-07, the market must be assessed relative to the economy (which has doubled in nominal terms). Moreover, the slower build-up, restrictions on redemptions and lower risks of direct wealth effects all suggest much more moderate amplification risks than in 2008.

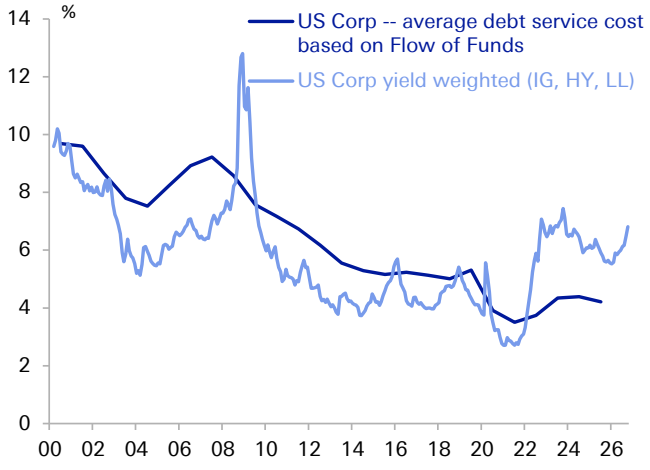
There are already signs that the private-credit growth cycle is slowing (see chart) and our credit strategists have noted warning signs including weaker new inflows into BDCs, increasing payment-in-kind interest and prospective refinancing pressures. That said, defaults remain limited thus far so these should not be seen as evidence of an imminent systemic event.

Whether more systemic risks could emerge will depend on issues of solvency, opacity and capacity for intervention.

- First, while redemption gates can limit immediate cash outflows and reduce the need to sell illiquid loans under pressure, they do not restore a borrower's ability to service debt or absorb underlying credit losses. This distinction between liquidity protection and solvency is therefore central to judging whether a slowing credit cycle remains orderly.
- Opacity remains a concern, both in financing structures and in the distribution of exposures among end-holders, including insurers and other non-banks. This can make losses harder to identify and potential amplification harder to assess.
- Policymakers' success in containing liquidity pressures may not translate into an effective response to credit events - liquidity provision alone cannot repair impaired balance sheets. Extending support to opaque non-bank exposures raises difficult questions about eligibility, loss allocation and the threshold for intervention.

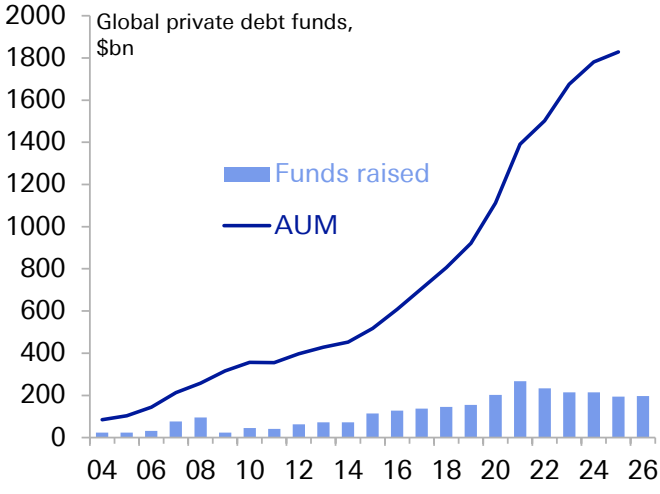
Quite aside from private credit, the fact that we are back in a rate hiking cycle from the major central banks also adds to the current risks. Indeed, higher rates bring down inflation, in part because they tighten financial conditions. But a consequence of this is that it often reveals vulnerabilities in the financial system, and makes a crisis more likely. For instance, it was a contributing factor behind the 2023 regional banking crisis, as banks had growing unrealised losses on their securities portfolio when rates began to rise again. So even if higher rates don't directly cause a crisis, they can often be a key contributor. Indeed, hiking cycles have regularly coincided with some kind of shock, as the experience of recent decades demonstrates.

Figure 13: Market costs of US corporate borrowing is well above existing debt service costs



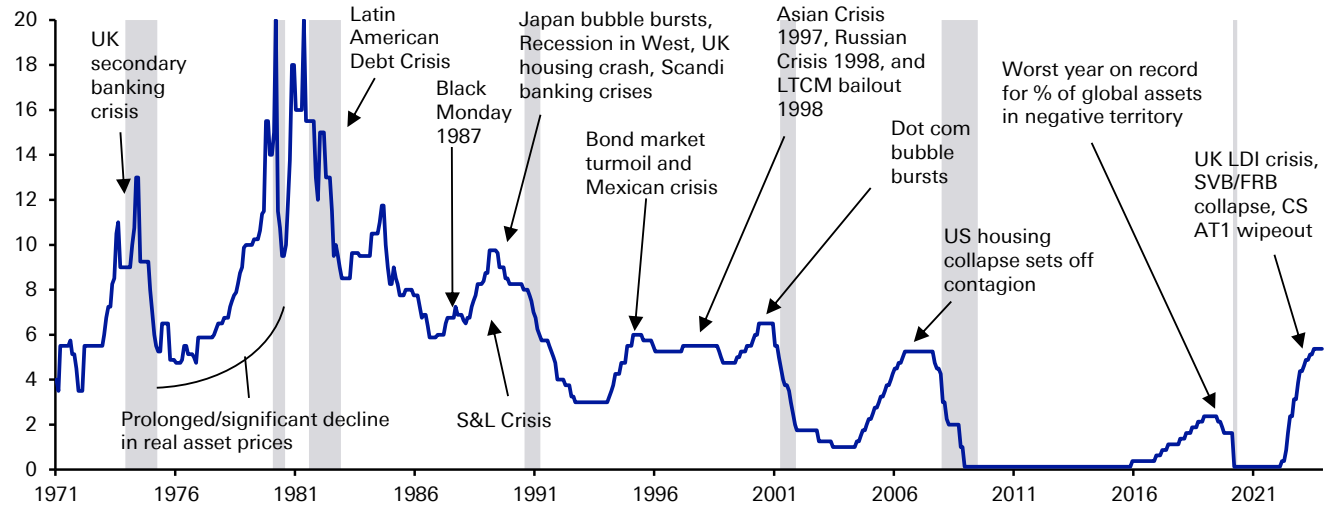
Source : Deutsche Bank Research, FRB, Bloomberg Finance LP, Haver Analytics

Figure 14: Growth of the private credit market has significantly slowed in 2025-26



Source : Deutsche Bank Research, Preqin

Figure 15: Federal Funds Rate and Crises - many crises have often coincided with hiking cycles



Source : Bloomberg Finance LP, Haver Analytics, Deutsche Bank

3. Stretched fiscal positions and possible NBFI amplifiers

As discussed earlier, the migration of debt towards sovereign balance sheets has helped cushion private-sector shocks, but this has come at the cost of higher fiscal risks. Higher public debt and sovereign yields have limited the room for fiscal stimulus, while persistent inflation has limited the room for monetary stimulus. This makes another round of coordinated macroeconomic support harder to deliver.

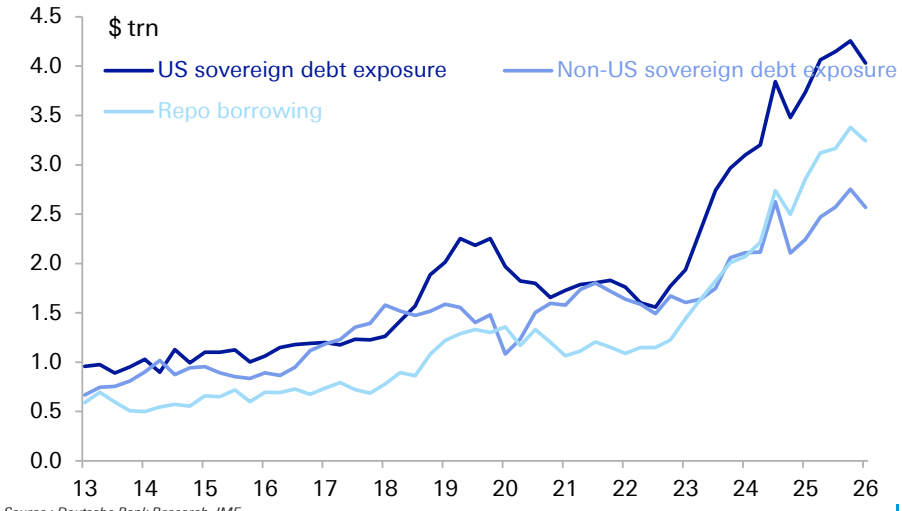
The issue with high public debt is that it if fiscal sustainability is questioned, that can raise sovereign borrowing costs, which can amplify existing private-sector vulnerabilities via a tightening in financial conditions and slower growth.

The financial risk posed by sovereign stress will also depend on the financial amplification channels. These were clear in the European sovereign crisis with the bank-sovereign loop, but the changing structure of sovereign markets suggests that the sovereign–NBFI feedback loop may now warrant increasing attention. This was visible with the 2022 LDI crisis in the UK, while a more recent concern is the run up in leveraged hedge fund positions in sovereign bond markets, with hedge funds' sovereign bond exposures more than doubling since 2022 (see chart below). As noted also by the IMF, this reflects greater issuance and a rising share of sovereign debt held by hedge funds.

The lessons from Black Monday, LTCM and the LDI crisis offer reassurance that intervention can contain market stress originating outside the banking system. But they are not a guarantee that future liquidity shocks will remain economically benign. The greater scale and interconnectedness of non-bank finance mean that we can't simply rely on historical examples today.

Looking forward, a combination of sovereign repricing, leveraged financial amplification and a more constrained policy response risks proving harder to resolve than past episodes.

Figure 16: IMF estimates that hedge funds' sovereign exposures have more than doubled since 2022



Source : Deutsche Bank Research, IMF

Conclusion

Whilst a major financial crisis has been avoided since the GFC and the subsequent sovereign crisis, the problem of financial instability has yet to be permanently solved. On the one hand, reforms to the financial system and the swift actions of policymakers have helped to prevent a repeat. But as we know from Hyman Minsky's warning, lengthy periods of stability can often lay the foundations for the next crisis, even if these take place slowly. Indeed, risks often shift rather than disappear.

Looking forward, there are several vectors of risk worth paying attention to. But financial crises often start in areas where people are paying little attention, or risks are assumed to be contained. Indeed, it is the areas where risk is underappreciated which often prove the most vulnerable.

As such, predicting where the next crisis might arise is inherently difficult. But the topic is a vital one, given the long-lasting damage that financial crises can cause when they do occur. Indeed, with the fiscal space for sovereigns increasingly constrained today, the margin for a future policy error is limited.

References

The end of the safety net: Life without the macroeconomic stabilisers. Deutsche Bank Research Institute paper, 28 September 2026.

Deutsche Bank Research (2017). Long-Term Asset Return Study: The Next Financial Crisis. September 2017.

Brunnermeier, Markus K. and Martin Oehmke (2013). "Bubbles, Financial Crises, and Systemic Risk". Handbook of the Economics of Finance, volume 2, part B, chapter 18.



Correia, Sergio, Stephan Luck and Emil Verner (2026). "Failing Banks". The Quarterly Journal of Economics

Diamond, Douglas W. and Philip H. Dybvig (1983). "Bank Runs, Deposit Insurance, and Liquidity". Journal of Political Economy

Financial Stability Board (2025). Global Monitoring Report on Nonbank Financial Intermediation 2025.

Greenwood, Robin, Samuel G. Hanson, Andrei Shleifer and Jakob Ahm Sørensen (2022). "Predictable Financial Crises". The Journal of Finance

Hernández de Cos, Pablo (2026). "The role of central banks in managing financial crises in times of rapid change". SUERF Annual Lecture, Vienna, 5 October. Bank for International Settlements.

Hofmann, Boris and Gert Peersman (2017). "Is there a debt service channel of monetary transmission?". BIS Quarterly Review

IMF Global Financial Stability Report, October 2026: Resilience, Risk, and Transformation. In particular, chapter 2, "Hedge Funds and Financial Stability".

Kindleberger, Charles P. (1978). Manias, Panics, and Crashes: A History of Financial Crises. New York: Basic Books.

Laeven, Luc and Fabian Valencia (2026). "Systemic Banking Crises Database: 1970-2025". IMF Working Paper No. 2026/094

Lucas, Robert E., Jr. (2003). "Macroeconomic Priorities". American Economic Review. Presidential Address to the American Economic Association.

Reinhart, Carmen M. and Kenneth S. Rogoff (2013). "Banking Crises: An Equal Opportunity Menace". Journal of Banking & Finance

Triffin, Robert (1960). Gold and the Dollar Crisis: The Future of Convertibility. New Haven: Yale University Press.

Van Nieuwerburgh, Stijn (2026). "Financing the AI Buildout". Brookings Papers on Economic Activity, Fall conference draft, 24–25 September.

Yale Program on Financial Stability. The New Bagehot Project. Yale School of Management.



Appendix

Figure 17: List of selected financial crises

Period	Crisis	Type of crisis	What happened?	Financial and economic impact	How did policy intervene?
1929-33	**Great Depression**	Banking	Stock market collapse followed by bank failures, deflation and a contraction in credit.	Adherence to the gold standard constrained policy responses and transmitted deflation internationally, while falling prices increased real debt burdens.	US federal deposit insurance, stronger federal oversight and the separation of commercial and investment banking.
1970s	**Oil Shocks & Stagflation**	Supply shock	Oil embargo in 1973 and Iranian revolution in 1979 caused soaring inflation and recession.	Led to high inflation, low growth, and tighter monetary policy by the 1980s.	Price controls in the early 1970s failed to control inflation. The Fed under Paul Volcker eventually brought inflation down with aggressive monetary tightening.
1982	**Latin American Debt Crisis**	Sovereign, Banking	Countries could not service dollar debts after aggressive Fed rate hikes.	Major sovereign debt crisis across emerging markets.	IMF adjusted programs and repeated rescheduling that kept banks rolling loans over.
1987	**Black Monday**	Market Liquidity	After a run-up of nearly 40% in 1987, stocks collapsed in a single session.	Global stock markets crashed over 20% in a single day. However, there was no deposit run and the real economy grew strongly the following year.	Fed Chair Alan Greenspan publicly reaffirmed his readiness to provide liquidity and act as the lender of last resort, stabilising the market panic.
1990-92	**Japanese Asset Bubble Burst**	Banking, Asset bubble	Property and equity bubbles collapsed.	Marked the start of Japan's "Lost Decades" of weak growth and deflation.	BoJ began easing in July 1991, reached zero rates in 1999 and began QE in 2001.
1992	**ERM / Sterling Crisis**	Currency, Balance of Payment	UK forced out of the European Exchange Rate Mechanism.	Demonstrated vulnerability of fixed exchange rates.	Exit from ERM and the pound floats freely.
1994-95	**Mexican Crisis**	Currency, Balance of Payment, Sovereign	Mexico nearly ran down its reserves trying to defend its crawling peso peg against capital flight. The peso was devalued in Dec 1994 and then floated.	A banking crisis and a deep 1995 recession followed. Argentina and other emerging markets faced capital flight.	The US organised a package including loans from the US Exchange Stabilization Fund and the IMF, plus an austere adjustment program.
1997-98	**Asian Financial Crisis**	Banking, Currency, Balance of Payment	Thailand, Indonesia, South Korea and others experienced currency and banking crises.	Led to deep recessions across Asia; spillovers reached Latin America and Eastern Europe in 1998.	Large IMF programs with the World Bank and ADB, focused on bank restructuring and fiscal tightening.
1998	**Russian Default & LTCM Collapse**	Currency, Balance of Payment, Sovereign	Russia defaulted on sovereign debt; hedge fund LTCM nearly failed as investors dumped risky and illiquid bonds in flight to quality.	Credit spreads widened sharply worldwide. LTCM's forced unwinding threatened fire sales across markets where dealers held the same trades. Russia suffered high inflation and a deep contraction.	In Sep 1998, the NY Fed brought together 14 banks and brokers who put $3.6bn into LTCM. The Fed also cut rates in Sep, Oct and Nov.
2007-09	**Global Financial Crisis (GFC)**	Banking, Asset bubble	US housing collapse triggered worldwide banking crisis.	Drove the biggest global downturn since the Great Depression of the 1930s.	Rates cut to near zero, QE, emergency credit facilities and dollar swap lines from the Fed. TARP bank recapitalization and stress tests in the US, nationalizations and guarantees in the UK and Europe. Then Basel III and Dodd Frank Acts.
2010-12	**European Sovereign Debt Crisis**	Sovereign, Banking, Balance of Payment	Greece, Portugal, Ireland and others faced debt sustainability concerns.	Threatened the survival of the euro.	EU/IMF programs and EFSF and ESM rescue funds. The ECB bought bonds and lent long-term to banks. ECB President Mario Draghi pledged in July 2012 to do "whatever it takes" to save the euro.
Mar-20	**COVID-19 Financial Panic**	Market Liquidity	Global dash for cash froze Treasury and credit markets.	Massive central bank intervention stabilised the financial system.	Massive support to households, firms and markets. Rates went to zero, central banks bought assets without preset limits and backstopped credit markets, governments ran large job-retention programs like UK furlough.
2022	**UK LDI Crisis**	Market Liquidity	Liability-driven pension strategies came under stress after gilt yields surged.	Revealed risks from leverage outside traditional banking.	BoE launched temporary, targeted purchases of long-dated gilts on financial stability grounds. The government reversed most of the fiscal measures.
2023	**SVB / Regional Banking Crisis**	Banking	Silicon Valley Bank and others failed after deposit runs.	Run pressure spread across regional banks. Banks had to borrow through the Fed's emergency programme, total amounting to over $750bn.	Regulators invoked a systemic-risk exception so all SVB and Signature depositors were protected. The Fed launched the Bank Term Funding Program, lending against collateral valued at par for up to a year.

Source : Deutsche Bank Research, Federal Reserve reports, IMF reports

Appendix 1

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Research

COP31 Preview: Cooperation Amid Fragmentation

SUSTAINABILITY | THEMATIC RESEARCH | OCTOBER 2026

COP31, taking place in Antalya, Türkiye from 9 to 20 November 2026, will test whether the 'Implementation COP' can turn climate ambition into finance delivery amid heightened geopolitical tensions. Corporates and investors should watch for signals that could shape policy and capital allocation, including commitments to grid investment, electrification, adaptation and carbon markets. We also expect AI's climate potential and rising electricity demand to feature prominently in discussions, with a focus on clean-energy expansion to support its growing power needs.

The report outlines the top 10 focus topics, discussions and outcomes for COP31 (see pages 1-3 for a summary)

Debbie Jones | Andrew Dixon

Deutsche Bank Research Institute

IMPORTANT RESEARCH DISCLOSURES LOCATED IN APPENDIX D.

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Antalya - 10 things to watch for at COP31

From 9 to 20 November, ~100,000 participants and nearly 200 Parties are expected at COP31 in Antalya, Türkiye. This is an 'Implementation COP' and will focus on climate finance delivery for mitigation and adaptation commitments. Yes, there will be geopolitical tensions and competitiveness concerns. But, for corporates and investors, the discussions will shape how private capital is mobilised, and how policies will affect de-risking efforts. We expect electrification efforts will feature prominently, driven by energy security concerns as well as by rising AI-driven power demand. Furthermore, AI's ability to help tackle climate change has the potential to play a bigger role in shaping the agenda of future COPs.

01 TÜRKIYE HOSTS, AUSTRALIA LEADS NEGOTIATIONS

Türkiye and Australia must stay aligned to drive progress

See page 6

- As host, Türkiye's Murat Kurum will preside over COP31 and its Action Agenda, which is intended to build cooperation between willing participants to drive climate action. Australia's Chris Bowen will run the formal negotiations that will seek consensus (or lack of objection) on finance delivery and existing climate commitments. Feedback following pre-COP meetings on the arrangement has been broadly positive, but concern remains over how any potential disagreement will be handled.
- This structure is the result of a compromise after both countries bid to host. Both countries want COP31 to focus on climate finance implementation and clean-energy growth, however, they remain coal-reliant (in different ways) with no national coal phase-out date, a criticism likely to continue from climate advocates at COP.

02 CLIMATE FINANCE

Success at COP31 requires notable progress on mobilising existing commitments

See page 7

- For COP31 to be successful it will need to deliver agreements to mobilise and channel climate finance. For context, at COP29 in 2024 it was agreed (as part of formal negotiations) that developed nations would commit $300bn in climate finance annually by 2035, with a call for $1.3tn from all sources. At present, more credible plans are needed to secure existing commitments and attract broader capital. Note the EU has already stated it will not make new finance pledges at COP31.
- Article 9 contains the main financing provisions of the Paris Agreement and has driven much debate during COP talks. Developing countries want focus on public funding from developed governments (Article 9.1), while developed countries prefer to discuss Article 9 in a broader context. This allows private capital to be included. COP30 launched a two-year work programme on Article 9, but June's Bonn talks stalled, so COP31 meetings will attempt to reach agreement.

03 ADAPTATION & RESILIENCE

Implications of a 1.5°C overshoot point to more adaptation financing

See pages 8–9

- The UN has warned that an overshoot of the Paris Agreement's 1.5°C goal is now unavoidable, at least temporarily. That has dampened morale for climate supporters ahead of COP but, at the same time, it has drawn more attention to the topic of adaptation finance, which is set to be on the formal negotiating agenda. Parties appear deadlocked over whether the COP30 goal to triple adaptation finance by 2035 (potential annual target of $120bn) should be explicitly referenced in the final COP31 decision on the Global Goal on Adaptation (GGA). This is important, as anchoring it to the GGA could signal finance flows.
- The EU Commission proposed the Climate Resilience law on 27 October. It needs Parliament and Council approval, but if approved, member states will be required to run national climate-risk assessments based on a 3°C-by-2100 warming scenario (i.e. the EU is planning for overshoot). Members would also need to adopt resilience plans, including details of costs and funding, with The Commission estimating the EU needs €70bn a year for adaptation to 2050, vs. €29bn of current spend.

Antalya - 10 things to watch for at COP31

04 **ELECTRIFICATION**

Türkiye to champion voluntary electrification commitments

See page 10

– Türkiye's voluntary Action Agenda includes a '35-by-35' target to lift electricity's share of final energy use from ~23% to 35% by 2035, alongside waste, building energy intensity and circularity targets. This sits outside the formal negotiations and commits only those that volunteer. China and the EU have voiced support, and the timing is right given global energy-security concerns, though some question the emissions impact if fossil fuels are used to generate the additional electricity.

05 **FOSSIL FUELS**

Phase-out likely not on the formal agenda, but workarounds are emerging

See page 11

– Brazil's COP30 presidency is to present a fossil fuel transition roadmap separately, outside the UN formal negotiating process. For perspective, COP28 in Dubai reached consensus on "transitioning away from fossil fuels in energy systems", but at COP30 in Belém (2025) calls from more than 80 countries for a transition roadmap faced opposition from other nations, so it was omitted from the final decision and moved to a separate track.

– Santa Marta is a second workaround for the lack of formal agreement on a transition roadmap. The Colombia- and Netherlands-led process is a 'coalition of the willing' of 57 participants (excluding the US, China, India and Russia) agreeing to voluntary action to move away from fossil fuels faster than consensus allows. At COP31, Colombia and the Netherlands want to bring the Santa Marta outcomes into the summit. The format is not yet settled, but the Australian and Turkish presidencies have said they are open to it. The process then heads to a second conference in Tuvalu, co-hosted with Ireland, in 2027.

06 **TRADE TENSIONS**

Trade's inclusion in the provisional agenda is contentious, with the potential for disagreement over trade to stall progress on wider agreements

See page 12

– The role of trade at COP is contentious, but China wants it discussed. On 21 October, Vice Minister Li Gao called 'unilateralism and protectionism' a 'new problem' undermining collective climate efforts. Specifically, he wants COP31 to discuss the free flow of green technology. This was widely read as being aimed at US and EU tariffs on Chinese cleantech.

– Despite a COP30 agreement to create three annual trade dialogues in 2026–28 (the first formal space under the UNFCCC to discuss trade) away from COP, developing countries are still pushing for trade to be a formal negotiating item in Antalya, and it is on the provisional agenda. They argue that climate-related trade measures such as the CBAM burden their exporters and question whether they are compatible with UNFCCC principles. The EU meanwhile claims they help prevent carbon leakage, with developed countries preferring trade discussions to focus on technical information-sharing without duplicating the WTO.

07 **GEOPOLITICS & MULTILATERALISM**

Consensus under strain, but 'coalitions of the willing' are rising

See pages 13–14

– Geopolitical fragmentation and the US exit from the Paris Agreement make consensus, historically central to COP decision-making, harder among nearly 200 nations. This has encouraged more voluntary action and "coalitions of the willing" to pursue more ambitious goals. If these strategies deliver more effective progress, their approach will shape future COPs.

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Antalya - 10 things to watch for at COP31

Appendix: COP terms (p.19) | Calendar (p.20) | Societal perspectives (p.21) | Disclosures (p.22)

08

AI & CLIMATE

AI's power demand in focus, with the Antalya pledge calling for low emissions infrastructure

See page 15

- The Antalya AI Pledge invites governments and companies to commit voluntarily to sustainable infrastructure and environmental transparency. Its call for zero- and low-emissions energy infrastructure leaves room for nuclear power, prompting discussion of its role in meeting data-centre demand alongside renewables. While we see binding commitments as unlikely, watch for corporate power commitments and technology partnerships.
- We expect AI's climate potential and rising electricity demand to feature prominently at COP31, prompting debate over data-centre energy use. We also expect AI to help delegations manage COP's heavy paperwork, from reviewing negotiating texts and submissions to comparing successive draft decisions. This should help negotiating teams prepare more effectively and make decisions faster, particularly smaller delegations with limited staff and resources.

09

NATURE & CARBON CREDITS

Efforts to boost demand and build credibility to help scale carbon markets

See page 16

- Carbon credit discussions will centre on government policies to boost demand and corporate engagement, with much of the formal UN architecture now in place. If carbon markets can scale, the benefits would be wide-ranging (e.g. supporting emissions targets and providing finance for developing countries). Credibility and greenwashing concerns remain key risks, however; we expect prices to diverge in favour of high-integrity credits.
- The Tropical Forest Forever Facility (TFFF), launched at COP30, plans a fixed-income portfolio to reward verified forest conservation. It now has ~$7.8bn of sponsor capital against a $10bn end-2026 target. In our view, fresh COP31 commitments would signal appetite to fund nature conservation and protection.

10

PACIFIC, OCEANS & SEA-LEVEL RISE

Australia to bring Pacific priorities into focus

See page 17

- Australia asked for a Pacific pre-COP in return for ceding the hosting. It has promised to keep championing Pacific interests at COP31. The Taku Pakasoa Declaration was adopted at that pre-COP on 6 October. 'Taku Pakasoa' is Tuvaluan for strength found in unity and collective action. It is a joint call to keep 1.5°C alive, calling for action on climate finance, resilience and oceans. Its intention is to give the Pacific a shared position going into COP31 and signal its policy priorities to other blocs. Other countries are invited to sign on. The climate-ocean nexus will also be a COP31 focus, including investable solutions.

COP31 should indicate what COP can deliver in a more fragmented geopolitical world. Key questions? Can the current structure help mobilise meaningful finance and investment, and do smaller coalitions and voluntary commitments deliver meaningful outcomes? Answers here could shape future COP agendas. Looking ahead, Ethiopia will host COP32 in 2027. There, we expect a continued focus on delivering on commitments and improving access to climate finance, reflecting African priorities. COP33 then falls to Asia-Pacific in 2028, with China reportedly considering a bid and South Korea potentially interested.

We should note that the second Global Stocktake, the Paris Agreement's five-yearly review of collective progress, begins at COP31 and concludes at COP33, with a technical assessment (2027) informing the final negotiations. Its outcome should inform countries' next national climate plans (NDCs, their emissions-reduction pledges), due in 2030.

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What will COP31 deliver?

Key focus topics in more detail:

Where agreement is possible.
What could hold it back.
Implications for investment.

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APPENDIX

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COP31's unique dual structure will need to bridge divides among negotiating groups and build consensus

As host, Türkiye's Murat Kurum will preside over COP31 and its Action Agenda. Australia's Chris Bowen will run the formal negotiations. This dual structure is the result of a compromise after both countries bid to host. Feedback post pre-COP meetings on the unique arrangement has been broadly positive, but concern remains over how any disagreement between the two countries would be handled should it occur. Both countries clearly want this year's COP to focus on climate finance implementation and clean-energy growth.

A unique leadership structure

COP31 will take place in Antalya, Türkiye. The setup is unusual, emerging as part of a deal between Türkiye and Australia after both countries bid to host the event. What began as a compromise could become an advantage, provided the two countries remain aligned. Their differing diplomatic relationships could help them bridge divides between negotiating groups and build common ground on climate action and finance.

How it will work

Murat Kurum, Türkiye's Minister of the Environment, will serve as COP31 President, overseeing the conference and Action Agenda. Australia's Minister for Climate Change and Energy, Chris Bowen, will serve as President of Negotiations, including drafting final texts, with authority to steer government-to-government talks. Pre-COP preparations have reportedly been positive. They will, however, both likely face criticism from climate advocates asking for a plan to shift from fossil fuels, notably coal, given efforts to position themselves as global climate leaders.

The Action Agenda includes targets for electrification, waste, buildings and circularity

This voluntary Agenda proposes 2035 targets of a 35% electricity share of final energy use (up from 23% currently), halving projected municipal waste growth, cutting building energy intensity by at least 25% and reaching at least 15% circular material use. While pledges are non-binding, they can signal policy and decisions that could support infrastructure investment.

Formal negotiations to focus on implementing existing commitments

Formal negotiations seek consensus (or lack of objection) on implementation pathways. The key need is guidance on how agreed climate-finance commitments will be funded and delivered, and on countries' roles. Adaptation will also be a focus, with countries aiming to reach agreement on how to advance the Global Goal on Adaptation after discussions stalled earlier this year. These formal talks remain separate from the voluntary Action Agenda.

FLAGSHIP INITIATIVES	ACTION AGENDA FOCUS Bold = quantified 2035 targets
Clean Energy Transition & Electrification	**Electricity share of final energy: 35% by 2035**, from 23% now. Includes commissioning of the IEA to deliver special reports to map out pathways.
Zero Waste	Circular waste solutions and methane reduction. **Halve projected waste growth by 2035.**
Climate-Resilient Cities	Resilient cities and buildings. **Cut building energy intensity by at least 25% by 2035.**
Green Industrialization	Low-carbon industry. **Raise circular material use to at least 15% by 2035.**
Dynamic and Resilient Health Systems	Build resilient, low-carbon health facilities. Strengthen risk assessment, early warning and workforce preparedness.
Oceans and Seas	Stronger international cooperation on blue carbon, ocean observation data, data sharing, finance and implementation and partnerships
Rio Synergy	Build on previous COP action. Pilot wildfire restoration and coordinate climate, biodiversity and land action.
Food Security	Integrate food, water and climate planning, expand finance for women farmers, develop youth skills and scale resilient agriculture.
Climate Implementation Bridge (CIB)	Turn national priorities into investable projects and connect them with finance. Actions: prepare projects and strengthen national capacity.
Youth and Education	Climate literacy and young people's participation. Develop curricula, train educators and build green skills.

Sources: COP31 Action Agenda, IEA and Eco-Business

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Climate finance delivery will be a central test for COP31

June's Bonn preparatory talks reportedly made little progress and exposed persistent finance divisions, leaving COP31 to turn both the $300bn and $1.3tn-by-2035 goals into a more credible delivery plan. This is a big ask and requires clearer public/private roles, consistent tracking mechanisms, more concessional finance, risk-sharing and guarantees that can help reduce the cost of capital. New large-scale finance commitments appear unlikely, with the EU already stating it will not make new pledges.

Climate finance: needs, commitments and progress

The image below explains the various climate finance estimates and goals.

Total estimated climate investment need for developing countries (excluding China)
(annual, by 2035)

$3.2 trillion
per year by 2035

~$1.9 trillion
Domestic finance (within developing countries)

$1.3 trillion
International climate finance (from abroad) – aligned with the NCQG $1.3tn goal

Source: Independent High-Level Expert Group on Climate Finance (IHLEG)

$1.3 trillion a year by 2035

COP31 focus: first implementation update on the Baku-Belem roadmap to $1.3T. This total is inclusive of the $300bn NCQG target.

Context: COP29 set the New Collective Quantified Goal (NCQG), calling on all actors to scale climate finance for developing countries to at least $1.3tn a year by 2035. No country has an assigned share, and private capital is expected to count.

At COP31: first implementation update on the Baku-Belem roadmap to $1.3T, based on "measurable progress" against targets including:

- a global monitoring framework to track progress in climate finance mobilization, identify bottlenecks, and highlight emerging solutions.
- clear pathways for public, private, domestic and international finance;
- Expanding operability, comparability, and interoperability of climate-finance data; and
- action on grants and concessional finance, debt and fiscal space, private-finance costs, capacity and system reform.

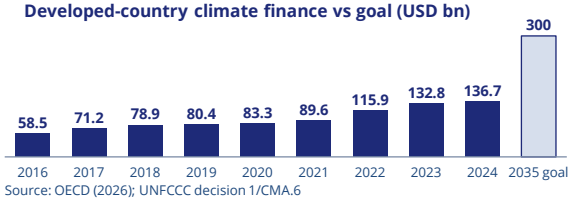
Are the NCQG and IHLEG aligned? IHLEG's $1.3tn external-finance estimate (within the $3.2tn needed excl. China) does align with the NCQG $1.3tn goal and underpins the Baku-to-Belém Roadmap. However, they are built differently: political target vs needs-based estimate.

$300 billion a year by 2035

COP31 focus: what counts and who contributes. This amount contributes towards the $1.3tn NCQG target from all actors.

Context: Developed countries agreed to lead mobilisation of at least $300bn a year by 2035, counting public finance and the private finance it mobilises; no country has a fixed share.

At COP31: Focus turns to how to deliver. Developing countries are pushing for the development of action plans to chart a pathway to $300bn and for the Article 9 work programme to be included on the formal agenda (with emphasis on Article 9.1 – which explicitly covers developed country finance provision).



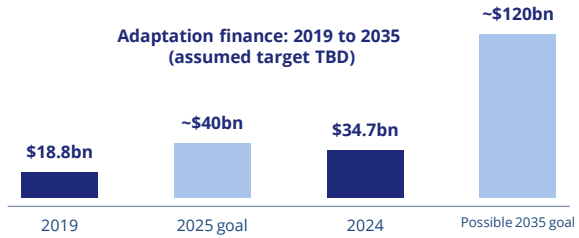
Source: OECD (2026); UNFCCC decision 1/CMA.6

Adaptation finance

COP31 focus: embedding and tracking the tripling pledge

Context: COP30 called for adaptation finance to at least triple by 2035, yet this was relatively vague; it did not specify a base year or a target figure (some have suggested $120bn, i.e. 3x the 2025 goal).

At COP31: Parties need to resolve disagreement on whether to anchor the pledge in the Global Goal on Adaptation (GGA) and put a finer point on the agreed Belém Adaptation Indicators that will track it, after disagreement stalled the June Bonn talks.



Loss and Damage Fund

COP31 focus: guidance, replenishment and access

Why it matters: The Fund is the only dedicated channel for irreversible climate impacts, yet it has no annual funding target and resources remain far below demand.

At COP31: Focus will be on its long-term funding strategy, with the first replenishment cycle set to begin in 2027. $342m made available in the first funding window, which offers $5–20m grants with at least half reserved for SIDS and LDCs; but the voluntary nature of payments, disputes over who pays and sensitivity around "compensation" make major new pledges in Antalya unlikely.

Pledges and received funds (as at end-25)

~$820 million
pledged by 27 partners

~$440 million
received by the fund

Hurdles to these targets

Grants vs loans
Developing countries want grants (especially for adaptation), citing debt burdens and historical responsibility; funders favour loans and blended structures that include private capital.

Pressure on development banks
The US is pushing MDBs (e.g. the World Bank), to drop their climate focus, according to the World Resources Institute (WRI). Developing countries want MDB reform to deliver more investment and faster disbursement.

Competing budget pressures
Donors are diverting aid to fund domestic priorities such as defence. There is also the reality that climate is now a lower priority for some governments.

Reliance on private capital
The IHLEG estimates ~½ the $1.3tn could come from private sources. Private capital is looking for regulation, insurance and more guarantees to reduce risk.

Sources: IHLEG (2024), OECD climate finance data, Fund for Responding to Loss and Damage, World Resources Institute (WRI), Baku to Belém Roadmap to 1.3T.

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Adaptation will be one of the most important, and more divisive, negotiating topics

Adaptation is also set to be on the formal negotiating agenda at COP31. UN warnings that an overshoot of the Paris Agreement's 1.5°C goal is now unavoidable (at least temporarily), have dampened morale for climate activists. As it relates to the COP agenda on adaptation, parties appear deadlocked on whether the COP30 goal to triple adaptation finance by 2035 should be included in the Global Goal on Adaptation (GGA) text. This is important, as anchoring it to the GGA could signal to investors the scale and timing of future public and blended finance flows.

The role of adaptation in the COP process

The Global Goal on Adaptation (GGA) was established under the Paris Agreement to enhance adaptive capacity, strengthen resilience and reduce vulnerability to climate change. At COP30 last year, countries adopted 59 indicators (still being refined) to measure progress against the goal. They also agreed a goal to triple adaptation finance by 2035. The pledge was vague: it set neither a concrete figure nor a baseline year (a potential annual target of $120bn would be triple the previous goal of c.$40bn by 2025).

Why GGA talks were deadlocked at the June climate meetings

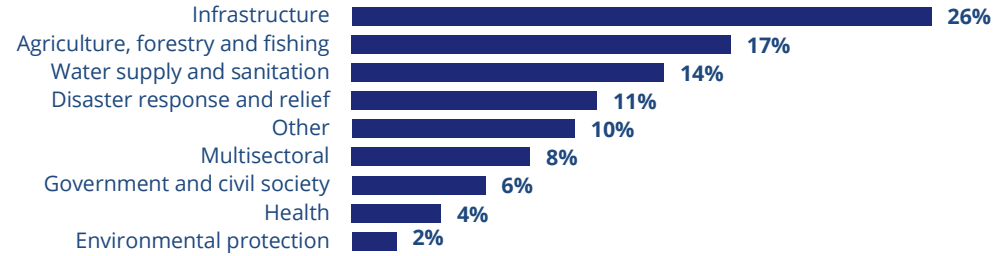
Finance was the main sticking point at the June Bonn meetings. Developing countries wanted the COP30 goal to triple adaptation finance by 2035 referenced in the GGA text; countries including Canada, Norway and Japan opposed. Parties also split on whether the indicators task force should be expert- or party-led, so no GGA text was agreed and the decision passes to COP31. Moving the Adaptation Fund to serve the Paris Agreement exclusively (funded by a 5% share of proceeds from Article 6.4 credits) stalled in Bonn over the composition of its board; talks will resume at COP31.

What is needed at COP31 to unlock adaptation progress

Adaptation is on the provisional COP31 agenda, including the tripling adaptation finance goal (though this may get blocked). Some parties want it explicitly referenced in the final COP31 GGA decision; this should become clearer when the agenda is adopted as COP31 opens on 9 Nov. At issue: the tripling pledge still lacks a baseline and named contributors. At COP30, developed nations resisted new finance obligations, while the EU and many others pushed for a roadmap to transition away from fossil fuels, which a group of mostly oil- and gas-producing countries blocked. The result was a looser tripling goal and no agreed fossil-fuel roadmap. Developing countries are unlikely to give ground, as adaptation relies far more on public funding than mitigation, so a repeat of COP30's finance vs. mitigation debate is plausible.

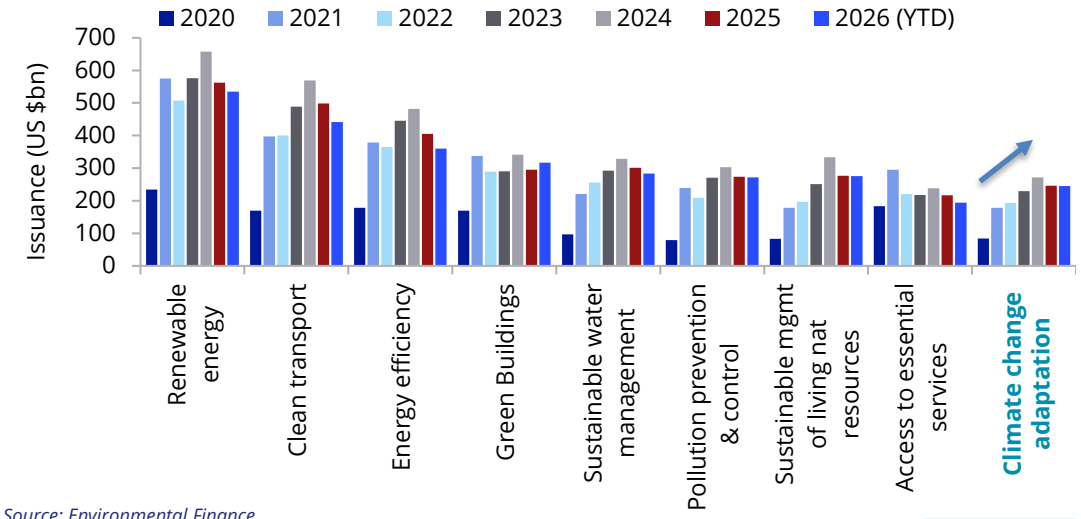
Sources: United Nations, UNEP, Carbon Brief, European Parliament (COP30 outcomes briefing, December 2025)

Sectoral distribution of MDB-funded adaptation projects (2019-2023)



Source: UNEP

Global sustainable bond issuance by use of proceeds (top 9, 2020-2026 ytd)



Source: Environmental Finance

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The EU's proposed Climate Resilience Law plans for a 3°C world, while reiterating commitment to net zero 2050

The EU Commission's recently proposed Climate Resilience Law would set a reference global warming trajectory of 3°C, providing a benchmark for member states' adaptation planning. The EU states it remains committed to reaching net zero by 2050, but some at COP will likely view the proposal as accepting overshoot. With the EU also pointing to an estimated need for EUR70bn in annual adaptation investment, this provides an interesting backdrop as COP31 seeks to resume adaptation negotiations which stalled earlier this year, primarily due to finance.

What the Commission proposed

On 27 October the Commission presented its EU Framework for Climate Resilience, anchored by a proposed European Climate Resilience Law. With current global policy commitments pointing to around 2°C of warming by 2050 and 3°C by 2100, the law would set a common reference trajectory of 1.5°C in 2030, 2°C in 2050 and 3°C in 2100, with this regarded as the "baseline assumption". Member states would be required to use it in national climate risk assessments and adopt a single national resilience plan by end-2030.

Why it matters: the EU is planning for overshoot

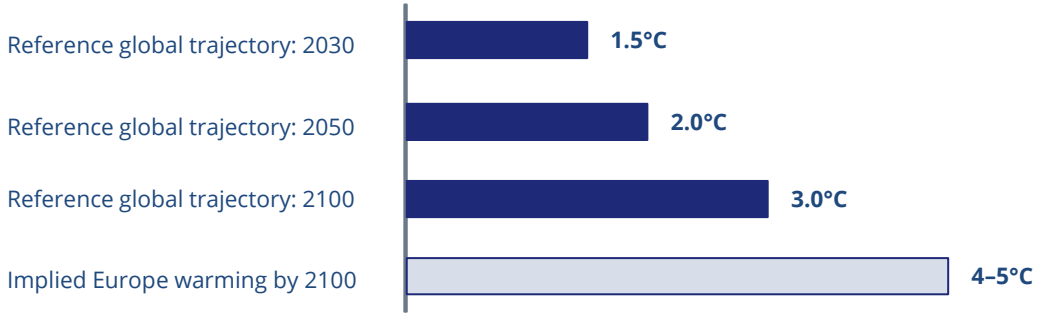
The EU highlighted warnings that overshoot of the 1.5°C goal is now "unavoidable" (even if just temporarily) as necessitating policy action, even as the EU says it is still committed to climate neutrality by 2050. The Commission links the plan to Europe's hottest summer on record and estimates annual economic losses from climate extremes of €50bn over the past four years, adding that output losses caused by climate change go far beyond this. It says EU adaptation needs are ~€70bn a year through to 2050, with Europe warming at 2x the global rate.

What it means for COP31

The proposal demonstrates that warnings on the implications of a significant overshoot of the Paris Agreement global warming limits are moving from scientific theory to real government policy. While the opening line of the new EU framework emphasized it remains committed to reaching net zero by 2050, some may view a 3°C planning baseline as accepting overshoot. The headline €70bn annual EU adaptation finance requirement also provides an interesting backdrop heading into Global Goal on Adaptation talks given they are stalled largely due to disagreement over explicit links to finance in the GGA text. For context, the UNEP estimates annual adaptation needs for developing countries of $310–365bn by 2035. Ultimately, we expect adaptation focus to continue growing as warming impacts are increasingly felt and societal calls for relief grow louder. Corporates will need to consider potential strategic impacts, while some investors have already begun raising their baseline warming forecasts.

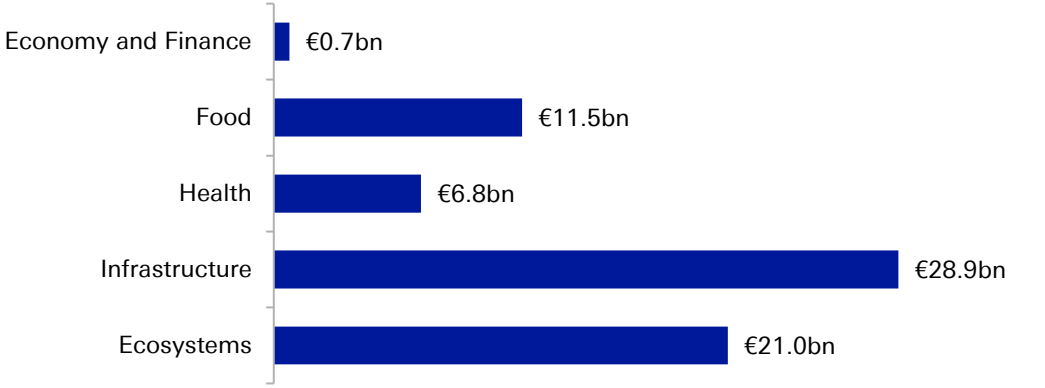
Sources: EU Commission, United Nations, UNEP, DG CLIMA

Common EU warming reference trajectory (°C above pre-industrial)



Source: EU Commission

EU adaptation investment needs by area (€bn a year, to 2050)



Source: DG CLIMA

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Türkiye is championing a voluntary pledge to lift electrification from 23% to 35% by 2035

Electrification is expected to be at the centre of this "implementation COP", targeting a rise in electricity's share of global final energy use from 23% to 35% by 2035. We expect support for this action given it is being positioned alongside energy security, but we also expect discussion and debate over how it is delivered (e.g., the role of renewables, nuclear and natural gas), alongside concerns over clean tech dependency in supply chains. China and the EU have expressed support but not yet committed.

Some may brand this the electrification COP
The Action Agenda pledge aims to raise electricity's share of global final energy from 23% today to 35% by 2035. The pledge was launched in Bonn this past June and formalised as the "35-by-35 Global Electrification Pledge" in New York on 21 September.

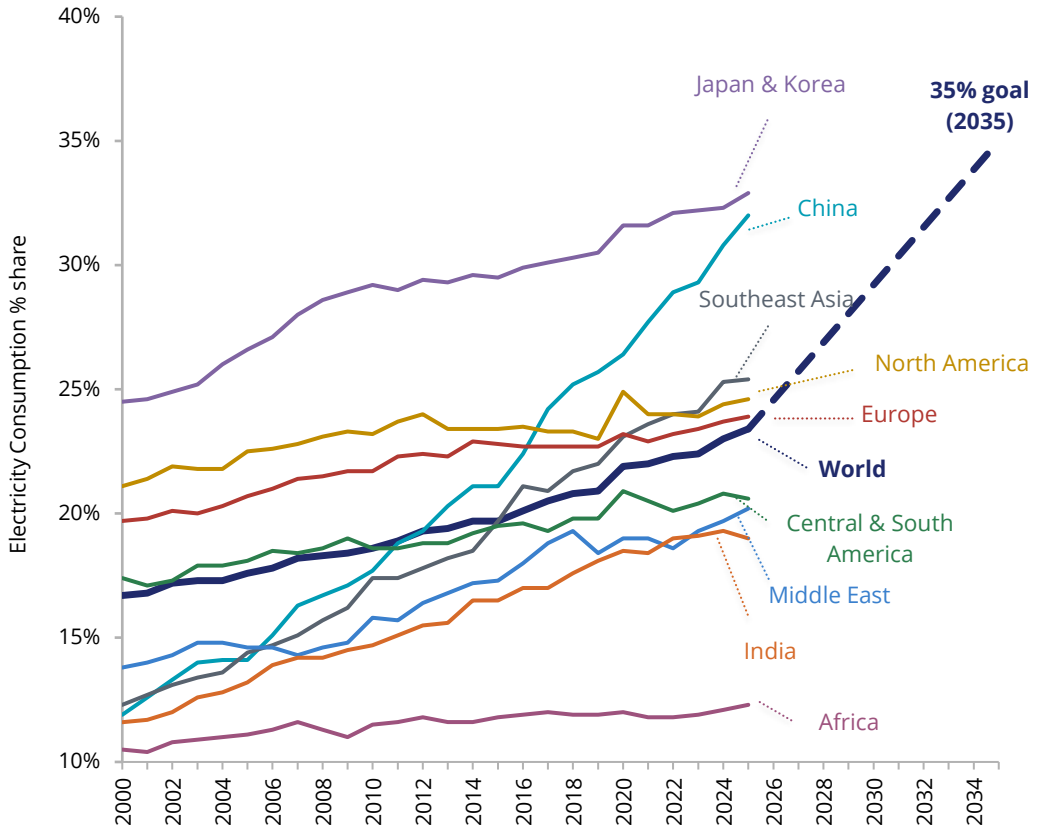
China and the EU have signaled support, but have not yet signed the pledge
Electrification fits modern day energy-security goals making it a viable target for this year's COP. China's cleantech dominance means it has a further incentive to support initiatives that seek to increase the pace of global electrification. It also aligns closely with China's existing development trajectory. While the 35% goal is based on Paris-aligned pathways supported by IEA and IRENA analysis, the numerical target itself does not require the additional electricity to come from non fossil fuel sources. This has invited some criticism, though it may make it easier for some nations to sign the non-binding pledge.

Alignment with the voluntary renewables pledge
Recall that COP28 called for tripling global renewable power capacity by 2030 in COP28's formal Global Stocktake decision, equivalent to around 11.2 TW. IRENA's September 2026 assessment puts capacity at 5.15 TW at end-2025. Record additions of 693 GW in 2025 remain below the 1,200 GW needed annually in 2026–30. The 35-by-35 pledge complements this supply goal by shifting transport, buildings and industry towards electricity, enabling renewable power to replace fossil fuels. Its emissions benefit still depends on cleaner generation.

We view the goal positively, but delivery needs grid enhancements.
IRENA counts ~2,500 GW of renewable, storage and large-load projects awaiting connection and says annual investment to strengthen grid infrastructure and enhance system flexibility must rise from ~$500bn in 2025 to around $1tn through 2035, and $1.2tn thereafter. That said, signatories signal support, but domestic delivery plans will determine investment.

Sources: UNFCCC, COP31 Presidency, IEA, IRENA

Global electrification rates



Source: IEA

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The fossil fuel transition roadmap track is growing outside the formal COP process

COP30 failed to agree a fossil fuel transition roadmap, despite support from more than 80 countries. Brazil's COP30 presidency is therefore developing a roadmap outside formal UN negotiations to be presented ahead of COP31. Separately, Colombia and the Netherlands convened a 'coalition of the willing' of 57 countries in Santa Marta in April 2026 to explore practical steps towards a fossil fuel phase-out. This parallel forum allows willing countries to cooperate without needing consensus from all COP parties.

Evolution of international fossil fuel transition efforts

Dec 2023 | COP28, Dubai
First Global Stocktake calls on countries to transition away from fossil fuels in energy systems in a just, orderly and equitable manner.

Nov 2025 | COP30, Belém
80+ countries back a fossil fuel roadmap, but it is left out of the final decision; Colombia and the Netherlands announce Santa Marta while Brazil promises to develop a global roadmap

24–29 Apr 2026 | Santa Marta, Colombia
First conference: 57 countries and 1,000+ stakeholders, plus a science pre-conference of ~400 academics. France becomes first country to present national fossil fuel phase-out roadmap.

Mid-2026 | London Climate Action Week (Jun 2026) and UNGA (Sep 2026)
Report with the 10-point 'Santa Marta Vision' delivered to the COP30 presidency at London Climate Action Week. Germany unveils national fossil fuel phase-out roadmap at UNGA.

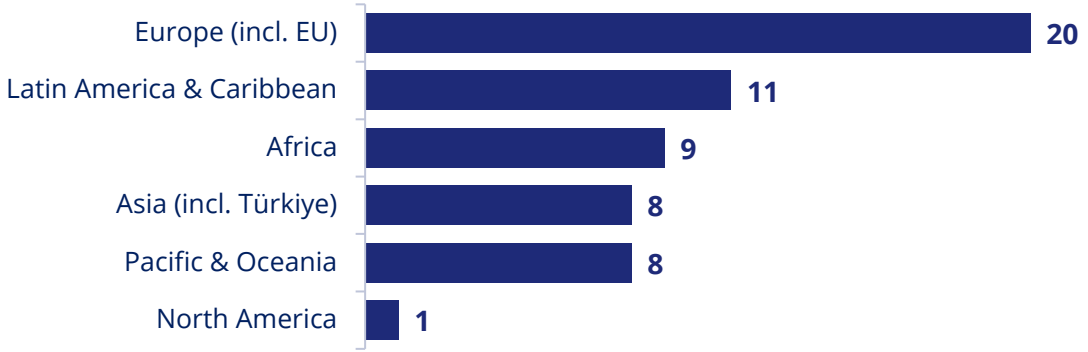
Nov 2026 | COP31, Antalya
COP30 presidency reports on its roadmap prior to the event; COP31 host said in June they were open to reflecting Santa Marta, format not settled

2027 | Second conference, Tuvalu
Co-hosted with Ireland; proposed focus is to plan, finance and transform the real economy

The first conference pinpointed 5 pathways for the fossil fuel transition process

1. **Collective action for the transition away from fossil fuels through global partnership**
2. **Accelerate global progress, spread the spirit of Santa Marta, and achieve system-wide coherence**
3. **Develop or strengthen national and regional roadmaps to transition away from fossil fuels**
4. **Overcoming fossil fuel-related macroeconomic dependencies, scaling up transition investment, & addressing barriers in the international and domestic financial architecture**
5. **Aligning producers and consumers towards decarbonised trade balances and supporting green economic transformation**

Who took part: 57 participants (incl. the EU) by region



SANTA MARTA CO-HOSTS	Colombia and the Netherlands ran the 1st conference (Apr 2026); Tuvalu and Ireland host the 2nd (2027)
OTHER NOTABLE ATTENDEES	the UK, France, Germany and the EU, alongside climate-vulnerable states such as the Marshall Islands and Vanuatu
PRODUCER ATTENDEES	Canada, Norway, Brazil, Nigeria, Angola, Australia and Mexico, major fossil fuel producers, attended
DID NOT ATTEND	US, China, India and Russia, none of which took part (and were reportedly not invited)

Sources: Carbon Brief, Santa Marta conference report

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Trade could resurface at COP31 as a challenge to collective agreement; talks have been requested

Trade tensions could feature at COP31 as governments weigh climate ambition against industrial competitiveness. Developing countries fear that measures such as the CBAM disadvantage their exporters, while the EU says they prevent carbon leakage. COP30 agreed three annual dialogues on trade's role in climate cooperation (no binding outcome), creating for the first time a formal space for trade discussions in the UNFCCC process, but The African Group and others would like it on the formal COP31 agenda.

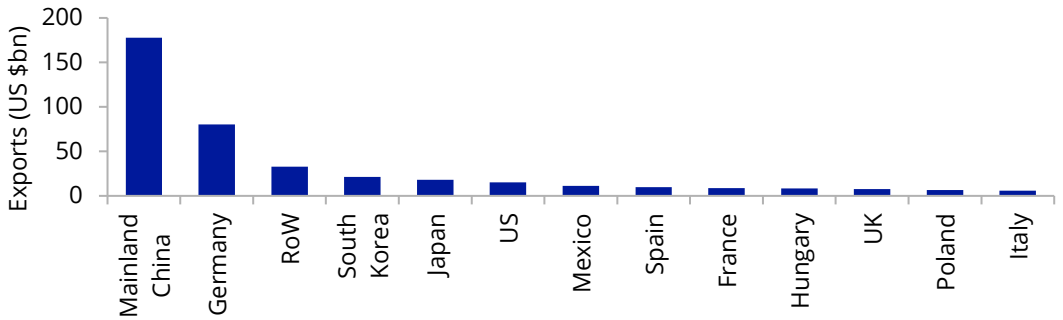
Do trade discussions belong at COP?
Developing countries have long pushed back against climate-related trade measures such as the EU's Carbon Border Adjustment Mechanism (CBAM) and Deforestation Regulation (EUDR). They argue such measures are protectionism disguised as climate policy and disproportionately impact poorer countries. Developed countries view such measures as necessary to level the playing field for domestic firms in cases where foreign competitors face lower environment-related compliance costs relative to domestic industries, while also arguing that COP is not the appropriate forum to discuss trade matters. Despite this, a European Parliament briefing noted trade is a point of "exposure" for the EU at COP31.

COP30's agreement on trade dialogue was notable, but it may not be enough
COP30 agreed three annual climate-trade dialogues at the mid-year sessions in 2026-28, with a high-level event in 2028. The first, in Bonn (13 June), was calmer than expected but exposed the divide: developing countries argued that climate-linked trade measures such as CBAM raise costs and restrict market access and called for impact assessments within the UNFCCC, while the EU defended them as responses to environmental externalities. With trade also raised in other Bonn rooms, it is far from clear the dialogue will contain the issue in Antalya.

The African Group of negotiators has requested a Trade dialogue on the formal agenda.
China has already said it intends to use COP31 to oppose cleantech trade barriers, with its vice minister of the Ministry of Ecology and Environment saying the event should discuss "in depth" the "new problem of unilateralism and protectionism undermining global collective efforts" on climate. The African Group is also pushing for trade to be on the formal agenda. Ultimately, trade talks could prove more consequential at COP31 than in previous years given inflationary pressures and concerns over business competitiveness, but it is hard to assess the potential impact other than possibly causing delays in agreement. However, even if the impact is limited, the issue could shape negotiations elsewhere outside of the COP process.

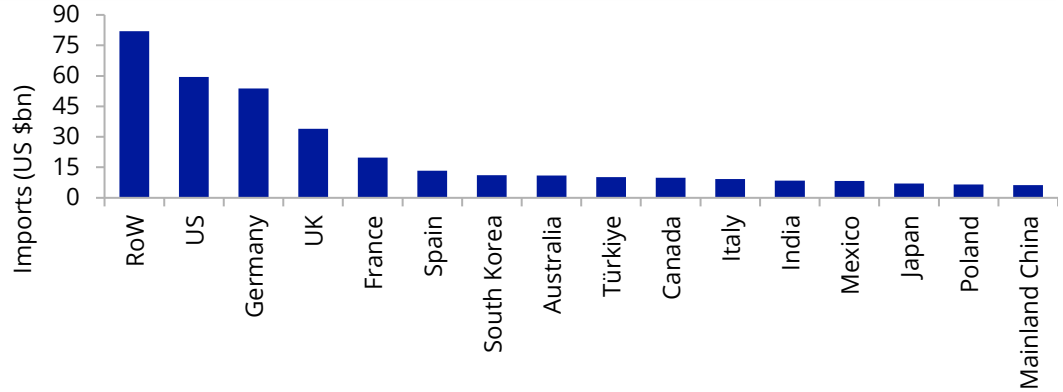
Sources: European Parliament, Carbon Brief, Reuters

2025 Clean tech exports by country (US $bn)



Source: BNEF

2025 Clean tech imports by country (US $bn)



Source: BNEF

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Multilateralism is at a crossroads

Geopolitical fragmentation and the US exit from the Paris Agreement make consensus among nearly 200 nations harder, encouraging "coalitions of the willing" (outside of the formal COP negotiations) to pursue more ambitious climate goals. COP31's success or failure in securing multilateral agreement or credible voluntary pledges could shape future COPs by showing which approaches deliver progress. The US retreat from the Paris Agreement will also continue to have far-reaching implications for climate finance. Overall, the EU and China will be expected to lead.

The consensus agreement principle under strain

The United Nations Framework Convention on Climate Change (UNFCCC) provides the multilateral framework for COP, the forum where countries negotiate collective climate decisions. The process is anchored by the view that all countries must act on climate change, but the extent of their efforts depends on national circumstances. This puts the onus on developed countries to lead. The question is whether this principle still holds against a more fragmented backdrop and transactional diplomatic context as national budgets prioritize security and business competitiveness. With the COP process still separating developed and developing countries by their 1992 UNFCCC classifications, developed countries are continuing to push for wealthier developing nations such as China and Saudi Arabia to play a greater role on finance. This has been a major point of contention at recent summits.

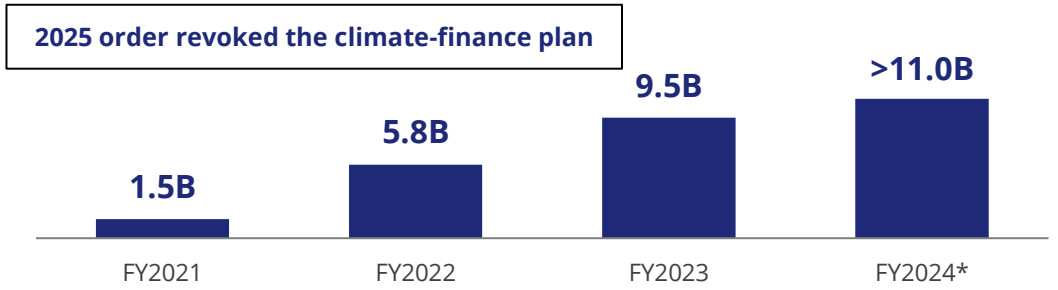
The US stepping back has far-reaching implications

The US skipped COP30 and left the Paris Agreement. It will formally exit the UNFCCC in February 2027. This retreat has weighed on broader climate-finance decisions, both from a monetary sense and through the loss of negotiating power. While cities, states and businesses from the US continue to support COP efforts (e.g., groups like America Is All In), the federal administration's absence has put more pressure on the EU and China to lead.

What does an evolving process look like?

Because formal COP decisions require consensus (or lack of objection), progress often stalls or moves slowly. COP30 saw Brazil move items where there was strong, but not unanimous, support such as the fossil fuel roadmap, to separate streams outside the formal COP process. Countries disappointed with the lack of stronger mitigation action also launched a separate summit on transitioning away from fossil fuels, held this past April in Colombia. While COP is the flagship global forum, these actions demonstrate a shift towards "coalitions of the willing" for issues where global agreement is not possible.

US international public climate finance (US $bn, fiscal years)



Source: Carbon Brief, White House Press Release

COP30 in Belém, where consensus held and where it did not

Notable decisions agreed by consensus	Not agreed
Adaptation finance: triple by 2035 (albeit vague)	**Fossil fuels:** not mentioned in final deal, transition roadmap pushed outside the UN negotiating process
Just transition: new Belém Action Mechanism focused on the fairness of the transition	**Deforestation:** roadmap to halt and reverse it pushed outside the UN process
Adaptation goal: 59 indicators to track progress, finance link still contested	**NDC gap:** no binding response to weak national targets
Gender: New gender action plan agreed	**Finance duties:** developed countries' obligations left to a two year-work programme
Support for implementation: Global Implementation Accelerator and Belém Mission to 1.5°C	**Trade measures (e.g. CBAM):** dialogue agreed, but no measures to resolve concerns

Sources: EU Council, EU Parliament, Carbon Brief, UN, WRI

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Much will rest on the approach of the EU and China

With the US absent, the EU and China are the two actors with the greatest diplomatic and economic leverage with which to broker compromise agreements between the different factions at COP31. This will require careful diplomacy, with their interests not fully aligned in some areas; discussions around trade and climate finance may be particularly challenging. At the same time, both have reiterated their continued support of climate action and multilateral decision making; a successful COP outcome would back up these commitments.

Key COP Groups: Membership and Example Countries

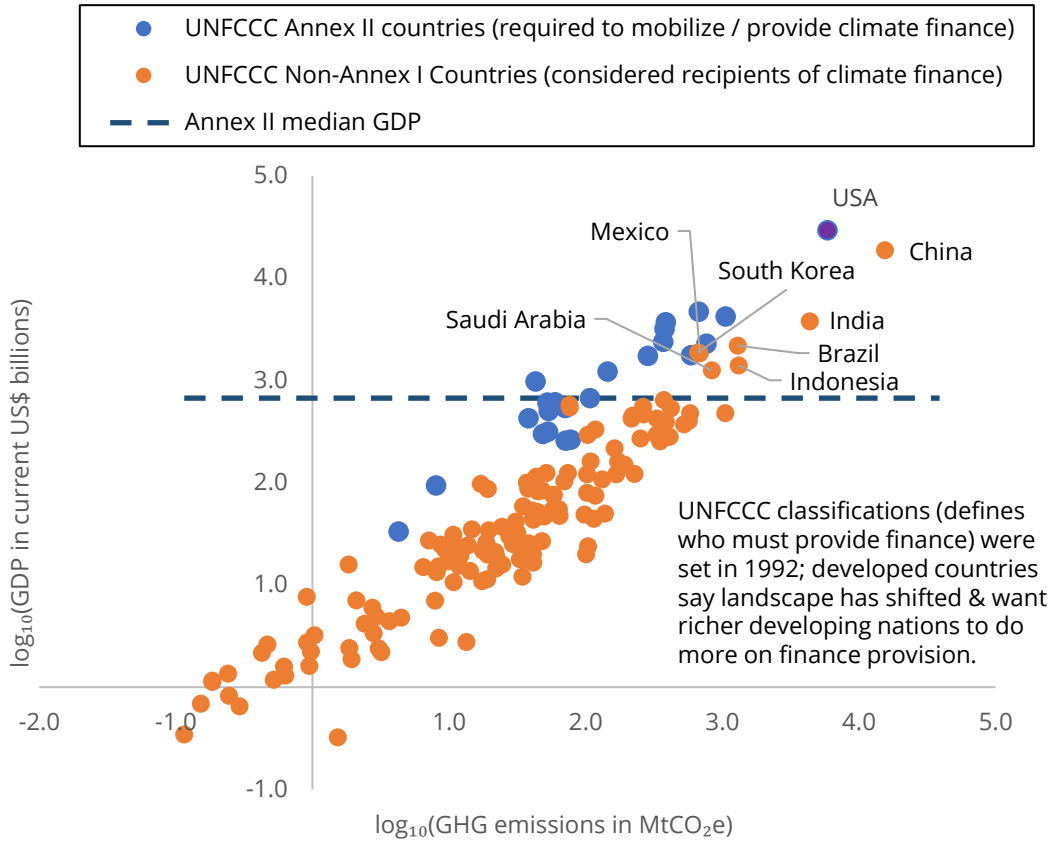
Memberships can overlap (i.e. countries can align with more than one group to best suit their interests)

54 countries	**African Group of Negotiators**	Members include Algeria, Egypt, Ethiopia, Nigeria, South Africa
7 countries	**Independent Association of Latin America and the Caribbean (AILAC)**	Members include Chile, Colombia, Costa Rica, Peru, Panama
39 countries	**Alliance of Small Island States (AOSIS)**	Members include Jamaica, Singapore, Cabo Verde, Papua New Guinea, Suriname
22 countries	**Arab States**	Members include Saudi Arabia, Iraq, UAE, Morocco, Qatar
52 countries	**Coalition for Rainforest Nations**	Members include Democratic Republic of Congo, Honduras, Brazil, China, India
27 countries	**European Union**	Members include Germany, France, Spain, Italy, Poland
134 countries	**G77 and China**	Members include Argentina, Indonesia, South Africa, Vietnam, Iran

Source: United Nations

GDP, Emissions and UNFCCC Classification by Country

(Wealthier Non-Annex I countries are labelled, as is the USA)



Source: World Bank, EDGAR

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The Antalya AI Pledge puts data-centre power demand on the agenda

We expect AI's climate potential and rising electricity demand to make it a prominent COP31 topic, prompting debate over COPs' role in data-centre energy use. The Antalya AI Pledge invites governments and companies to commit voluntarily to sustainable infrastructure and environmental transparency. Its call for zero- and low-emissions energy infrastructure leaves room for nuclear power, prompting discussion of its role in meeting data-centre demand alongside renewables.

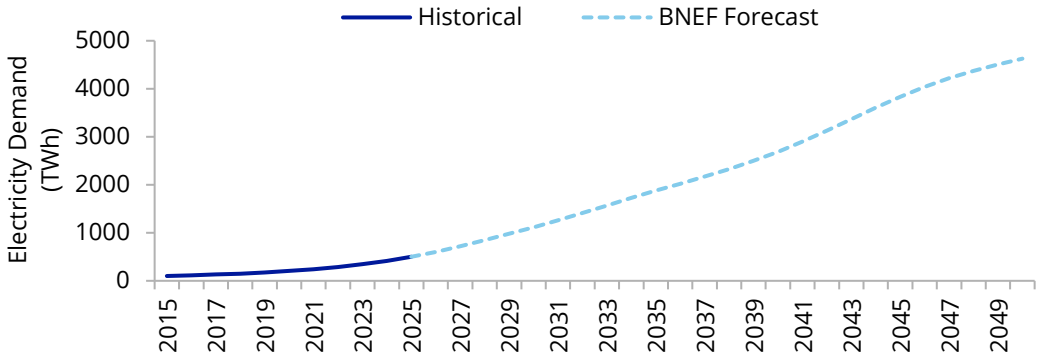
The Antalya AI Pledge seeks to build stakeholder support for sustainable AI buildout
Türkiye is encouraging stakeholders (inc. corporates) to commit to the pledge, asking signatories to "declare our shared ambition to make AI part of the climate solution. This means sustainably powered, efficiently designed, transparently measured and available to all". Signatories will commit to: 1) Power and build AI infrastructure and hardware sustainably, 2) Design, manage or procure AI responsibly across its lifecycle, 3) Support assessment and reporting of the environmental impacts of AI systems, 4) Strengthen financial support and technical infrastructure for AI for climate action.

The AI for Clean Technologies Initiative seeks to highlight AI climate solutions
The initiative plans to develop AI-enabled pilot projects (e.g. smart energy systems) and then link the projects with industry partners and investors, aiming to demonstrate AI's potential to contribute to sustainability goals. For grids, the IEA sees AI improving forecasting, congestion management and outage response, helping operators get more from existing lines.

The dialogue here will set the stage for AI discussions at future COPs
While the pledge is voluntary, many of these themes are already beginning to appear in early-stage regulatory demands, with transparency and disclosure receiving particular focus. There will be questions over how effective the pledge can be given the US has disengaged from the COP process, though California could send a strong signal if it commits to the pledge. We will be particularly focused on how the tech industry responds. Despite the trend of greenhushing, we expect many to send delegations to Antalya, with AI companies and investors viewing responsible practices and community engagement as key to avoiding backlash and delays to AI buildout. Note that responsible AI was also a hot topic at the Principles for Responsible Investment conference this year in October.

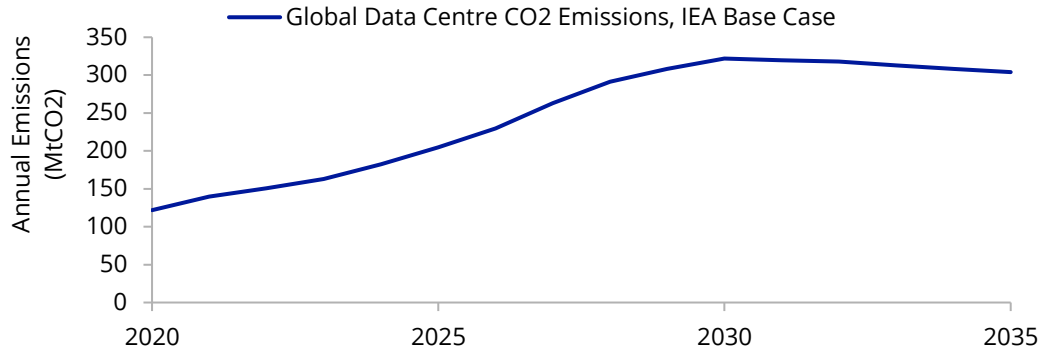
Sources: COP31 Presidency, IEA

Data-centre electricity demand (TWh)



Source: BNEF, Bloomberg Finance L.P.

Global data-centre CO_2 emissions, Base Case ($MtCO_2$)



Source: IEA

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Nature and carbon markets have key roles to play in supporting climate action

Carbon credit discussions will centre on government policies to boost demand and corporate engagement, with much of the formal UN architecture now in place. If carbon markets can scale, the benefits would be wide-ranging (e.g. supporting emissions targets and providing finance for developing countries). Credibility and greenwashing concerns remain key risks however; we expect prices to diverge in favour of high-integrity credits. The Tropical Forest Forever Facility (TFFF) now has ~$7.8bn of sponsor capital against a $10bn end-2026 target.

Voluntary carbon market development

Türkiye has stated a focus on carbon markets at COP31, particularly on government actions to boost demand, highlighting the potential for carbon markets to unlock more than $50bn in additional annual finance for developing nations. The Coalition to Grow Carbon Markets (made up of 13 governments) has announced it will unveil a "Policy Playbook" at COP31 to support market development. While much of Article 6 (carbon credits under UN process) has moved to delivery, countries will aim to reach agreement on the funding model for Article 6.2 at COP31. Further carbon market progress stands to have multiple benefits; it can support developed countries in hitting climate goals while reducing immediate decarbonisation requirements, while also providing finance and supporting ecosystems in developing ones.

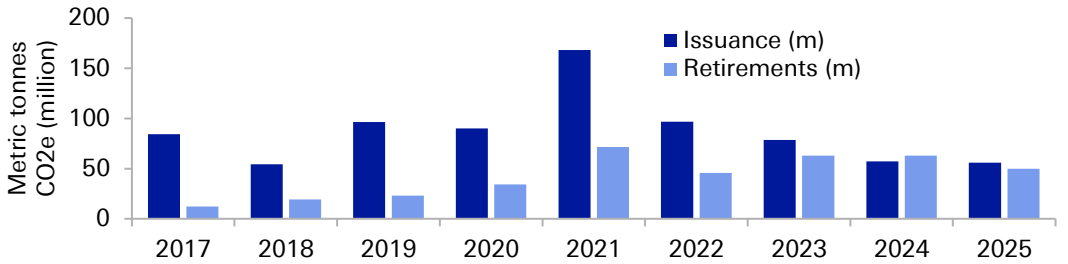
Understanding the pushback

Carbon markets have seen pushback over credit integrity (including regulatory scrutiny) and the risk that offsets delay domestic cuts. UN involvement is needed to enforce common accounting and integrity rules, such as the Article 6 rulebook finalised at COP29 and due for review in 2028, that stop the same reduction being claimed twice. Without that backstop, fragmented bilateral and voluntary schemes risk eroding the trust needed to scale the market.

Update on the TFFF

The Tropical Forest Forever Facility (TFFF), launched at COP30 last year, aims for $10bn of sponsor contributions by year-end. It pays tropical-forest countries per hectare conserved, discounts payments for each hectare lost or degraded, and directs at least 20% to Indigenous Peoples and local communities. Its Charter was adopted on 22 July 2026 and received a tailwind following the re-election of Brazilian President Lula, who is a key supporter. Shortly after Lula's re-election, the Netherlands pledged up to $500mn to the fund, bringing the total pledged to $7.8bn. Further financing commitments are key, with Norway's $3bn contribution conditional on the fund reaching at least $10bn by the end of 2026.

Nature Based Solution carbon credit issuance and retirements



Source: Bloomberg Finance L.P.

How the TFFF works

01 Pool public and private capital

$10bn end-2026 sponsor contribution target (~$7.8bn pledged), with a longer-term goal of $25bn. Sponsor capital aims to attract ~$100bn from private investors for a $125bn fund.

02 Invest capital to generate returns

The fund invests in a diversified fixed-income portfolio. Returns support investor payments & long-term forest conservation funding.

03 Verify forest conservation

Transparent satellite monitoring verifies forest cover in each participating country (eligibility: deforestation below 0.5% a year).

04 Pay countries for conserved forests

Countries earn ~$4 per hectare of standing forest, less $400–800 per hectare deforested; at least 20% goes to Indigenous Peoples and local communities.

Sources: Tropical Forest Forever Facility, Coalition to Grow Carbon Markets

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Oceans will be a priority item, with Australia promising to champion Pacific interests

Australia's ask for ceding the hosting was a Pacific pre-COP, and it has promised to continue championing Pacific interests at COP31. The Taku Pakasoa Declaration, signed at the pre-COP, reflects Pacific priorities, calling for faster emissions cuts, more adaptation finance and financial system reform. The climate-ocean nexus will also be a focus, including investable solutions.

Championing Pacific interests is a key focus for Australia

As part of the hosting compromise, Australia co-hosted the official pre-COP with Fiji and Tuvalu (5–8 Oct), attended by countries including Germany, the UAE and India (though some questioned the low number of international leaders who attended). Australia has pledged to keep championing Pacific interests as President of the Negotiations.

Taku Pakasoa Declaration reflects Pacific Priorities

Signed on 6 Oct at the pre-COP, it calls for faster emissions cuts, a substantial increase in climate finance (particularly for adaptation in capacity-constrained countries), reform of the international financial architecture, and for elevating the ocean-climate nexus. It also welcomes the 35% electrification goal.

The challenges of rising sea levels are gaining increasing prominence

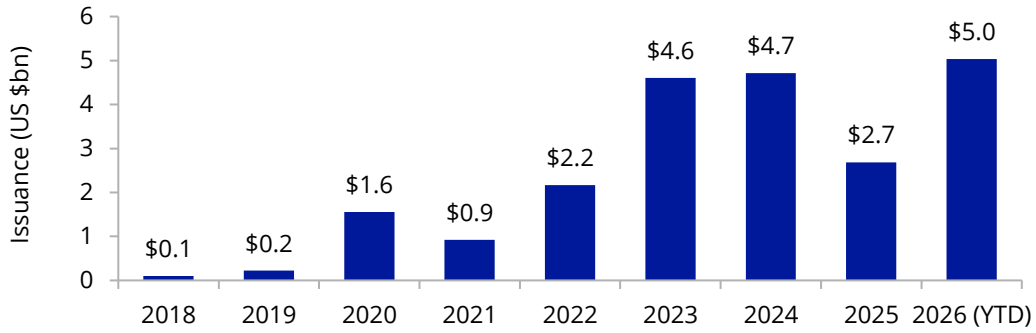
Sea levels in the region are rising faster than the global average, and half of Pacific Island infrastructure lies within 500 metres of the coast. The UN General Assembly adopted the UN Declaration on Sea Level Rise by consensus on 24 Sept; global sea levels rose nearly 6mm in 2024, the highest annual increase on record.

Oceans are key to climate stability, but more finance is needed

Pacific nations emphasise the critical role of oceans as a natural environmental regulator. They absorb 30% of all carbon dioxide emissions (largest carbon sink) and capture 90% of the excess heat generated by these emissions. Investor interest in sustainable ocean solutions is growing, with 2026 YTD blue bond issuance already an annual record, yet bond issuance supporting UN SDG 14: Life Below Water trails many other categories; scaling this up will be a focus at COP31. A commitment to delivering "tangible, bankable ocean outcomes" at COP31 is one of 13 points included as part of the Trabzon Ocean Call to Action unveiled by Türkiye, while the ORRAA has unveiled a Blue Climate Finance Pathway.

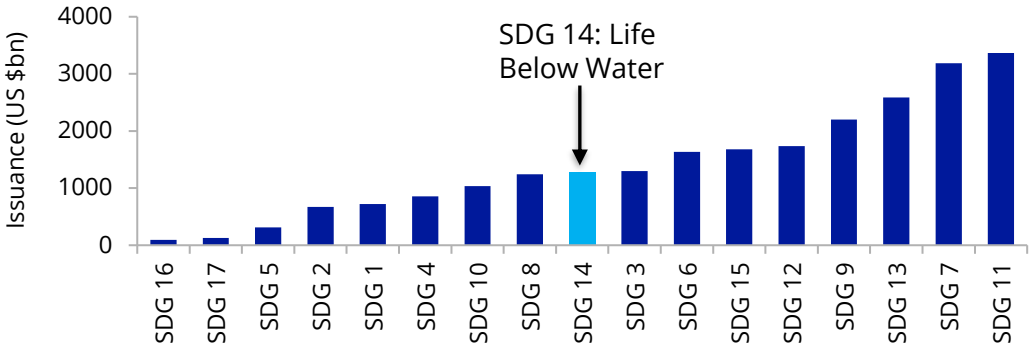
Sources: Taku Pakasoa Declaration (UNFCCC), UN Declaration on Sea Level Rise

Blue Bond issuance (US $bn)



Source: Environmental Finance. Note: Dates classified by settlement date.

Bond issuance supporting each UN SDG (US $bn, 2020-2026 YTD)



Source: Environmental Finance. Note: Dates classified by settlement date. Where bonds recorded contribution to multiple SDGs, the full value was counted in each category.

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Ethiopia will host COP32 in 2027, though questions over how COP will evolve are in focus

COP has shifted from setting targets to implementation. Ethiopia will host COP32 in 2027, where we expect a focus on delivering commitments and climate finance access, reflecting African priorities. COP33 falls to Asia-Pacific in 2028, with China reportedly considering a bid and South Korea potentially interested. COP31–33 will test delivery in a fragmented world, with a second Global Stocktake concluding in 2028. Its outcome should inform countries' next national climate plans (NDCs, their emissions-reduction pledges), due in 2030.

Sources: Official COP and UNFCCC documents (decisions, outcome texts and press releases) and the other sources referenced in this report. Status as at 29 October 2026; COP31 is upcoming.

1992–1997

1992 | Rio: UNFCCC signing

The Convention opens for signature and enters into force in March 1994, establishing the framework for international climate cooperation.

1995 | COP1: Berlin

The first COP adopts the Berlin Mandate, launching negotiations on stronger commitments for developed countries.

1997 | COP3: Kyoto

The Kyoto Protocol establishes legally binding emissions targets for developed countries, an early model centred on differentiated obligations.

2015–2022

2015 | COP21: Paris

The Paris Agreement sets a goal to keep warming well below 2°C above pre-industrial levels and pursue 1.5°C. National climate plans become central to the process.

2021 | COP26: Glasgow

COP26 decisions complete the Paris rulebook, with the cover text (Glasgow Climate Pact) including a call to phase down unabated coal power.

2022 | COP27: Sharm el-Sheikh

Countries agree to establish a loss and damage fund for particularly vulnerable developing countries. COP28 operationalises it in 2023. Funding pledges remain far below estimated needs. This gap is a continuing test of trust in climate-finance delivery.

2023–2024

2023 | COP28: Dubai

The first Global Stocktake calls for a transition away from fossil fuels in energy systems. It provides a collective assessment of progress and informs national climate plans.

2024 | COP29: Baku

The new finance goal calls for at least $300bn a year for developing countries by 2035, with developed countries taking the lead. It sits alongside a broader $1.3tn annual ambition from all sources.

Fossil-fuel follow-through

COP29's text omits the fossil-fuel transition call. Strengthening or even simply reiterating the call remains a contentious topic for nations at COP.

2025–2026

2025 | COP30: Belém

Countries call for efforts to at least triple adaptation finance by 2035 and agree to develop a just transition mechanism.

2026 | COP31: Antalya

Türkiye leads the conference and Action Agenda while Australia leads negotiations. The summit will test whether pledges can be turned into delivery and what mechanisms are most effective. The second Global Stocktake will begin.

Voluntary ambition

Failure to agree fossil fuel phaseout steps at COP30 led Colombia and the Netherlands to launch the Santa Marta process, with 57 nations participating. Brazil's voluntary transition roadmap will also be presented ahead of COP31.

2027

COP32 | Ethiopia

Ethiopia will host COP32 in Addis Ababa after securing unanimous African Group backing.

Delivery and funding access

We expect a focus on delivering commitments and improving access to climate finance, reflecting African priorities. Adaptation finance, loss and damage, and vulnerable countries' needs could also see focus.

Stocktake technical assessment

The technical assessment begins in June 2027, feeding evidence into the final political discussions rather than constituting the stocktake outcome.

2028

COP33 | Asia-Pacific

The Asia-Pacific region will host COP33. India withdrew its offer in April 2026. There is potential interest from South Korea, and China is reportedly considering a bid.

Second stocktake concludes

The second Global Stocktake concludes at COP33 in November 2028. Negotiations shape the final outcome, which informs the next round of national climate plans due in 2030.

Wider context

A November summit could coincide with the US elections. COP reform and the effectiveness of voluntary coalitions will remain important questions.

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Appendix A: Key COP terms used in this report

This appendix explains the main COP process terms and negotiating concepts used throughout the report. Definitions are summarised for non-specialist readers and reflect how each term is used in this deck.

TERM	DEFINITION
Action Agenda	Voluntary, non-binding pledges and initiatives launched alongside the formal talks. It is separate from negotiated decisions.
Annex II / Non-Annex I	UNFCCC country classification. Annex II covers developed economies; Non-Annex I covers most other countries, including China and India.
Article 6	Paris Agreement framework for international carbon markets. Article 6.2 covers bilateral trading of carbon credits between countries.
Article 9	Paris Agreement finance provisions. Article 9.1 covers developed countries' obligation to provide finance. COP30 set a two-year work programme for 2026-27.
Baku to Belém Roadmap to $1.3T	Roadmap to scale climate finance to $1.3tn a year by 2035 from all sources, inclusive of the $300bn NCQG.
Bonn sessions	Mid-year technical talks of the UNFCCC Subsidiary Bodies (e.g. SB64) in Bonn, Germany, which prepare the ground for COP discussions.
Brazil fossil fuel roadmap	COP30 Presidency's voluntary roadmap for transitioning away from fossil fuels, pursued outside the formal UN negotiations.
Consensus	How COP decisions are taken: adopted when no party formally objects, so a single party can theoretically block or dilute an outcome.
COP / COPs	Conference of the Parties: the UNFCCC's annual decision-making summit, where nearly 200 parties negotiate climate action.
COP Presidency	The host country's team that chairs the conference. For COP31, Türkiye's Murat Kurum is COP President and leads the Action Agenda.

TERM	DEFINITION
Final COP text	Decisions adopted by consensus at the close of the COP, setting its formal outcome.
Formal agenda	Negotiated items on the official COP agenda, decided by consensus. Separate from the voluntary Action Agenda.
Global Goal on Adaptation (GGA)	Paris Agreement goal to strengthen resilience. COP30 adopted 59 indicators; the link to finance is contested.
Global Stocktake (GST)	Five-yearly review of progress under the Paris Agreement. The second runs from COP31 to COP33 (2028).
Just transition mechanism	Belém Action Mechanism agreed at COP30, focused on fairness in the shift to a low-carbon economy.
Loss and damage fund	Fund set up at COP27 and operationalised in 2023 to support vulnerable developing countries.
NCQG	New Collective Quantified Goal (COP29): at least $300bn a year for developing countries by 2035, within a $1.3tn ambition.
NDC	Nationally determined contribution: each country's national climate plan, with emissions targets and actions, updated every five years.
Negotiating groups	Blocs that coordinate national positions, such as the G77 & China, the EU, LDCs and AOSIS.
Paris Agreement	2015 treaty to hold warming well below 2°C and pursue 1.5°C, with national climate plans (NDCs) as its core mechanism.
Pre-COP	Ministerial meeting ahead of the COP to test positions and build consensus. For COP31, Australia co-hosted a Pacific pre-COP.
President of Negotiations	COP31 role held by Australia's Chris Bowen, who drafts final texts and steers government-to-government talks.
UNFCCC	UN Framework Convention on Climate Change: the 1992 treaty behind the climate process. Its parties meet annually at the COP.

Sources: UNFCCC; COP30 and COP31 Presidency documents; and the other sources referenced in this report

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Appendix B: COP31 thematic days and calendar

COP31 runs from 9 to 20 November 2026 in Antalya. The COP31 Presidency has assigned each of the twelve conference days a thematic focus. Page references show where this report covers the related topic. The schedule is indicative and may change.

MON	TUE	WED	THU	FRI	SAT	SUN
9 Nov Day 1 **Food, Agriculture and Health** CONFERENCE OPENS	**10 Nov** Day 2 **Energy and Transport** *See page 10*	**11 Nov** Day 3 **Zero Waste** LEADERS SUMMIT DAY 1	**12 Nov** Day 4 **Resilient Cities and Built Environment** LEADERS SUMMIT DAY 2 *See pages 8–9*	**13 Nov** Day 5 **Finance and Trade** *See pages 7, 12*	**14 Nov** Day 6 **Children, Youth, Education and Skills**	**15 Nov** Day 7 **A Breath in Antalya** SOCIAL AND CULTURAL PROGRAMME
16 Nov Day 8 **Science, Industry and Technology** *See page 15*	**17 Nov** Day 9 **Ocean, Seas, Nature and Land Use (Rio Synergy)** *See pages 16–17*	**18 Nov** Day 10 **Human and Social Development**	**19 Nov** Day 11 **İmece: Enhancing Implementation**	**20 Nov** Day 12 **Final Negotiations** SCHEDULED CLOSING DAY		

KEY

- Thematic day
- World Leaders Climate Action Summit
- Rest day / cultural programme
- Final negotiations

See page: related topic in this report

5–8 Oct
Pre-COP
Ministerial meeting and Leaders' event hosted by Fiji, Tuvalu and Australia.

2–8 Nov
Pre-sessional period
Coordination and preparatory meetings ahead of the formal opening in Antalya.

9–20 Nov
COP31 in Antalya
COP31, CMP21 and CMA8, with SBSTA 65 and SBI 65 (Antalya EXPO Center).

11–12 Nov
World Leaders Climate Action Summit
Heads of state and government convene on the third and fourth days of the conference.

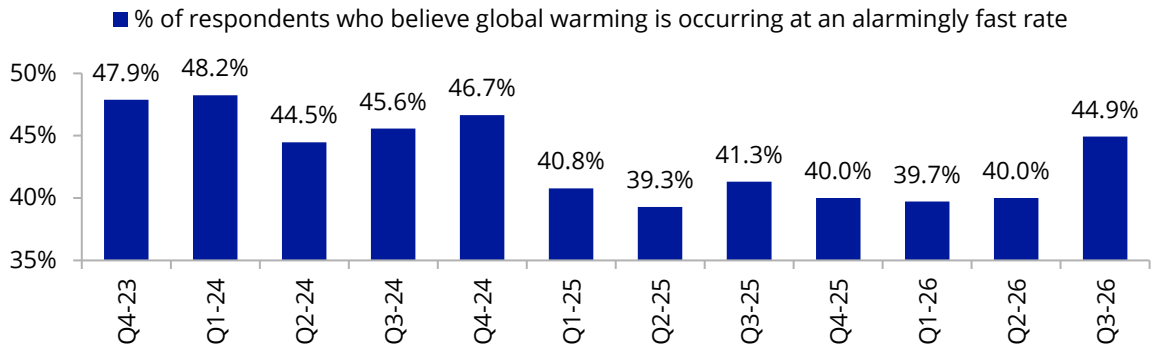
Sources: COP31 Presidency; UNFCCC COP 31 overview schedule (version of 7 October 2026)

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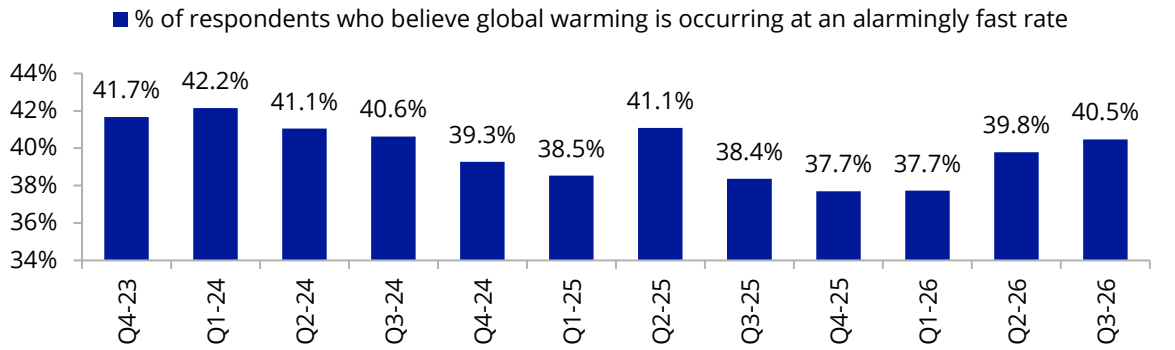
Appendix C: The level of alarm over global warming has begun to move higher (again)

As part of our monthly Climate & Sustainability Survey, we ask consumers in the UK, US, Germany, France, Spain and Italy for their views on global warming. In Q3 2026, we recorded a notable rebound in alarm regarding the pace of global warming, particularly in the UK and Europe following a summer of climate extremes. The level of alarm had begun to dip globally post the 2024 US presidential election in November.

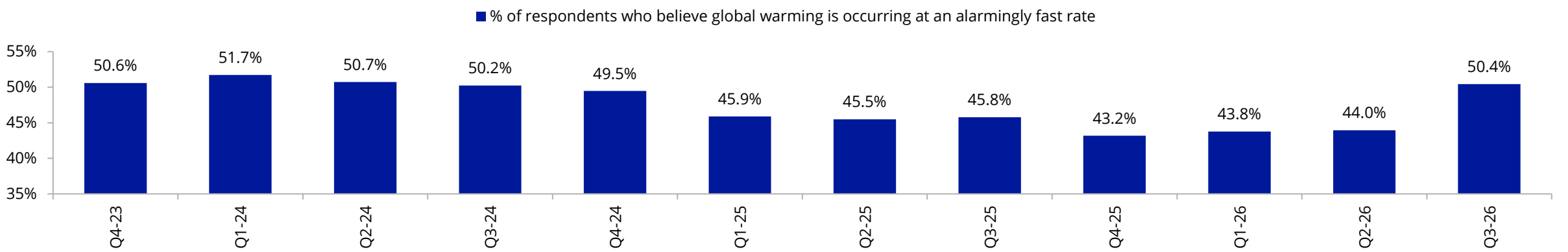
UK



US



Europe (GER, FRA, ESP, ITA)



Sources: dbDataInsights, Deutsche Bank Research.
Note: Data for both Q1-26 and Q2-26 contains only two months.

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Appendix D: Deutsche Bank Disclosures

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This Month in Geopolitics

November 2026: Key events, themes and trends to watch in the month ahead

30 October 2026

Dr. Helen Belopolsky | Miha Hribernik

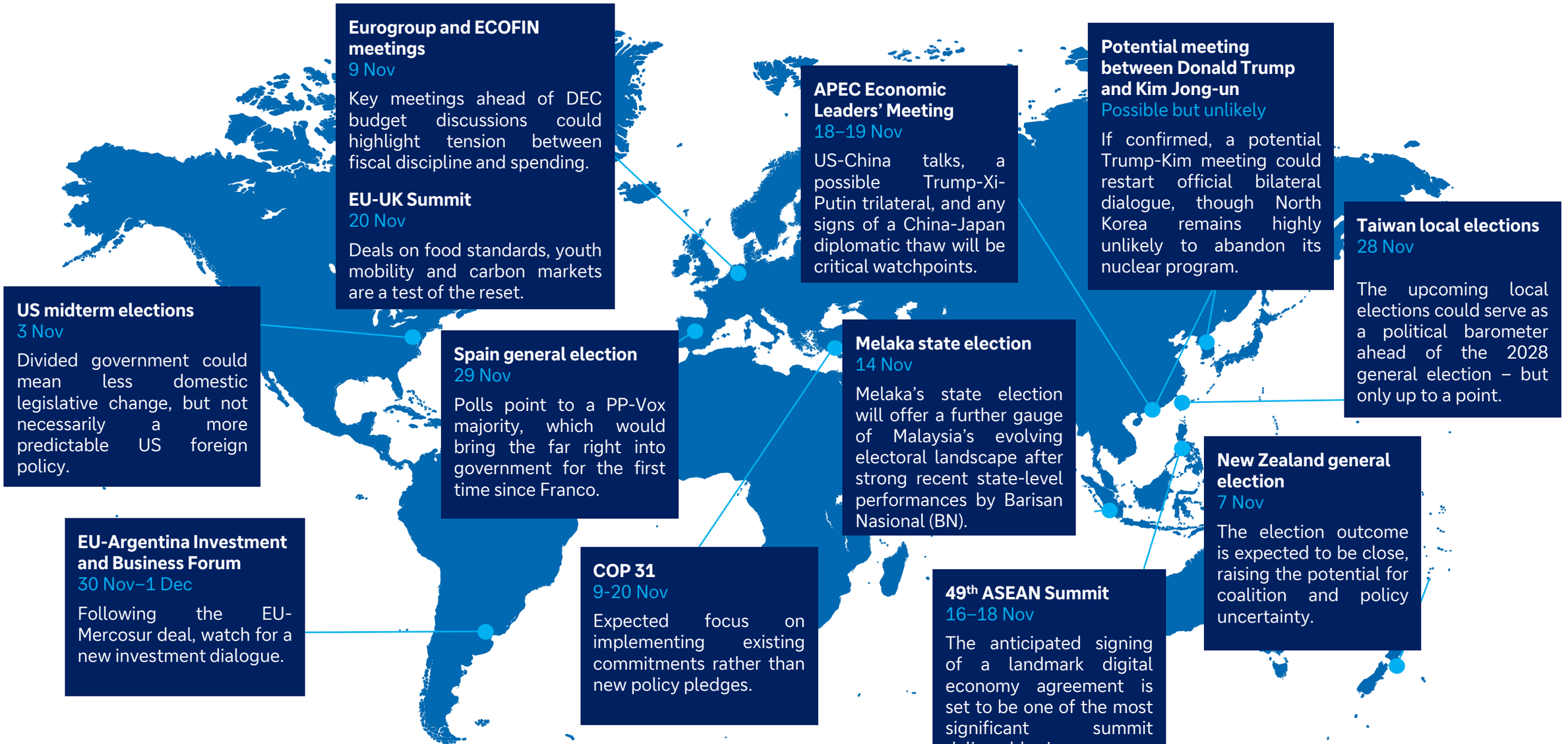
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IMPORTANT RESEARCH DISCLOSURES LOCATED IN APPENDIX 1.

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At a Glance: Geopolitical events to watch in **November 2026**

This Month in Geopolitics highlights key events, issues, and themes for the month ahead. The map below provides an at-a-glance summary, with detailed analysis provided in the subsequent slides. For a longer-term outlook (November 2026 – January 2027), please refer to our latest 90-day Geopolitical Risk Radar.



Eurogroup and ECOFIN meetings
9 Nov

Key meetings ahead of DEC budget discussions could highlight tension between fiscal discipline and spending.

EU-UK Summit
20 Nov

Deals on food standards, youth mobility and carbon markets are a test of the reset.

APEC Economic Leaders' Meeting
18–19 Nov

US-China talks, a possible Trump-Xi-Putin trilateral, and any signs of a China-Japan diplomatic thaw will be critical watchpoints.

Potential meeting between Donald Trump and Kim Jong-un
Possible but unlikely

If confirmed, a potential Trump-Kim meeting could restart official bilateral dialogue, though North Korea remains highly unlikely to abandon its nuclear program.

Taiwan local elections
28 Nov

The upcoming local elections could serve as a political barometer ahead of the 2028 general election – but only up to a point.

US midterm elections
3 Nov

Divided government could mean less domestic legislative change, but not necessarily a more predictable US foreign policy.

Spain general election
29 Nov

Polls point to a PP-Vox majority, which would bring the far right into government for the first time since Franco.

Melaka state election
14 Nov

Melaka's state election will offer a further gauge of Malaysia's evolving electoral landscape after strong recent state-level performances by Barisan Nasional (BN).

New Zealand general election
7 Nov

The election outcome is expected to be close, raising the potential for coalition and policy uncertainty.

EU-Argentina Investment and Business Forum
30 Nov–1 Dec

Following the EU-Mercosur deal, watch for a new investment dialogue.

COP 31
9-20 Nov

Expected focus on implementing existing commitments rather than new policy pledges.

49th ASEAN Summit
16–18 Nov

The anticipated signing of a landmark digital economy agreement is set to be one of the most significant summit deliverables in years.

In the spotlight for November 2026: US midterms and APEC Summit

US midterm elections (3 November)

A divided government could mean less change at home but more abroad. Third-party polling and betting markets forecast Democrat control of the House, and an increasing probability of flipping the Senate. 80% of respondents in our monthly poll now say living costs will influence their vote – the most since April. For markets, a divided government could mean less domestic legislative change, but not necessarily a more predictable US foreign policy. Losing Congress could leave the administration's geopolitical and geoeconomic toolkit largely intact.

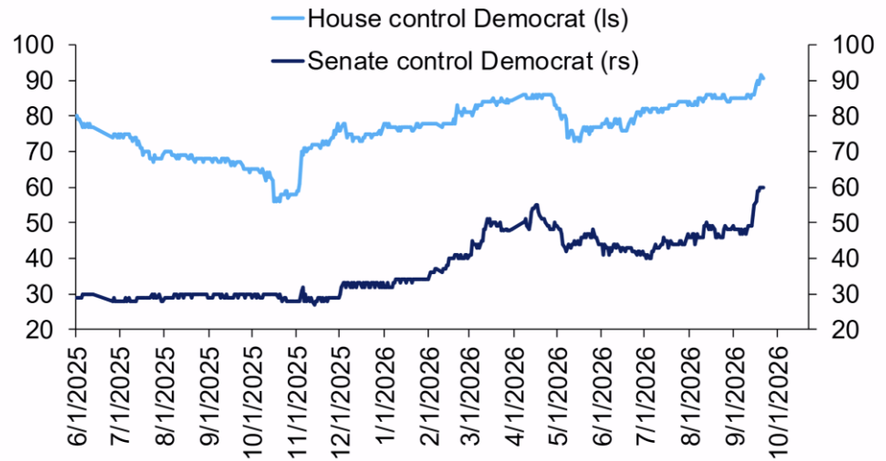
In our view, **weapons stockpiles, inflationary pressures and long-end yields may matter more than Congress in constraining the administration's choices** in 2027 and 2028. A Democratic House could dial up oversight of the administration, launch investigations, force votes on War Powers Resolutions, and use appropriations bills to press for restrictions on departmental funding. This may generate a steady stream of attention-grabbing headlines, but whether it decisively affects the event-driven risk that matters for markets (e.g. blocking a tariff or preventing a limited military strike) remains in doubt. If domestic policymaking becomes more challenging, foreign policy could become a top priority for the Trump administration.

APEC Economic Leaders' Meeting in Shenzhen (18-19 November)

US-China talks, a possible Trump-Xi-Putin trilateral meeting, and any signs of a China-Japan diplomatic thaw will be critical watchpoints. The priorities of the summit between representatives of 21 Asia-Pacific economies are set to include efforts to boost innovation, digital connectivity, supply-chain resilience and support for an open multilateral trading system. During the summit, presidents Trump and Xi are expected to meet for the third time this year. This meeting will provide signals on the state of China-US talks, including the operationalisation of new bilateral trade and investment boards, and on the likelihood of another extension of the Busan trade truce beyond January 2027. President Trump has also indicated he is open to a trilateral US-China-Russia meeting during the summit, possibly covering nuclear arms control and the Russia-Ukraine war.

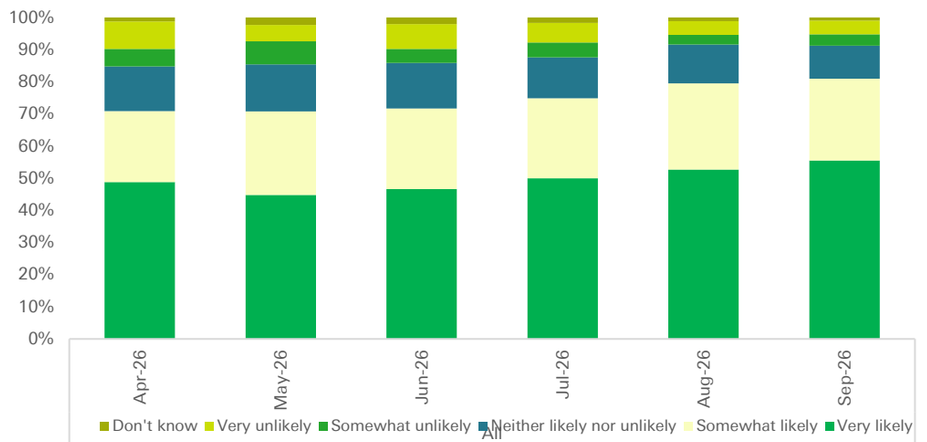
Investors will also closely watch whether Japanese Prime Minister Takaichi and President Xi meet on the sidelines, following reports that Tokyo is hoping to arrange a meeting. However, the bar for a formal breakthrough will likely remain high. As we noted in a report earlier this year, a full normalisation of relations and an easing of Chinese economic restrictions – which include rare-earth export controls – may require a lengthy period of diplomatic re-engagement and backchannel diplomacy.

Implied probability of Democrat control post 2026 mid-term election



Source: Kalshi, Bloomberg Finance LP. See: US Economic Chartbook: US midterm elections: Senate race heats up in the home stretch

US participants were asked: How likely or unlikely is it that current cost-of living concerns will influence your vote in the midterms?



Source: dbDataInsights. Responses collected April-September 2026 from an Online Nationally Representative sample of ~600 US respondents.

Key events, issues and themes to watch in November 2026

Listed in chronological order

New Zealand general election (7 November)

New Zealand's general election is shaping up as a close and fragmented contest. Both the ruling National and opposition Labour parties are polling at historically low levels, and smaller parties will likely determine the next government. In a closely contested vote, investors could see coalition uncertainty and a wider range of possible outcomes for tax, spending, infrastructure, energy and migration policy.

The National party is campaigning on a platform that would broadly preserve the current emphasis on fiscal restraint, lower regulation and private-sector investment. A Labour-led government could place greater weight on public services, housing and climate policy, alongside a proposed capital-gains tax on investment property. However, because any government will rely on minor parties for survival, the final policy mix will depend heavily on post-election negotiations.

Eurogroup and Economic and Financial Affairs Council (ECOFIN) meetings (9 November)

The meeting will be a test of the EU's ability to **translate the competitiveness agenda into practical reforms**. The focus will be on Europe's ability to restore competitiveness, mobilize investment and maintain fiscal discipline. The Eurogroup brings together Euro-area finance ministers and is expected to consider macroeconomic and fiscal developments, energy market pressures and progress towards the Savings and Investments Union. ECOFIN brings together all EU finance ministers, and will focus on financial integration, banking sector competitiveness, and economic coordination. To watch will be tensions between fiscal consolidation and the need to fund defence, energy security and strategic investment, alongside progress on integrating Europe's fragmented capital markets and banking sectors. A budget-focused Council meeting follows on 10 November.

COP 31 in Türkiye (9-20 November)

Implementation of existing commitments, rather than new policy pledges, is expected to be the primary focus of the next climate summit in Antalya. Building on the shift towards implementation at COP30, COP31 will focus on translating existing climate commitments into practical action, including through investment, financing and partnerships. For investors, the key watchpoint will be whether the summit can deliver tangible commitments that translate existing plans into binding funding pledges for critical infrastructure resilience, scaled-up renewable energy output, and climate finance.



Source: Jill Beckmann on Unsplash

Key events, issues and themes to watch in November 2026

Listed in chronological order

Melaka state election (14 November)

Melaka's state election will offer a further gauge of Malaysia's evolving electoral landscape after strong recent state-level performances by Barisan Nasional (BN). The contest will also be watched for its read-through to Prime Minister Anwar Ibrahim's Pakatan Harapan (PH), following setbacks in other state elections this year.

BN enters the election from a position of relative strength, while its electoral understanding with PAS and its allies could provide an early indication of how political alignments are evolving ahead of the next general election. However, Melaka's local dynamics mean the result should not be treated as a direct national forecast.

49th ASEAN Summit (16-18 November)

The anticipated signing of a landmark digital economy agreement is set to be one of the most significant summit deliverables in years. The 49th ASEAN Summit in Manila will focus on economic integration, energy security, supply-chain resilience, maritime security, the crisis in Myanmar, and relations with external partners.

For markets, the formal signing of the ASEAN Digital Economy Framework Agreement (DEFA) is the key event to watch. Designed as a legally binding treaty, DEFA aims to harmonize regional digital trade rules and cross-border data flows. Successful implementation is expected to double the projected size of the region's digital economy by 2030, boosting it from a projected baseline of US$1 trillion to US$2 trillion (up from approximately US$300 billion today).

EU-UK Summit (20 November)

The second EU-UK Summit will be an important **test of whether the post-Brexit relationship** can move towards tangible outcomes. Prime Minister Andy Burnham will seek to finalize agreements on food and agricultural standards, youth mobility, and linking emissions trading systems. Burnham has framed relations as "not good enough." To watch will be whether we see concrete deals versus declarations, and whether leaders are prepared to make the concessions necessary to deepen ties.



Source: istockphoto

Key events, issues and themes to watch in November 2026

Listed in chronological order

Taiwan local elections (28 November)

The upcoming local elections could serve as a political barometer ahead of the 2028 general election – but only up to a point. Local elections in Taiwan are primarily driven by municipal 'bread and butter' economic issues; geopolitics and cross-Strait relations have historically not been at the forefront of voters' concerns. While this divergence limits the elections' predictive value for the January 2028 general vote, the results can provide a sense of the domestic balance of power.

Elections in Taiwan's six special municipalities – Taipei, New Taipei, Taichung, Tainan, Taoyuan, and Kaohsiung – will serve as particularly important watchpoints. Some recent polling suggests a potentially close contest in Taipei between incumbent mayor Chiang Wan-an (Kuomintang) and legislator Shen Po-yang (Democratic Progressive Party).

Spain general election (29 November)

Spain's snap general election could mark a **major turning point for Spanish politics** and the EU, potentially ending eight years of Socialist-led government under Pedro Sánchez and bringing the conservative People's Party (PP) to power with support from the far-right Vox. Polling suggests PP is ahead and a PP-Vox alignment is the only one projected to reach a majority. A rightward shift in one of the EU's largest economies could occur just as Europe enters a wider electoral cycle, potentially complicating consensus on migration, climate policy, Ukraine and European integration.

EU-Argentina Investment and Business Forum (30 November – 1 December)

The Forum comes at a **pivotal moment in EU-Latin American relations**. It is the EU's first big commercial push into Argentina since the EU-Mercosur trade deal began to be provisionally applied on 1 May. The agreement opens new opportunities for trade, investment and market access. The EU is Argentina's largest source of foreign direct investment. Around 800 government, business and financial leaders will focus on energy, critical minerals, agriculture and infrastructure. The EU's Global Gateway investment agenda provides a framework for mobilizing public and private capital. While a key outcome will be the launch of a structured EU-Argentina investment dialogue, to watch will be the ability of Argentine officials to allay concerns around policy stability and regulatory predictability.



Source: Nestor Barbitta on Unsplash

Key events, issues and themes to watch in November 2026

Listed in chronological order

Potential meeting between Donald Trump and Kim Jong-un (possible but increasingly unlikely at time of writing)

A potential Trump-Kim meeting in November could restart official bilateral dialogue, though North Korea remains highly unlikely to abandon its nuclear program. South Korea's National Intelligence Service said in late September that it saw a roughly **50% chance** of US-North Korea talks around the APEC Summit in Shenzhen, although **more recent reporting suggests the prospects may have diminished**. If talks do resume, they would only mark the first step in a long and highly uncertain diplomatic process.

As we noted earlier this year, North Korea is highly unlikely to agree to denuclearization. Unless renewed negotiations sidestep denuclearization in favor of a risk-reduction or coexistence framework, dialogue is likely to stall, not least given Pyongyang's constitutional pledge to continue expanding its nuclear arsenal. This would make even a negotiated freeze on further expansion difficult to achieve.



Source: istockphoto

Appendix 1

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Research

The Next 90: Your Geopolitical Risk Radar
Nov 2026–Jan 2027

October 2026

Miha Hribernik | Dr. Helen Belopolsky

Deutsche Bank Research Institute

IMPORTANT RESEARCH DISCLOSURES LOCATED IN APPENDIX 1.

Deutsche Bank
Research Institute

The Next 90: Your Geopolitical Risk Radar | Nov 2026–Jan 2027

A geopolitical calendar mapping the next 90 days

NOVEMBER

Early (TBC)	Ukraine/US/ Russia	Trilateral peace talks expected in Abu Dhabi
3	US	Midterm elections (House, Senate, gubernatorial and state legislatures)
4	South Africa	Municipal elections
4-5	Spain	Ibero-American Summit
7	New Zealand	General election
9	EU	Eurogroup and Economic and Financial Affairs Council (ECOFIN) meetings
9-20	COP31	COP31 in Türkiye
14	Malaysia	Melaka state election
16-18	ASEAN	49th ASEAN Summit in Manila
16-18 (TBC)	Quad	Quad Leaders' meeting in Manila
18-19	APEC	APEC Economic Leaders' Meeting in Shenzhen; Trump and Xi to meet on sidelines
20	EU/UK	EU-UK Summit
27	France	S&P sovereign credit rating review
28	Taiwan	Local elections
29	Spain	General election
29	OPEC+	Ministerial meeting
30-1 Dec	EU/Argentina	EU-Argentina Investment and Business Forum in Buenos Aires
TBC	US/North Korea	Possible meeting between Donald Trump and Kim Jong-un

DECEMBER

First half of Dec	China	Politburo meeting followed by annual Central Economic Work Conference
3-4	OSCE	Ministerial Council in Switzerland
4	US	Section 232 tariffs (15% ad valorem) and minimum import prices on polysilicon, derivatives and solar products set to enter into force
5	The Gambia	Presidential election
5-6	Doha Forum	Qatar's flagship dialogue themed "Redefining Global Trust"
10-11	EU	Eurogroup and Economic and Financial Affairs Council (ECOFIN) meetings
11	US	Stopgap funding expires; shutdown risk
14	EU	Foreign Affairs Council meeting
14-15	G20	G20 Summit in Miami
17-18	EU	European Council Summit
31	OPEC+	Expiry of the current group-wide production framework
TBD	US	US Congress is set to adopt annual-must pass defence legislation

JANUARY

1	G20, G7, EU, ASEAN, BRICS+	New rotating presidencies of the EU (Lithuania), G7 (US), G20 (UK), ASEAN (Singapore) and BRICS+ (China)
1	US	Stricter US defence-procurement rules take effect for covered critical materials sourced from China, Russia, Iran and North Korea, including tighter waiver requirements
1	Canada/US	50% US tariff on Canadian automotive imports is set to enter into force
3	US	Convening of the 120th United States Congress
10	China/US	Busan trade truce expiry
16	Nigeria	General election
18-22	WEF	World Economic Forum annual meeting in Davos
27	Kyrgyzstan	Presidential election
30	Germany	Presidential election

Note: The calendar contains selected key events and is not intended to provide an exhaustive list.

Appendix 1

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LatAm's silicon dividend: tailwinds from the AI investment boom in the U.S.

October 28, 2026

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Key messages:

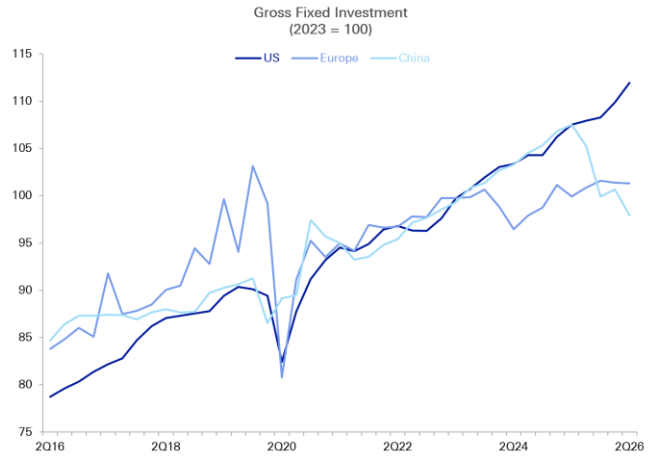
- U.S. investment has broken away from Europe and China since 2023, powered by AI-related equipment, software and research spending.
- The AI CapEx boom demands commodities and merchandise that LatAm can supply.
- Latin America's export gains are real but concentrated. Copper, computer and electronics products provide the clearest links to the U.S. AI buildout.
- Chile is the region's most direct copper beneficiary. Mexico captures the windfall stemming from increases hardware demand and trade-diversion. In these cases, the AI-related tailwinds have undergirded appreciable improvements in external accounts.
- The cases for nickel and zinc illustrate that the dividend is not widespread. U.S. imports of the former have mostly returned toward pre-pandemic levels, while for the latter bilateral gains partly reflect supplier substitution.
- While expanding AI use, supply shortages and incentives can extend the CapEx boom, its financial viability poses downside risks.

Another 'American exceptionalism' story

Investment has become one of the sharpest dividing lines among the world's core economic blocs. Since 2024, U.S. gross fixed capital formation has firmed up appreciably. Meanwhile, investment in Europe has broadly stalled while in China's it has retreated, the drivers of which are beyond the scope of this report (Figure 1).

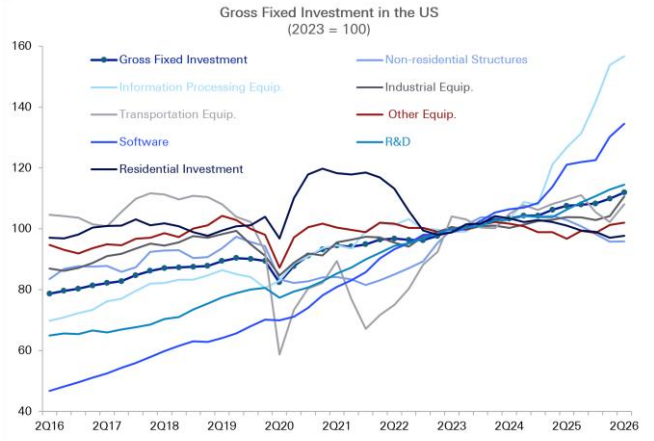
The U.S. divergence has largely been an AI story: a burst of spending on computing power, digital infrastructure and intellectual property has offset weakness in conventional construction and housing. Indeed, total U.S. fixed investment is nearly 12% above its 2023 level. Within it, investment in information-processing equipment has risen by roughly 57%, software outlays by about 35%, and research and development by around 15%. By contrast, non-residential structures and residential investment sit below their 2023 levels (Figure 2).

Figure 1: The global investment divergence



Source: US Bureau of Economic Analysis, Eurostat, National Bureau of Statistics of China, Haver Analytics and Deutsche Bank Research

Figure 2: AI-related categories lead U.S. investment



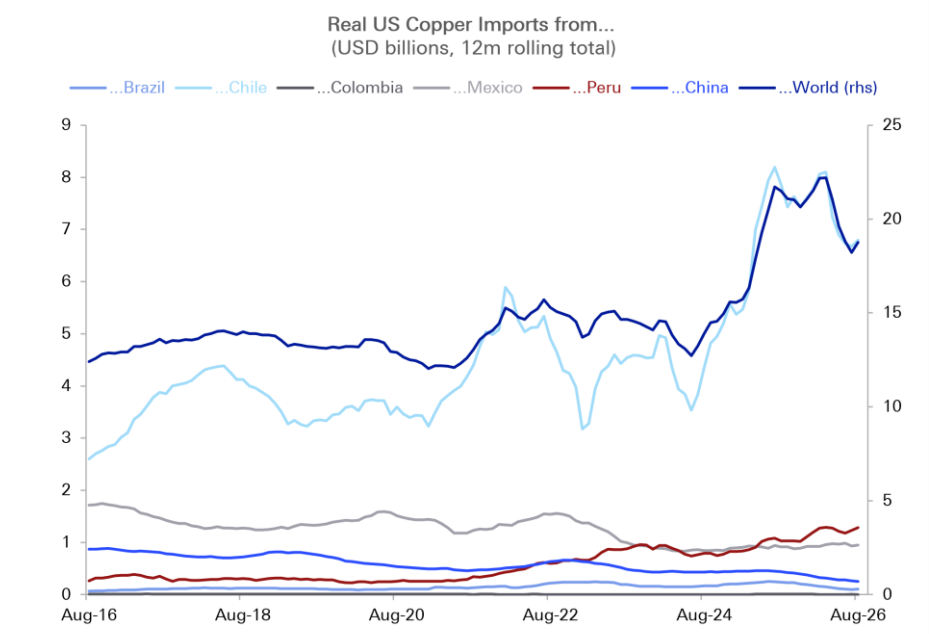
Source: US Bureau of Economic Analysis, Haver Analytics and Deutsche Bank Research

Latin America's commodity channel

AI is not merely a semiconductor story. Chips require specialized minerals; data centers require electricity and cooling; electricity requires grids, transformers and storage. That chain pulls in copper, aluminum, silver, tin, lithium and selected semiconductor inputs, along with natural gas and uranium. Latin America has meaningful exposure to several of these inputs, but copper has emerged as the commodity with the cleanest and most economically significant regional link.

Copper is used throughout the AI infrastructure stack, including power cables, substations, transformers, server boards, heat sinks and cooling systems. Rising U.S. copper demand has lifted imports from Chile and Peru, particularly the former (Figure 3). The effect matters beyond merchandise export volumes. Favorable terms-of-trade have helped Chile generate a current-account surplus for the first time in roughly fifteen years (0.2% of GDP as of Q2 2026 on a 4-quarter rolling basis) and have allowed Peru to log the largest surplus in about forty-five years (4.5% of GDP as of Q2 2026 on a 4-quarter rolling basis).

Figure 3: Copper epitomized the commodity link

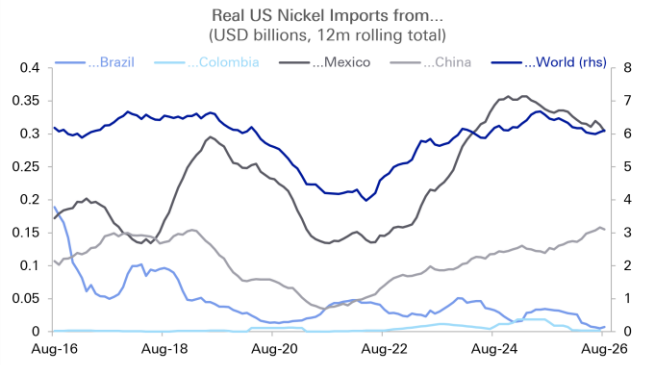


Source: U.S. Census Bureau, Haver Analytics and Deutsche Bank Research

Not every metal is copper

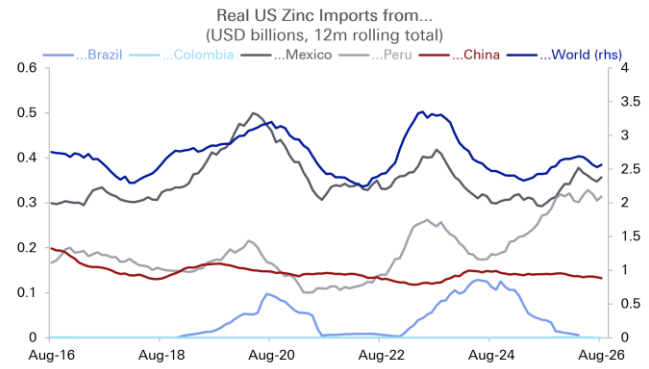
The broader mining story demands more caution, though. U.S. nickel imports have recovered toward pre-Covid levels, but the upswing has not displayed the same persistence as copper. Mexico has captured a disproportionate share of the bilateral improvement, yet that gain looks more like normalization than a sustained AI-driven break (Figure 4). Zinc tells a similar story. U.S. real imports stand shy of their pre-pandemic levels, and stronger purchases from Mexico and Peru partly reflect a reallocation away from Brazil rather than a decisive increase in U.S. demand (Figure 5).

Figure 4: Nickel: recovery, not a boom



Source: U.S. Census Bureau, Haver Analytics and Deutsche Bank Research

Figure 5: Zinc: a substitution story



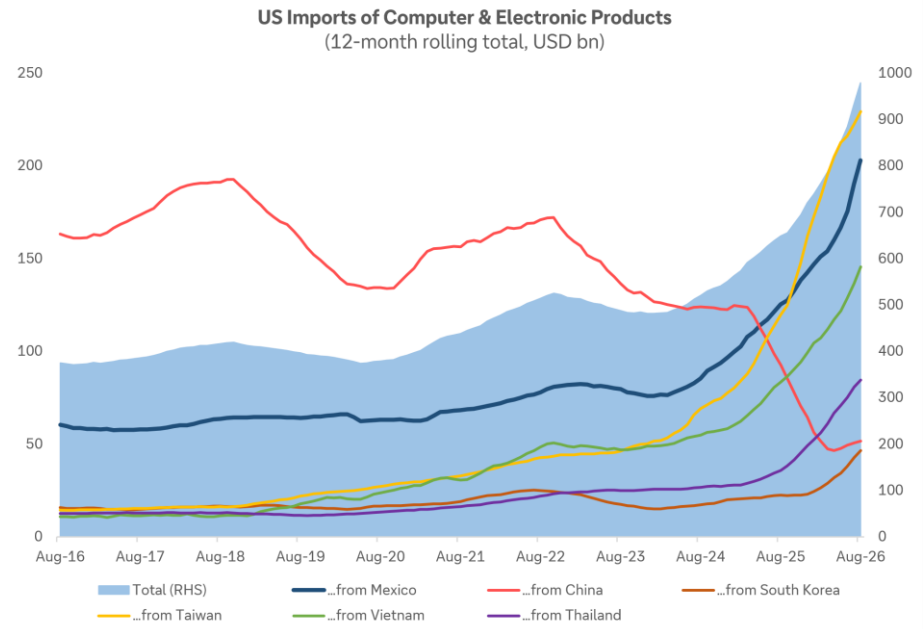
Source: U.S. Census Bureau, Haver Analytics and Deutsche Bank Research

Mexico's hardware advantage

The larger non-commodity prize is in computers and electronics. U.S. imports of these products have sky-rocketed, while the supplier mix has shifted away from China. Mexico and various Asian economies have gained share on the back of this double-whammy impulse (Figure 6).

This helps explain why Mexico's non-automotive manufacturing exports, notably electronics, have outpaced automotive shipments to the U.S. and pressures from higher oil prices. The resulting non-oil trade surplus has offset pressure from the oil balance and helped Mexico generate a balanced current account (0% of GDP as of Q2 2026 on a 4-quarter rolling basis). The external boost is meaningful, although it should not necessarily be interpreted as *bona fide* evidence of nearshoring, in our view.

Figure 6: U.S. computer & electronics imports going into overdrive



Source: U.S. Census Bureau, Haver Analytics and Deutsche Bank Research.

How much runway remains?

Quantifying or mapping out a timeframe for the AI investment outlook is beyond the scope of this piece or our expertise. Nevertheless, we'd venture to argue that the capex cycle appears more likely to mature gradually than to stop abruptly. One reason is that the investment case is shifting from frontier-model training toward enterprise inference. Capacity that once served development can increasingly run commercial workloads for companies. That creates a path to revenue even if artificial general intelligence (AGI) crystallizes later than expected. It also spreads demand to a much larger corporate customer base.

A second reason is scarcity. Graphics processing units (GPUs) remain tight, while power availability, grid connections, cooling equipment, construction capacity and data-center complexity limit how quickly supply can respond. Bottlenecks may therefore extend the cycle. The constraint is especially relevant for copper



and power infrastructure, where projects have long lead times and permitting hurdles.

A third reason is strategic. AI development resembles a prisoner's dilemma at both the corporate and national levels. Each participant may worry about costs, safety and control, yet slowing unilaterally risks surrendering technological, commercial or national security advantages. The same tension is visible in calls for guardrails alongside continued spending. Competition can therefore sustain capex even when short-term returns are uncertain.

We acknowledge the risks, nonetheless. Higher borrowing costs raise the hurdle rate for an infrastructure buildout financed partly through debt. More important, revenues must eventually justify the installed capital base. The bearish case is not that AI lacks value; it is that infrastructure spending may run ahead of monetization, compromising some companies' repayment capacity. Public opposition to the development of AI infrastructure and/or adoption of AI technologies could also slow down the technology's roll-out.

The regional balance

Summing up, the AI investment boom in the U.S. is delivering a dividend to parts of Latin America, not the region as a whole. Chile and Peru benefit through copper prices and export volumes. Mexico benefits through electronics manufacturing and the replacement of Chinese supply. For Brazil and Colombia, the AI dividend is rather muted. Where materialized, furthermore, the dividend supports trade and current accounts more clearly rather than improving the domestic investment.

These dynamics should persist while U.S. AI investment remains supply-constrained and strategically protected. If widespread adoption advances and the strategic race intensifies, the impulse can last. If financing costs rise further or monetization disappoints, the cycle may narrow. Either way, copper and Mexican electronics remain the region's most visible gauges of the U.S. AI investment appetite.

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Swing State: Navigating Midterm Volatility

October 23, 2026

A client-facing version of this report is available at the Deutsche Bank Research Portal. The following Deutsche Bank Research analysts contributed to this report: Brett Ryan (econ), Binky Chadha and Parag Thatte (equity strategy), Benjamin Soff (data centers), Daniel Knauff (software), George Hill (healthcare), Omotayo Okusanya (REITs), Peter Abramowitz (office REITs), Mark DeVries (consumer and specialty finance), Nicole DeBlase (MI/EE), Corinne Blanchard (clean tech), Richa Harnain (transports), and Scott Deuschle (aerospace and defense).

With the 2026 US midterm elections approaching, we look at key investor debates surrounding the event and legislative priorities of the next Congress. Despite the uncertainty that midterms can bring, the market appears fairly confident on expectations for the overall results. While a surprise Republican-majority victory in both chambers is still possible, consensus is converging around current polling indicating Democrats are likely to flip the seats needed for a House majority and may win a Senate majority. Historically, an incumbent president has lost seats in 20 of the last 22 midterms. Investor expectations are in line with this. Polymarket, for example, is currently above 90% for Democrats gaining control of the House and above 60% for Democrats gaining control of the Senate.

With this set of expectations, key investor debates instead focus on the legislative agenda for hot-ticket policy issues. Namely, these hinge on AI and data centers, affordability (affecting healthcare, housing, and education loans), and economic and fiscal policy. Debates also surround possible influence of the new Congress on the war in Iran and, by extension, energy prices. In this report, we cut through the noise on what the next Congress could reasonably advance from these agendas, what the impact would be on affected sectors, and how to address market uncertainty – and perceived certainty – into the elections.

Historically, midterm elections have proven to be a check on administration policies. The number of seats lost in the midterms by the incumbent party is tied to the president's approval rating going into midterms, which in turn tends to align with consumer confidence. President Trump's ~38% approval rating, at a record low for his tenure, would suggest a little over a 45-seat loss for the Republican Party (in the House and Senate combined). That said, polling in the Senate races that will determine control of the chamber largely remains within the margin of error, and in 2024, Trump won in at least five of the seven core toss-up states. Thus, the Senate remains a much closer call than the House race.

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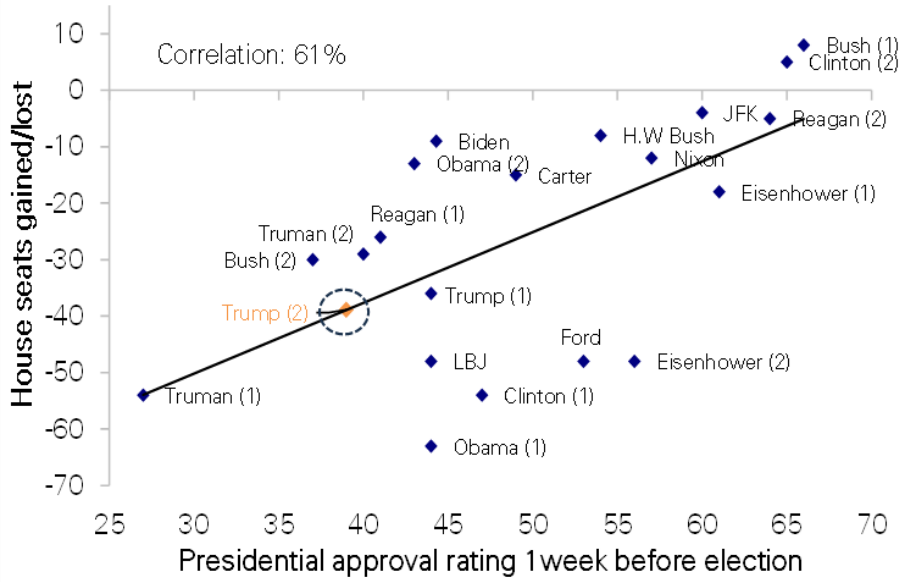
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Parag Thatte
Strategist

Brett Ryan
Senior US Economist

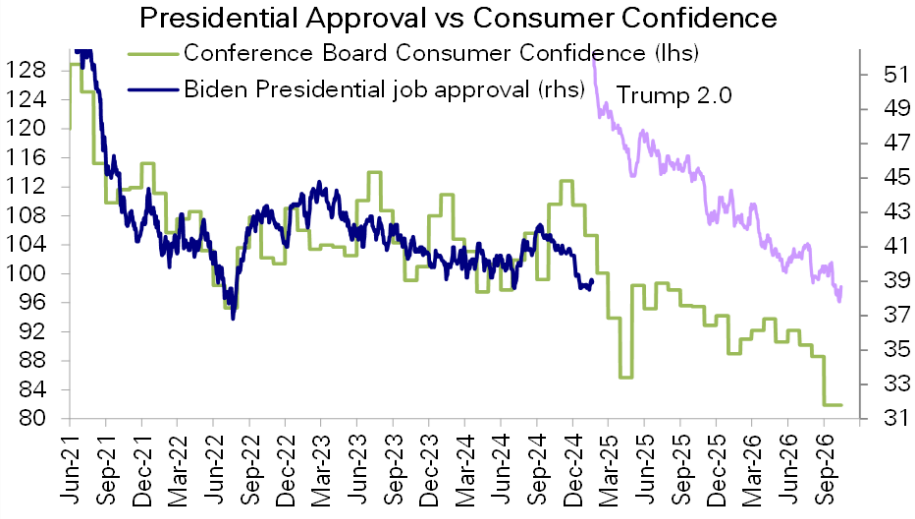
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Figure 1: Presidential Approval vs. House Seats



Source: Real Clear Politics

Figure 2: Presidential Approval vs. Consumer Confidence



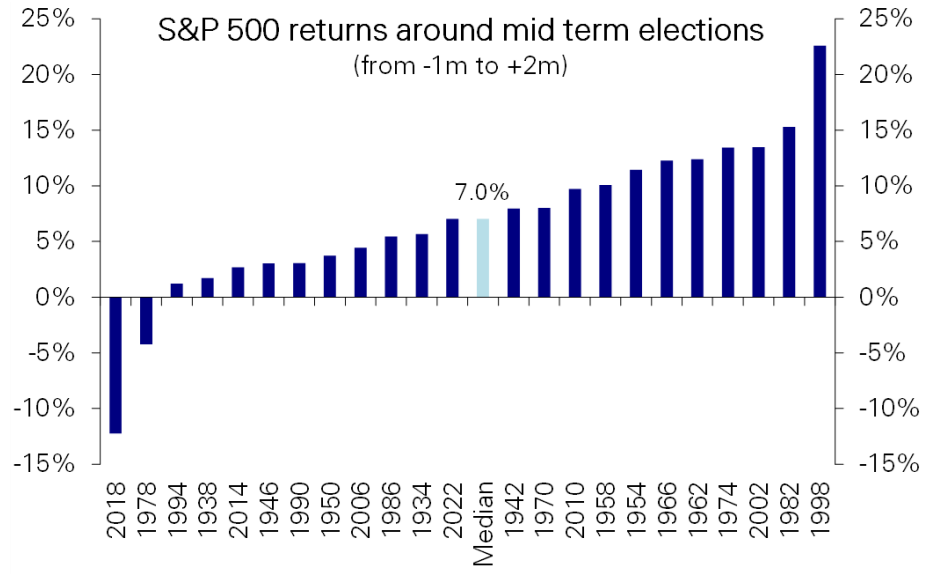
Source: The Conference Board

Whatever the outcome of the midterms, the legislative agenda will still face constraints from the White House. It is all but impossible for either party to gain a veto-proof majority in either chamber, minimizing the likelihood of a large shift in fiscal or regulatory policy. The most immediately obvious impact of Democrats taking the Senate would be the ability to slow or block executive branch appointees, including at the Federal Reserve Board of Governors or the Supreme Court, should any members step down before their terms end.

Without garnering a two-thirds majority in both chambers, Democrats lack a clear path to unwind the 2025 tax cuts or enact AI-related regulatory legislation – at least, not without President Trump's explicit approval. With respect to the near-term fiscal policy outlook, the most important period for investors may be the "lame duck" session, the period after the November 3 election and before the new Congress convenes on January 3. Recall that current federal government funding expires on December 11, and the Trump administration is asking for a nearly 60% increase in defense spending, the largest since the Korean War. In addition, the debt limit will become binding again around the middle of next year. In short, the actions during the lame duck session could determine whether the new Congress is seated amid a government shutdown, a contentious battle over the federal budget, and the debt limit looming in the background.

In terms of the market reaction to midterm elections, the risk-reward profile has typically been favorable, with 21 of the last 23 midterm years seeing positive returns in the fourth quarter, with an average gain of 7% in the S&P. The gains are notable even after adjusting for typical positive year-end seasonality.

Figure 3: S&P Returns and Midterms



Source: Bloomberg Finance LP

It would not be unusual for the market to go sideways as investors hedge potential volatility and uncertainty, as we have seen previously ahead of other well-telegraphed pivotal events. This has historically been followed by a rally as the results became clear and uncertainty dissipated, regardless of the result itself. In the current context, if election polls and prediction market probabilities continue being lopsided, the market may not treat the election as an uncertain risk event going in. Uncertainty could still ramp up after, and the typical midterm election rally may be pushed out. It is notable that the one prominent exception to the midterm rallies was in 2018, when the market sold off sharply after the elections as uncertainty spiked around trade war 1.0 as well as Fed policy.



We note that the midterm elections also coincide with the third quarter earnings season, during which we expect companies to deliver growth of 34%. For next year, we expect growth to be a still-robust 17% with risks to the upside. Eventually, earnings tend to be the anchor for equities and have been for several years despite numerous domestic political as well as geopolitical shocks.

Across sectors, most have rallied around past midterms in keeping with the market, but there isn't a consistent pattern of out- or underperformance. This suggests that for the initial conditions for each sector, specific policies and positioning played a bigger role. For that reason, we turn to a closer analysis of key legislative and policy debates impacting exposed sectors.



AI and Data Centers

AI and data centers have become a campaign talking point for candidates from both the Democratic and Republican parties. This has spurred investor questions on whether to expect a national regulatory framework constraining data center buildouts. Some of the headlines surrounding this debate overestimate what Congress can accomplish through national legislation compared with what is possible at a state level, particularly during a presidential administration supportive of AI and data center acceleration. Still, the industry is responding to a political backdrop that is increasingly scrutinizing AI and data centers.

The current administration has prioritized advancing AI and digital infrastructure development. The White House's AI Action Plan, released July 2025, included a set of executive orders supporting the rapid buildout of AI data centers by reducing certain federal regulatory and permitting burdens. The administration did not preempt state laws on data center infrastructure development and federal permitting orders generally do not override state zoning and utility rules.

In March 2026, the White House announced the Ratepayer Protection Pledge, under which large data center operators would voluntarily commit to covering the cost to upgrade generation, transmission, substations, or other necessary infrastructure to service these developments. The industry's response to these proposals has been largely constructive. The Data Center Coalition (DCC), an industry group representing major hyperscalers and data center owners and operators came out in support of the initial version of the Pledge, while arguing that large-load customers across all industries also should cover the costs to upgrade energy infrastructure. According to the DCC, "the data center industry remains committed to being responsible partners where we operate, which includes paying our full energy costs and investing in local jobs and workforce development."

Congress voted in September 2026 to codify the principles present in the Pledge via the Ratepayer Protection Act. The bill, which only required states to "consider" these standards, passed overwhelmingly in the House (417-3), but failed to reach 60 votes in the Senate (57-43). Critics noted the bill stopped short of imposing mandatory nationwide rules.

Looking forward, we expect a Congress that is more focused on consumer protection might look to codify a more stringent version of these rules, requiring large-load customers to bear the full incremental costs associated with generation, transmission, and distribution upgrades. However, even if such a framework were passed into law, we believe the overall economic impact to large data center operators would be relatively modest, for two reasons.

For one, the White House would maintain a veto over bills that include language it views as overly burdensome towards AI infrastructure development. Two, data center operators generally underwrite projects only when they have a high degree of confidence in securing power and, where necessary, may already be embedding the cost of upgrading energy infrastructure into their investment plans. We expect developers to be capable of passing on sustained increases in power infrastructure costs to their customers, which, for many 100 MW+ developments, are the major investment-grade hyperscalers.



Though data center development is receiving increasing national attention, it is still largely a local issue, and the most consequential elections will be at the state and local levels. We have seen a number of state-driven actions on data centers in recent months, including:

- In July, New York State's governor issued an executive order to establish a temporary moratorium on data center development while the state develops a standardized framework of rules for key issues (including responsibilities for power and energy infrastructure upgrades; additional power generation; clean energy; water use, emission, and waste management; and state/local financial incentives). The New York legislature also passed a one-year moratorium on Department of Environmental Conservation permits and approvals for 20 MW+ data center developments.
- In August, Texas' governor directed the state's public utility commission to conduct a comprehensive audit of every data center advancing through the ERCOT (Texas's grid operator) interconnection process. The audit will cover several notable areas including public financial support, electricity demand and supply, water use, community impacts, and ownership and control.
- In September, California's governor passed a comprehensive regulatory package, covering electricity, water, reporting, and environmental review. Most of these provisions take effect January 1, 2027, and phase in through 2027-2028, but the package does not ban data centers or impose a statewide moratorium on development activity.
- Oregon, Pennsylvania, New Jersey, and other states have also established orders to create standards and frameworks governing data center development (but without imposing broad moratoriums).

At the more local level, a large number of moratoriums have been passed on new data center development. Data collected by SemiAnalysis, an independent research firm, shows that, as of September, more than 300 local data center moratoriums have been adopted nationwide across counties, municipalities, townships, and other local governments.

However, these state and local moratoriums appear thus far to be less impactful to near- and medium-term development activity than headline coverage might otherwise suggest. According to SemiAnalysis, only about 2.3 GW of the 38 GW of data center capacity planned or under construction for 2027 has faced delays as a result of the New York state executive order (0.8 GW) and/or local moratoriums (1.5 GW).

We believe the primary reasons that moratoriums are, in reality, less likely to impact near-term development activity are (1) that these moratoriums largely impact new projects and applications (rather than existing developments in progress), (2) there may be flexibility to relocate or rezone early-stage projects in response to moratoriums, and (3) data center developers are increasingly partnering with local communities to gain their support via investments and other programs (as we wrote about recently). Potential regulatory risk has not yet slowed capex spending patterns, with capex spend still expected to climb from



$0.5 trillion in 2025 to $1.3 trillion by 2027; data centers typically represent ~20-30% of the total bill of materials for new AI capacity.

Although the practical impact on near- and medium-term development pipelines appears less pronounced than headlines might suggest, all this is not to say that data centers will not face increasing local opposition following the midterms. Indeed, in our conversations with industry participants, there is an acknowledgment that greater public scrutiny of data centers is not going away. Additional moratoriums, zoning restrictions, permitting requirements, or utility-related conditions could affect the pace, cost, and overall feasibility of future projects.

Affordability Agenda: Healthcare, Housing, and Education Loans

The affordability agenda has become a centerpiece of many Democratic as well as Republican campaign platforms. This platform is impacting investor debates across healthcare, housing, and student loans. In healthcare, the push among some Democrats for expanded health coverage under the Medicare for All banner is placing pressure on managed care stocks, but meanwhile, companies with exposure to Medicaid and the Affordable Care Act (ACA) could see a supportive political backdrop. On housing, investor debates on the impacts of the affordability agenda have continued since passage of legislation in July 2026 that caps the number of single-family homes for-profit institutional investors can purchase. Affordability of higher education is likewise part of the conversation, leading investors to debate which levers Congress could pull that would affect student loan companies.

Healthcare

In healthcare, we see the upcoming midterm elections as a sentiment event for managed care stocks, not a policy event. With the presidential veto in place through 2028, we do not expect the government to enact direct healthcare legislation during the next Congress. Forecasts place the probability of a change in House control at roughly 79%, with both chambers changing in about 44% of simulations. Although investor skittishness around Medicare for All (M4A) rhetoric is high, we see the likelihood of M4A replacing private coverage as a near-zero probability in this timeframe.

M4A has no realistic legislative path before 2029, but the rhetoric weighs on managed care organization (MCO) stock multiples by raising perceived long-term risk rather than affecting near-term earnings. Polling from Kaiser Family Foundation, a health policy research organization, shows support for M4A at a decade-plus high, and the 2018–2019 cycle suggests that increased campaign attention can pressure MCO multiples even without a direct policy impact. We expect similar sentiment-driven pressure through year-end and potentially into the 2028 primary cycle but would not view it as a change in near-term fundamentals.

For Medicare Advantage, we see a headline risk over earnings reality. The near-term earnings impact for affected companies should be limited, because the US Centers for Medicare & Medicaid Services (CMS) has finalized a 2.48% Medicare Advantage (MA) rate increase for 2027, versus 0.09% in the advance notice. A change in House control would instead increase focus on 2028 payment policy, risk adjustment, and prior authorization. Coding reform already has bipartisan support: sponsors estimate that the Cassidy-Merkley No UPCODE Act could generate up to $275 billion of savings over ten years, while MedPAC estimates coding intensity will add roughly $40 billion to 2025 Medicare Advantage costs relative to traditional Medicare. In our view, the more immediate consideration is potential multiple pressure rather than a measurable change to company earnings.

However, the next Congress could benefit Medicaid funding and ACA policy. Medicaid and the ACA appear relatively better positioned, although any benefit



would likely be reflected first in sentiment and lower perceived policy risk rather than near-term estimate revisions. Medicaid community engagement requirements take effect in most states by January 1, 2027. Although repeal remains unlikely given the presidential veto, a change in House control could shift the year-end funding debate toward restoring or preserving federal Medicaid support, which could ease state budget pressure and support rate-setting. The ACA policy backdrop could also become more supportive, although we see limited scope for enhanced premium subsidies to return before 2029 after the three-year extension passed by the House stalled.

Housing

Turning to the housing sector, the likely midterm outcome of a split government would result in more gridlock. Historically, political gridlock in terms of passing new regulation has been a tailwind for the real estate sector, reducing regulatory risk at the federal level. In the scenario of a split Congress, in which the Democrats have a majority in the House but not the Senate, bills with bipartisan support have a greater likelihood to become law. Two key areas in which this bipartisan support exists are housing affordability and consumer protection.

Among real estate investment trusts (REITs), we see Residential as among the most impacted subsectors, particularly as it relates to the affordability debate. Regardless of the election outcome, housing affordability has become front and center on the political agenda across both parties. Federal legislation is more likely to target supply-side solutions, as legislation restricting rent increases is typically handled at the local and state level. We think Multifamily would be the most likely subsector impacted, as a push to expand rental housing supply could lead to eased zoning restrictions, particularly in supply-constrained coastal markets that have historically benefited from higher barriers to entry. That said, this would likely take years to materialize. With a Democratic majority, Congress could voice opposition to the influence of large institutional landlords, but we see this strictly as a sentiment risk to stocks.

Single Family For Rent (SFR) is less exposed to near-term legislative shocks, as the sector was a focus of the 21st Century ROAD to Housing Act passed in July 2026 over the summer. The measure impacts institutional owners' ability to buy homes off the MLS but contains specific carve-out provisions that will allow them to keep developing build-to-rent (BTR) product and buy excess inventory from homebuilders, the sector's preferred acquisition channel in recent years. The ROAD to Housing Act also contained provisions that could theoretically ease the production of manufactured homes, the main business segment for Manufactured Housing (MH) REITs. The first-order impact is likely more flexibility in home designs that could lead to further densification of MH communities, although we note emerging investor debate about the long-term impacts of increased MH production on competitive supply for a property type that has historically been supply constrained, mainly due to NIMBY-ism.

Parallel to the conversation on affordability, it is clear that, regardless of which political party may be in power following the midterms, a major focus remains addressing the US fiscal deficit and its continued growth. We continue to pay attention to potential budget cuts or sequestration to slow government spending and reduce the deficit, which could have an impact on real estate. Key areas to watch include cuts to healthcare spending (which would be a negative for skilled



nursing focused REITs, given Medicare is a major revenue source for their tenant base) and cuts that would impact Office REITs with large government exposure.

We also worry that an elevated federal deficit and the elevated Treasury borrowing needs today create a risk that interest rates will remain higher for longer. This puts pressure on real estate valuations and cost of financing for REITs to drive external growth via acquisitions and development. A split Congress is likely to result in more difficult budget negotiations and increased oversight, but it remains unclear what the ultimate outcome would be to address the budget deficit issue.

With real estate being more dependent on local market dynamics, we believe the more likely regulatory impact would come from local regulation. This could show up on the ballots during the midterms, with increased possibility of rent control legislation targeted at residential real estate as well as self-storage, as cities and municipalities focus on affordability issues. There is also growing legislative pushback at the local level against development of real assets such as data centers given the large power requirements, as discussed in detail elsewhere in this report.

Education Lending

Midterm campaign platforms on education affordability have a bearing on the consumer and specialty finance sector. Much of the policy exposure in this sector is through the executive branch bodies, not Congress, however. This includes, for example, regulatory authorities such as the FDIC (Federal Deposit Insurance Corporation), the CFPB (Consumer Financial Protection Bureau), or the FHFA (Federal Housing Finance Agency, which regulates Fannie Mae and Freddie Mac). That said, historical data shows that student loan company stocks can still directionally align with midterm party outcome and expectations in Congress, although the sample size is small.



Mixed Impacts of Economic and Fiscal Policy

Democratic and Republican candidates have aimed to distinguish themselves from their opponents along a range of economic and fiscal policy matters, including tariffs, tax, labor, and the budget. In reality, the outcome of the midterms, regardless of which party prevails, is likely to present a mixture of relief and restrictions affecting businesses. In some cases, investor fears over sweeping economic policy changes are overblown for topics over which Congress realistically has limited influence, presenting more headline risk than a risk to fundamentals. In considering these, we look to industrials that are particularly sensitive to key investor debates, spanning multi-industry/electrical equipment, clean tech and alternative energy, transport, and defense.

For the multi-industry and electrical equipment (MI/EE) sector, historical midterm election cycles suggest a generally constructive backdrop heading into election day, with positive returns in six of the past seven elections. On average, the MI/EE group has gained ~5% during the three months preceding historical midterm elections and ~8% during the final month prior. Relative outperformance versus the S&P 500 has been less consistent; we observed outperformance in only four of seven cycles, though the group still generated average excess returns of ~250bps over the preceding three months and ~270bps over the preceding month.

Overall, the data supports a favorable absolute return backdrop ahead of midterms, but significant stock-level dispersion suggests company-specific fundamentals and factor leadership have historically mattered more than the election cycle alone. We mapped out election results scenarios across key policy pillars for our coverage (tax policy and capital investment; trade, tariffs, and supply chains; energy, grid, and power infrastructure; AI and data center regulation; and the regulatory environment). We focus on two key scenarios: one in which Republicans maintain control of both chambers, and a divided government that sees the Democrats controlling one or both chambers of the legislative branch.

Under a divided government, we expect stock-level dispersion would likely narrow, with globally exposed companies receiving some trade policy relief and HVAC/automation companies benefiting from a potentially more supportive regulatory backdrop. However, should Republicans maintain control of Congress, our scenario analysis suggests this would be favorable for electrical equipment and power infrastructure manufacturers, due to a combination of pro-investment tax policy, faster infrastructure deployment, and a lighter AI regulatory framework. That said, globally exposed companies could face incremental tariff and retaliation risks.

Looking to the clean tech and alternative energy sector, the midterm elections could represent a positive tailwind. The next Congress might be positioned to pursue targeted extensions or modifications to certain clean-energy tax credits. The introduction of H.R. 8477, which proposes to restore several clean energy tax credits rolled back in late 2025, seems to signal that some, albeit a limited number, of Republicans could be supportive of this initiative. If the Democrats gain a majority in the House but not the Senate in the midterms, it is plausible that Democrats could find sufficient Republican support to advance



legislation extending clean-energy credits, though they would still face significant legislative hurdles. A midterm outcome in which Republicans retain control of both chambers is likely to maintain the status quo for the remaining existing tax credits, especially for those tied to domestic manufacturing. While those could potentially see further extension, we think this is a limited outcome.

While we see a low probability of a major change in, or support for, the residential tax credits that expired at the end of 2025 (25D), we envision a scenario in which indirect residential tax credits (48E) could be extended, which would help boost part of the demand. Softness in demand is expected to remain going into 2027, still impacted by the phaseout of the 25D tax credits, the eventual roll-off of safe-harbor benefits, and a more constrained financing environment.

From a manufacturing tax credit standpoint, there could be additional tailwinds if we were to see any manufacturing tax credit extension or positive modification (although lower probability, in our view). This would also likely improve utility-scale project economics if broader project-related credits are preserved, though upcoming elections could cause a slowdown in new booking activity until legislative uncertainty clears.

For Transports, the midterms would be most relevant for the trucking supply thesis. Truckload stocks could have a modest negative reaction to a Democratic majority if investors expect some rollback of the Trump administration's constraints on drivers, which has tightened supply. We would separate perception from fundamentals, however. The election of a Democratic Congress would not automatically by itself change leadership at the Department of Transportation (DOT) and the Federal Motor Carrier Safety Administration (FMCSA), the executive branch entities driving much of the current push for restrictions on drivers.

One possible impact of the midterms for Transports is if a change in Congressional control contributed to a faster resolution of the war in Iran. Any resulting lower energy prices would be a positive for Transports. We view this as a lower-probability outcome, however, given the constraints on a divided government.

With clashes over the budget likely looming, aerospace and defense could get caught in the crosshairs. While defense spending has historically enjoyed bipartisan support, it has grown politicized following the war in Iran, making it more difficult for a split Congress to pass such a budget. The defense budget has experienced significant growth over the last few years, rising from $900 billion in FY2024 to $1 trillion in FY2026. The Trump administration would like to see this trend accelerate: in April 2026, the president issued a budget request that called for a $1.5 trillion budget for the Department of Defense in FY2027. With a continuing resolution likely into year-end, the outcome of the midterm elections will determine whether the next Congress can pass such a budget. If Republicans hold the House and Senate, we think there is a reasonable prospect of a budget of $1.2 trillion or more passing into law. Defense investors would see that as a substantial tailwind for beaten-up defense stocks. On the other hand, this would be unlikely in a split Congress, in our view. Given all this, we see the outcome of the upcoming midterm elections as key to determining how US defense stocks will perform into year end and beyond.



Taking Stock of the Known Unknowns

Although Republicans could maintain control of both chambers, consensus is converging around polls pointing to a Democratic majority in the House and possibly the Senate. The perceived certainty has allowed the investor debate related to midterms to focus not only on who will win but how they will govern. In some cases, headlines are misrepresenting what a Congress constrained by the executive will be able to accomplish. In other areas, legislation that is expected to be passed will need to be taken together with offsetting factors. Whatever the outcome, the political backdrop will continue to influence the market and investor debates in the year ahead.

Appendix 1

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STABLECOINS: What next for crypto assets?

October 20, 2026

Marion Laboure | Camilla Siazon

Deutsche Bank Research Institute

IMPORTANT RESEARCH DISCLOSURES LOCATED IN APPENDIX 1.

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STABLECOINS: What next for crypto assets?

Institutional adoption of stablecoins continues to incrementally progress in 2026. While the frenzy from GENIUS has diminished, stablecoins' links to the financial system have continued to deepen, and more policymakers are seriously considering their use cases. The market cap has hovered around ~$300bn over the past 12 months, a sign that the stablecoin market is maturing. Stablecoin payments, while still marginal relative to stablecoins' crypto-native purpose, are also growing, rising from around $103bn in adjusted transaction volume in 2022 to $375bn in 2026 YTD.

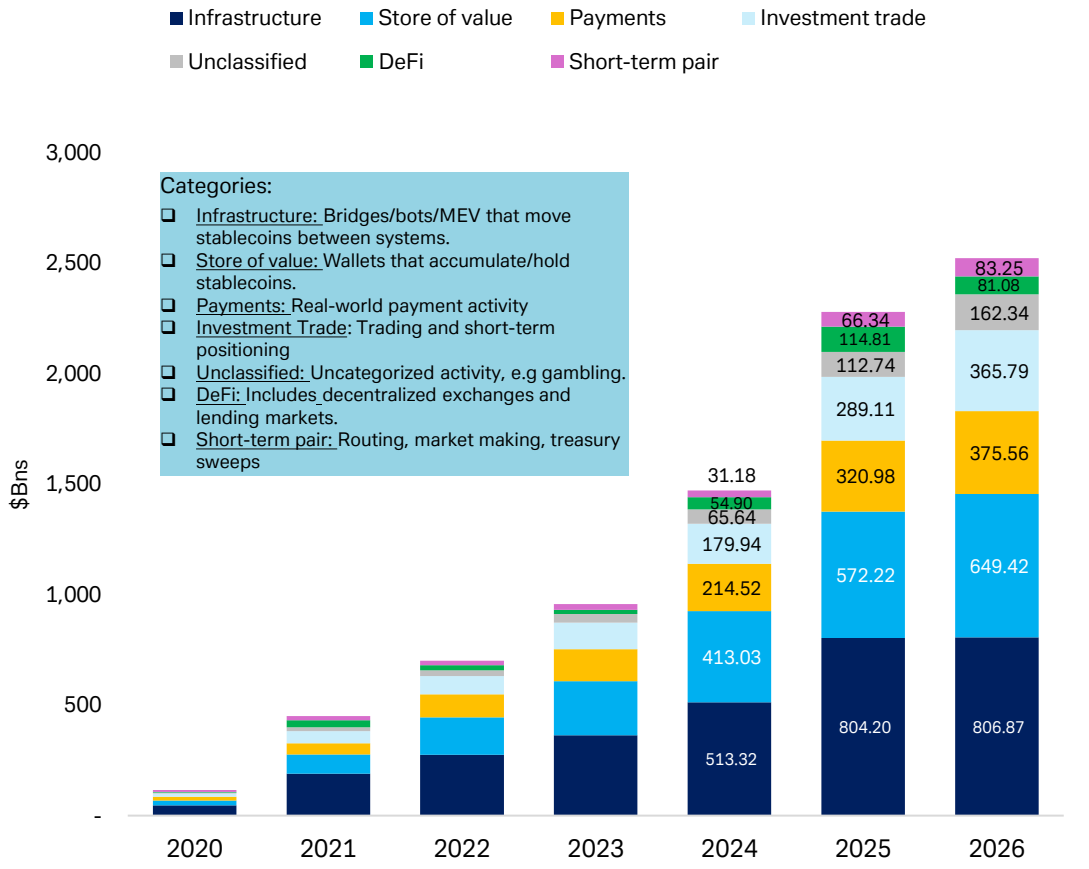
For now, institutional adoption remains concentrated in cross-border payments, treasury management, 24/7 liquidity provision, collateral, and real-time settlement.

The next big question is whether stablecoins will evolve to become merely a payments technology, or a genuine disruptor to the global banking system. As we previously established, the US promotion of stablecoins remains a geopolitical tool to ensure USD dominance. Bloomberg also reported that the Trump administration is considering joint ventures with private actors to promote USD stablecoins overseas, in a bid to boost the dollar's dominance and source demand for US Treasuries. Last week, Circle announced plans to expand into 40 new markets in Southeast Asia, LatAm and the Middle East. 99% of stablecoins (by market cap) remain pegged to USD and are largely backed by US Treasuries.

Looking ahead, the key debate to watch is whether US regulations will allow for yield-bearing stablecoins. Research shows that consumers say policymakers should prohibit crypto companies from offering interest-like rewards for holding stablecoins, which will threaten to siphon off the bank deposits that drive local lending. The US CLARITY Act failed in Sep after lawmakers failed to agree on whether the broader crypto market structure would support such yield-bearing stablecoins.

In this chartbook, we revisit the stablecoin market a year after the GENIUS Act was signed. We look at: (i) how adoption has evolved, (ii) the emergence of stablecoins as financial market plumbing via liquidity provisioning and collateral management, and (iii) the key regulatory debates that will determine the industry's next direction.

Payment volume has a +66% CAGR



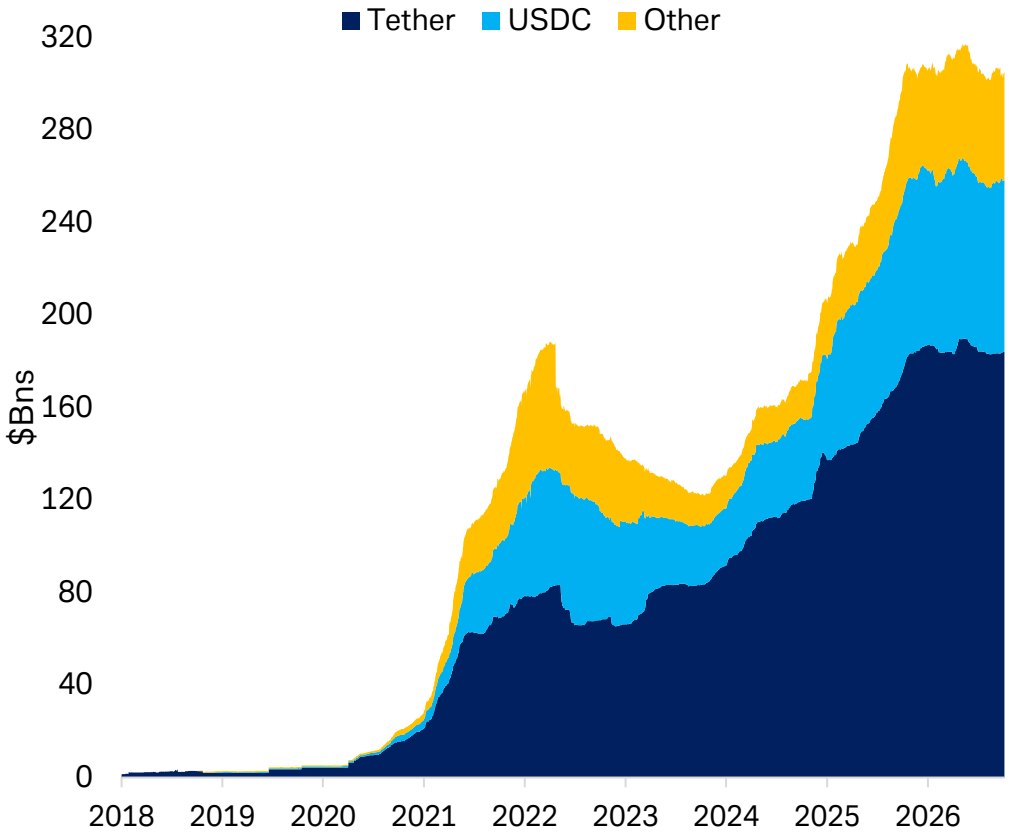
Source: Deutsche Bank Research, VisaOnChainAnalytics, Allium data. Data shows stablecoin transaction volume (adjusted). October 19 2026.

REFRESHER: What are stablecoins?

- Definition: Digital tokens pegged to "stable" assets (e.g. USD or gold), issued and transferred on a blockchain, which records ownership and transactions.
  - Fiat-backed stablecoins are considered fully "collateralized": USD stablecoin with 1mn coins in circulation would be backed by $1mn in cash and equivalents held in reserve by the stablecoin issuer.
- Types: Fiat-, commodity-, crypto- and algorithmic-backed.
  - ~99% of stablecoins market valuation is pegged to the USD.
- Market cap: $5bn (2020) to ~$310bn (2026)
  - Concentrated in USDT (~60%) and USDC (~23%).
- Benefits: 24/7 transfers & settlement, lower financial intermediaries, programmability (e.g. escrow, conditional release, automated reconciliation), and dollar-based store of value.
- Use cases:*
  - Crypto trading (~77%) remains dominant.
  - Payments (5-12%) growing, cross-border transfers and remittances.
  - Store of value (17-33%) especially in emerging markets.
  - Other: (~2%-8%): on/off ramps and tokenized RWA settlement.
- Alternatives: CBDCs, tokenized deposits, traditional financial architecture

**Estimates on market share taken from BCG analysis.*

Stablecoins' reached a plateau of ~$310bn in October 2026



Source: Deutsche Bank Research, Haver Analytics, Bloomberg Finance LP. Shows stablecoin total market cap. Updated October 19 2026.

ADOPTION OVERVIEW: Most stablecoin volume is still not driven by real economic activity

Stablecoin transaction volume:

- ~$100trn (unadjusted), ~$2.42trn (adjusted) in 2026 YTD, according to Allium.
- Chainalysis places more bullish estimates of $28trn in 2025; projected to reach $719trn by 2035 on organic growth alone.

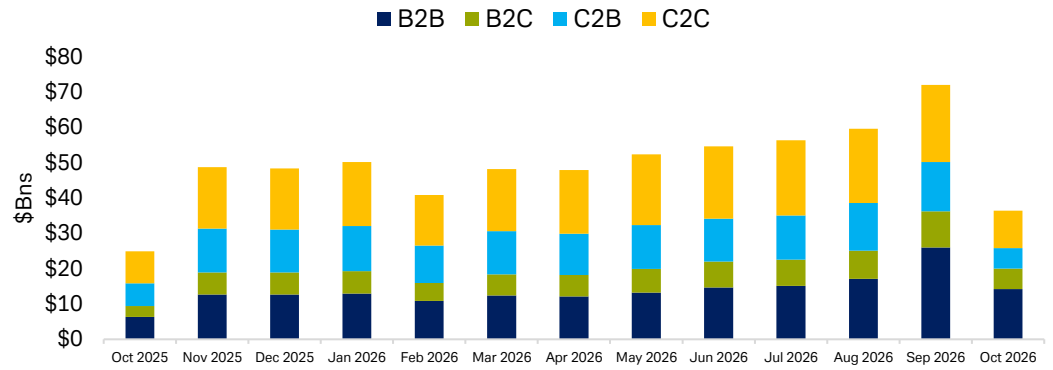
Stablecoin payments:

- Stablecoin payment volume grew from $151bn in 2022 to $375bn 2026 YTD (according to VisaOnChain Analytics).
- Businesses account for the majority (58%-64%) of payments, led by B2B payment volumes.
- Allium Sep report showed largest business payment categories include service fees ($56bn), payroll ($43bn) and service payments ($32bn).
- Case study: Payments volume across Visa's >160 stablecoin-linked card programs across consumer, business and commercial activity grew ~200% y/y.

Strongest adoption in remittances, B2B payments and dollar access in countries with unstable currencies or weak banking systems.

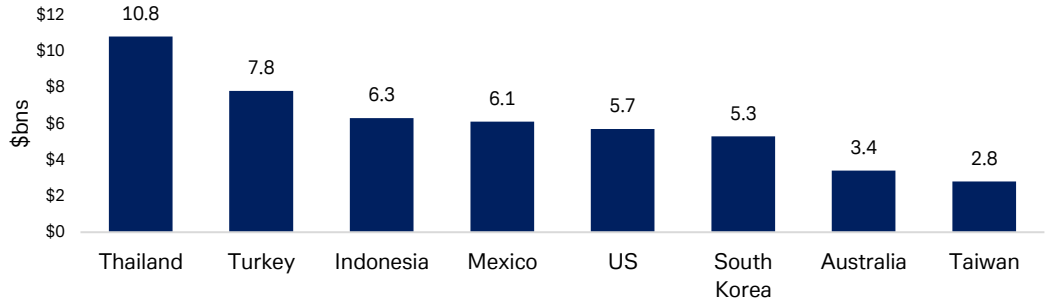
- Benefits for cross-border use cases: Instant settlement, <$0.10 per transaction, borderless, transparent, 24/7, no intermediaries. Remittance fees are lower (0.5-3.0%) than the global fee average (~6.6%).
- Stablecoins represented 0.31% of fiat cross-border payment value in 2025. Marginal portion but registered +64% growth increase vs fiat (+9%) according to Allium. B2B payments had the highest cross-border share (43%).
- A 2025 IMF study also shows transaction flows reached ~7.7% of GDP in Latin America and ~6.7% in Africa and the Middle East.

Stablecoins payment use case is steadily growing



Source: Deutsche Bank Research, Allium. Shows payments volume by use case. Data updated October 19 2026.

Cross-border use cases



Source: Deutsche Bank Research, Allium. Shows top 8 countries by received payments from Jan to August 2026. Only the receiving end is attributed, so a $100 payment counts once, not twice. That attribution covers 13.5% of stablecoin payments.

CONSUMER/RETAIL PAYMENT: Still on the periphery

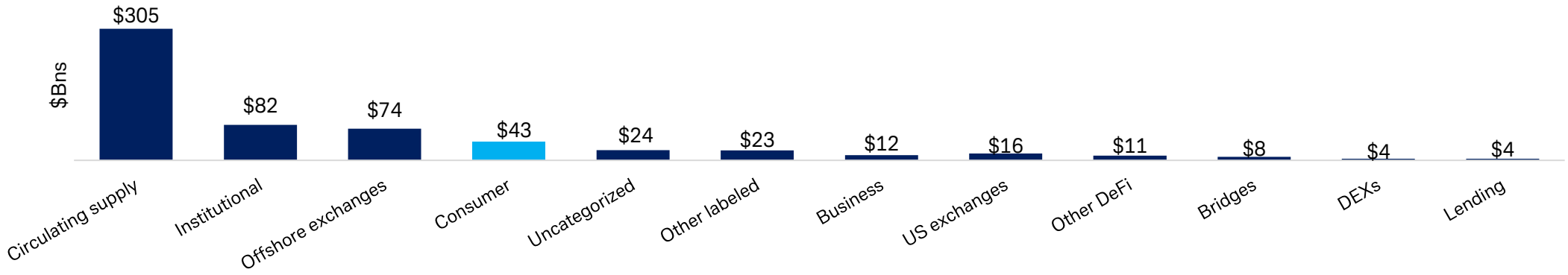
Consumer payments: Not yet there

- Retail adoption is still the most distant use case, especially amongst developed-market consumer credit card spend. A recent ABA survey found that only 24% of consumers think that stablecoins could provide meaningful benefits.
- Stablecoins still look more like a debit instrument than a credit product. They do not naturally replicate the consumer value proposition of credit cards (i.e. rewards, revolving credit, dispute rights, chargebacks, fraud protection and convenience), so there is limited incentive for developed-market consumers to use them.
- Limited scope for payments also evident in wallets (see chart below). Very little of the stablecoin supply is being held as balances in consumer wallets.

Still, the number of retail payments is incrementally growing, with cross-border, emerging market use cases strongest

- Retail transaction volume: $2.6bn (2020) to $71bn (2024); $66bn YTD in 2026.*
  - Strongest use in: (i) economies with hyperinflation; (ii) poor banking access; or (iii) high remittance fees.

The amount of stablecoin supply in consumer wallets stands at ~14%



Source: Deutsche Bank Research, Allium data. Shows circulating supply by holder type (Sep 2026). Categories calculated as a percentage of total circulating supply. Note that wallets account for money being held vs payments (money moving). Business wallets in this case refer to those holding smaller balances with quick turnover (i.e. payment services, merchants, gaming), while institutional wallets hold larger balances and turn them over quickly, i.e. financial-services firms, treasuries, and wallets. Note: *Retail sized transactions are defined as transactions of $250 or less, according to Visa OnChain Analytics.

CASH/LIQUDITY MANAGEMENT: Stablecoins are evolving as a major tokenized contender

Stablecoins can help treasury management and settlement

- Stablecoins have a strong use case for settling transactions for FX, treasury operations and collateral management i.e. intra-day repos.
- This is because stablecoins' 24/7 settlement can address operational frictions in cross-border transactions.

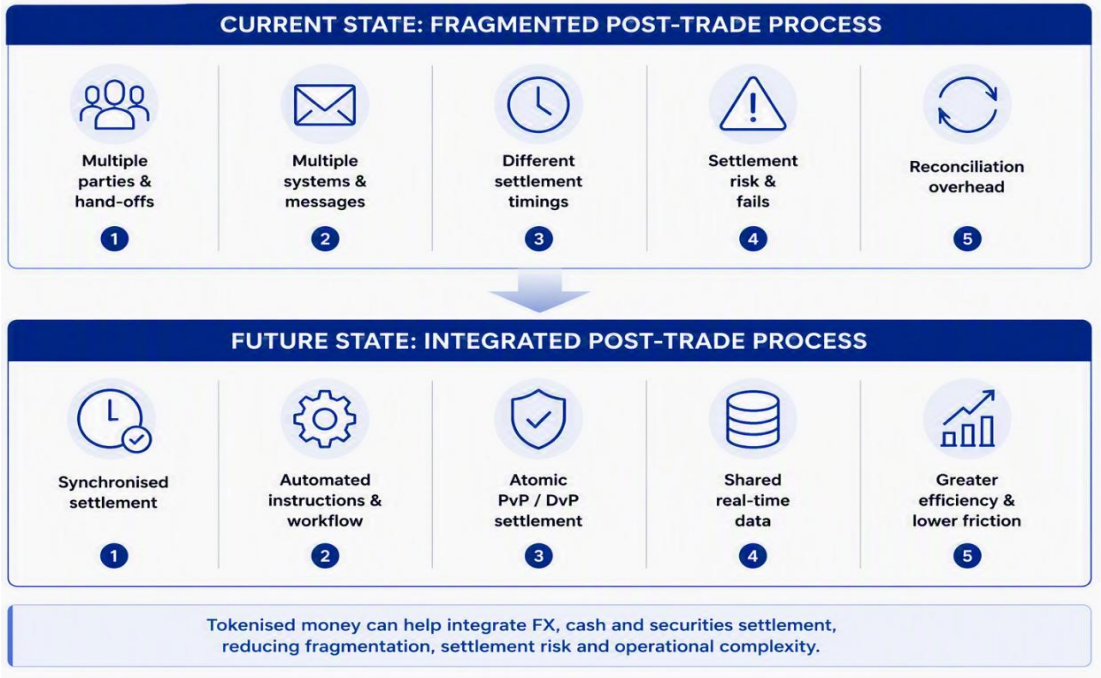
Poses significant risk to cross-border B2B transactions for banks

- Compared to traditional banks, stablecoins offer substantial relief from transaction costs. Our equity analysts estimate total customer processing fee savings to increase from 0 to ~$70bn in 2035, if consumers switch from traditional banks to stablecoins.

Stablecoins is not a replacement solution to FX settlement risk

- FX settlement requires two currencies to be exchanged simultaneously. Currently, stablecoins only serve as an intermediary between the two currency legs, resulting in a "stablecoin sandwich".
- Overwhelming concentration of USD-stablecoins also limits stablecoin ability to support widespread payment-versus payment settlement across currency pairs.
- Stablecoins also face fragmented interoperability/liquidity issues and regulatory uncertainty across issuers/networks/jurisdictions.
- For more, read our FX report here.

A more integrated post-trade process



Sources: Deutsche Bank Research, taken from DB's Tokenised money and the future of FX and Stablecoins and agentic commerce: new rails, old frictions.

INSTITUTIONAL ADOPTION: Corporates, card networks, and banks are integrating stablecoins

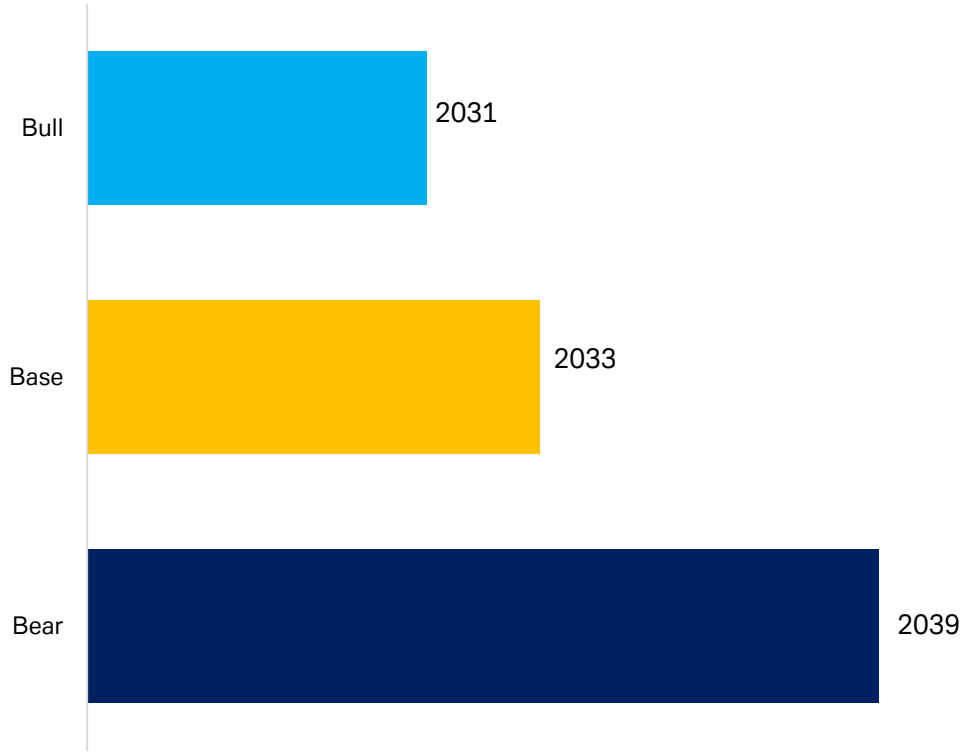
Institutions are racing to capture stablecoin market share

- Card networks: Mastercard expanded its settlement capabilities to include stablecoins, using USDC and Paxos-issued stablecoins.
- Institutions: Exploring stablecoins via developing in-house permissioned blockchain experiments and/or exploring issuance of proprietary stablecoins.
  - >140 companies and financial institutions helped launch Open USD (OUSD) in Sep 2026. OUSD reached $799mn of supply in 2 weeks. Signals shift toward consortium-backed/network-led digital dollar models.

Are stablecoins posing a threat to traditional payment systems?

- Bullish: Stablecoins to surpass traditional rails by the 2030s (Chainalysis).
  - ~17% of stablecoin-linked card volume is occurring across Visa's business and commercial card programs. Shows growing adoption of stablecoins for settlement, treasury management and cross-border commerce.
  - Key concerns for banks: Fee pressure from stablecoin adoption → bank outflows → higher reliance on wholesale funding → pressure on net interest margins.
- Base case: However, Deutsche Bank equity analysts see more limited near-term risk. Visa and Mastercard to remain dominant especially in advanced markets.
  - KYC, AML, sanctions screening, wallet monitoring, fraud controls and jurisdiction-specific limits remain essential for banks.

Stablecoin transactions could match Visa and Mastercard transactions sometime between 2031 and 2039 (Chainalysis).



Read: Stablecoins and agentic commerce: new rails, old frictions and State of the Private Markets.

INSTITUTIONAL ADOPTION (CONT): Banks have started to embrace stablecoins with varied implementation approaches

FORGE SOCIETE GENERALE GROUP

Launched EURCV and USDCV – EUR and USD-pegged stablecoins.

AllUnity® — galaxy, DWS, FLOW TRADERS

Launched EURAU and planning to launch CHFAU – EUR and Swiss franc-pegged stablecoins.

ODDO BHF

Launched EUROD – EUR-pegged stablecoin with reserves directly on balance sheet.

qivalis — SEB, Raiffeisen Bank, ING, Sella, Danske Bank, BBVA, CaixaBank, KBC, BNP PARIBAS, DZ BANK, UniCredit

Consortium of European banks planning to launch a EUR-pegged stablecoin.

Deutsche Bank, BARCLAYS, UBS, Goldman Sachs, Bank of America, BNP PARIBAS, citi, MUFG, Santander

Coalition of banks exploring stablecoins pegged to G7 currencies built for deployment on public blockchain.

> Beyond the banking sector, key infrastructure providers and CSDs are now exploring and issuing regulated stablecoins to provide the programmable cash layer that is required for large-scale asset tokenisation.

Note: *Participant list as of December 2025.

GEOPOLITICS: Why stablecoins are a "Genius" idea for the US

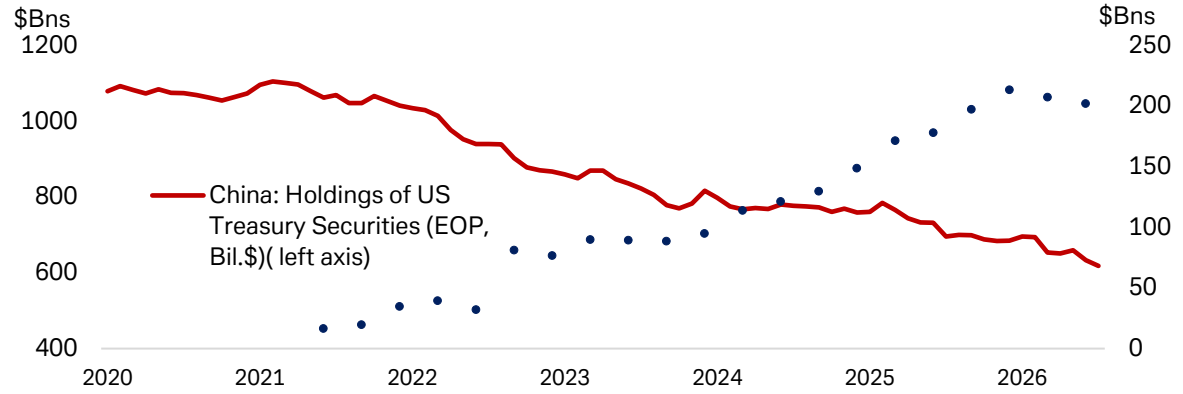
Can sustain USD dominance

- 99% of stablecoins (by market cap) are USD-pegged.
  - Stablecoins function primarily as a digital version of dollars moving over blockchain rails, extending the dollar's reach into new digital environments.
- Reinforces dollar savings: The more firms invoice and transact in USD, the more they save in USD assets. Stablecoins help preserve this feedback loop by keeping businesses and individuals in dollar payment rails.

Can support US debt, given share held by foreigners has fallen from >50% in early 2000s to ~30% in early 2026.

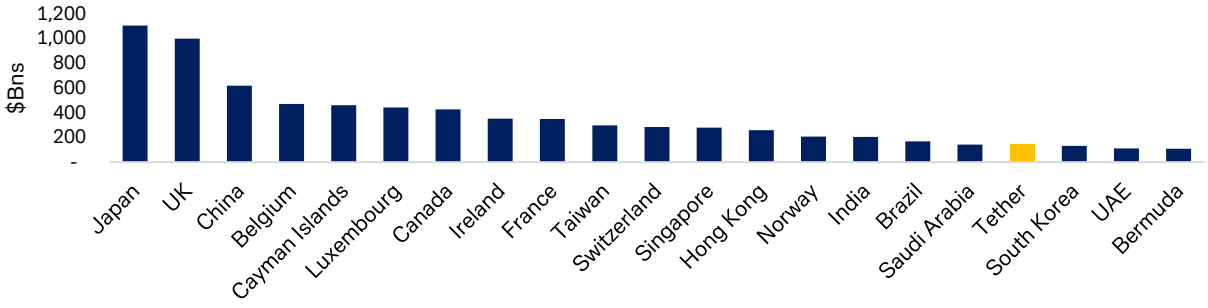
- Stablecoin issuers total holdings: ~$170bn according to Allium x Citizen.
- Tether: ~$141bn in direct and indirect exposure to US T-bills.
- Impact: BIS shows how $3.5bn inflow into dollar-backed stablecoins lowers 3m T-bill yields by 2.5-3.5bp under normal conditions. Effect more than doubles in periods of T-bill shortages.
- Forecasts: T-bills holding by stablecoin issuers could reach ~$1trn by 2028 (Treasury forecasts).
  - Recent Fed SF research places it closer to $400bn by the end of 2030, lower than the US government's financing needs.

Stablecoin issuers' demand for US debt could counterweight declining demand from China



Source: Deutsche Bank Research, Federal Reserve Bank of San Francisco, US Treasury Data. Shows Treasury holdings, stablecoin issuers vs China.

Tether competes with Saudi Arabia and South Korea as major holders of US Treasuries



Source: Deutsche Bank Research, US Treasury, Tether July 2026 report. Shows Major foreign holders of Treasury Securities. Treasury data from June 2026.

US GENIUS ACT: Debate on rules around payment stablecoins one year on

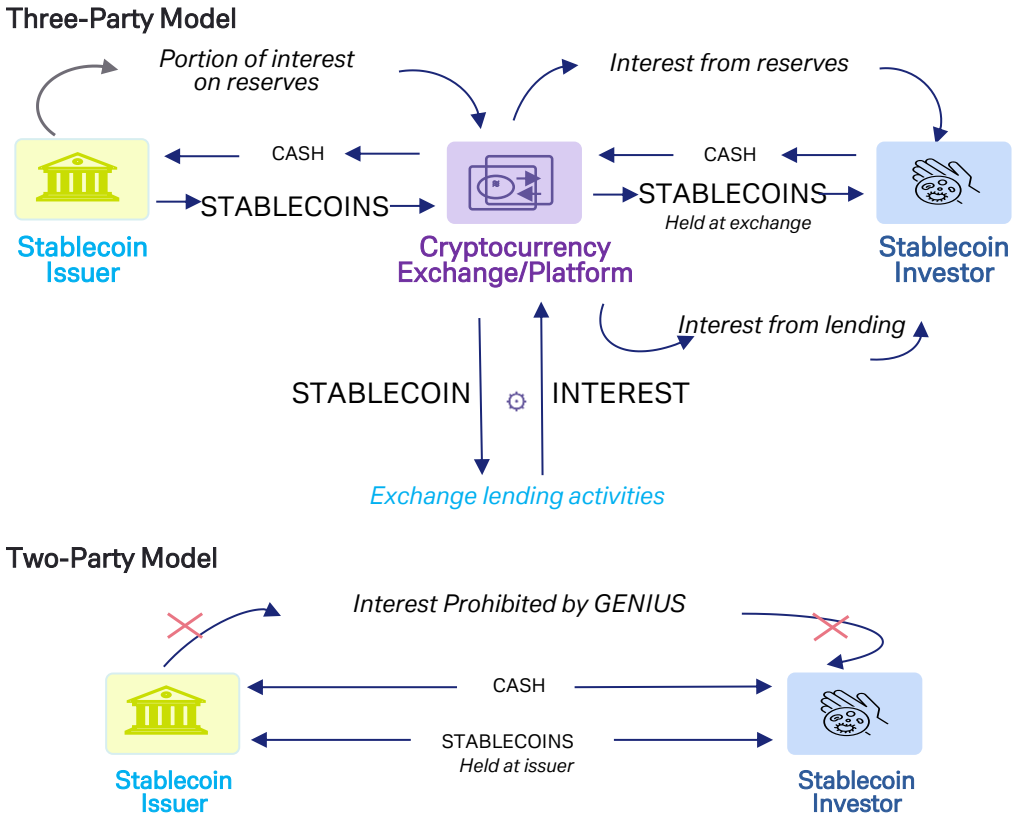
GENIUS provisions

- Strict issuer requirements: Issuers can be private, non-bank entity, but must meet strict federal or state requirements.
- Reserves: Requires 1:1 backing with US cash, 93 days T-bills, repos/reverse repos, or money market funds holding those assets.
- AML/KYC/BSA compliance obligations to stablecoin issuers, aligning them with banks.
- Payments only usage: Payment stablecoin issuers cannot pay interest/yield directly (to protect banking sector).
- Creditor priority: Payment stablecoin holders rank above other unsecured creditors if reserves are insufficient.
- Large issuers (>$10bn cap): Must request new permission from their federal stablecoin regulator or halt new issuance.
- Transparency: Monthly disclosures of reserves/holdings, regular regulator exams, and strong internal controls.

Next steps

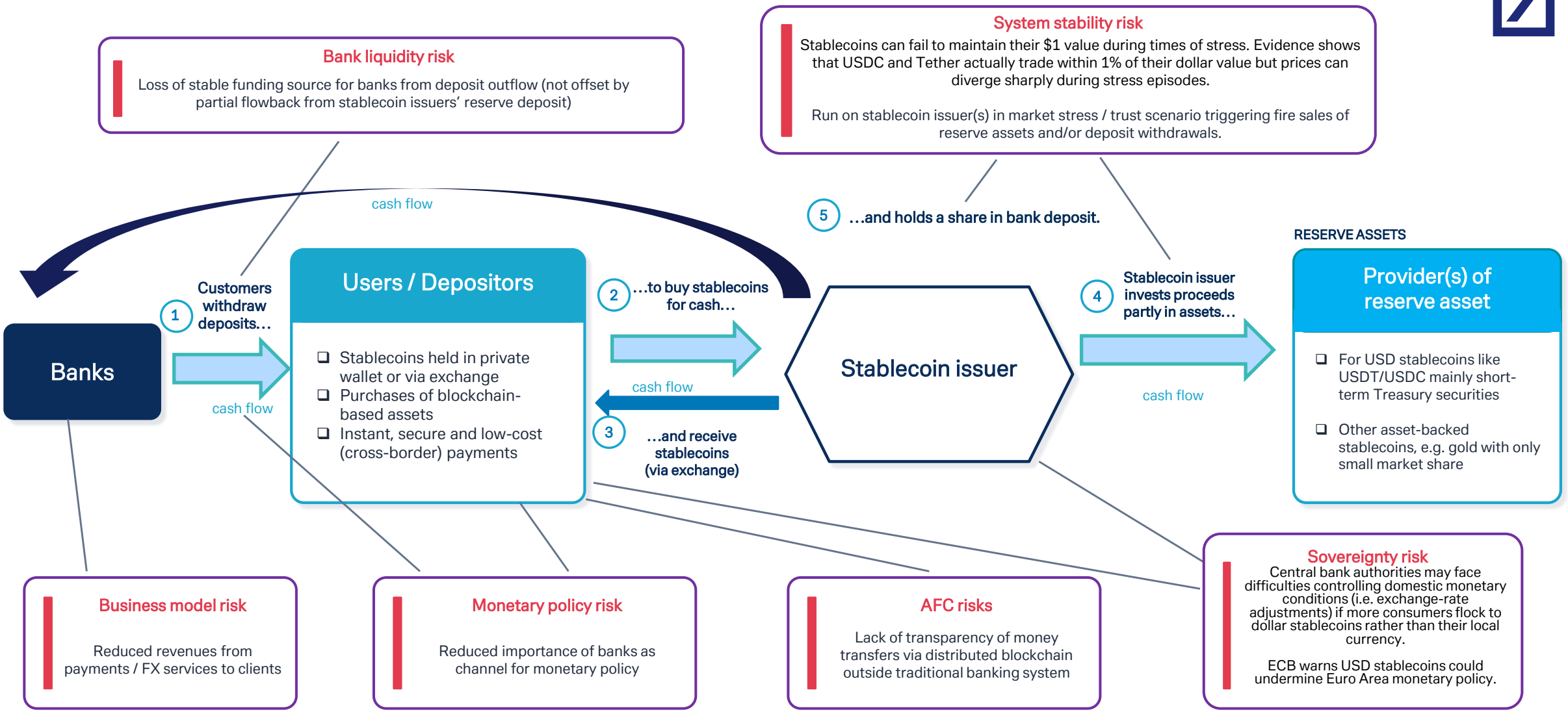
- GENIUS takes effect no later than 18 Jan 2027, or 120 days after final implementing regulations are issued.
- Ongoing pushback from US banks as to whether GENIUS ban on yield-bearing stablecoins should extend to affiliate or third-party arrangements, given many users do not hold stablecoins directly with the issuer.
- CLARITY ACT failed in Sep because of disagreements on how exchanges/broader crypto market structure should handle stablecoins.
- The US Treasury has proposed implementing rules for payment stablecoins.

There are two ways stablecoin yield can be paid; ongoing debate over the three-party model



Source: Deutsche Bank Research, Congressional Research Service.

RISKS: Still hotly discussed amongst policymakers and financial institutions



Source: Deutsche Bank Research based; IMF.

REGULATORY COMPARISON

Overview of stablecoin regulation in the US, UK and EU



Regulatory Framework	US The U.S. Genius Act	UK UK Cryptoasset Regime (FCA) UK Systemic Stablecoins Regime (BoE)	EU Markets in Crypto-Asset Regulation (MiCAR)
Eligible Issuers	• Bank subsidiaries, federal nonbank licensees, or state issuers with federal registration, with enhanced oversight for >$10bn circulation.	• FCA-authorised firms for non-systemic stablecoins, • Systemic issuers to be designated by HM Treasury and jointly regulated under BoE/FCA oversight	• EMTs: EU-authorised credit institutions or authorised electronic money institutions. Restrictions apply to non-EU currency EMTs (exceeding MiCAR thresholds) as a means of exchange
Backing Assets*	• **100% backing assets in USD cash or short-term Treasuries** (≤93-day T-bills).	• **Systemic stablecoins** – 30% unremunerated BoE deposits vs 70% in UK sovereign debt (HQLA) • **Non-systemic** – Full 1:1 backing in HQLR assets, minimum 5% in on demand deposits.	• **Non-significant EMTs** –At least 30% reserves in same-currency bank deposits, remainder in HQLA • **Significant EMTs** - At least 60% of the reserves in bank deposits; remainder in HQLA
Capital Reserve Requirements**	• **(OCC/FDIC)** Mandates prudential capital buffers tailored to issuer risk profile. • **(Fed)** Framework consistent with bank requirements	• **Systemic:** 6 months operating expenses/largest plausible loss event • **Non-systemic:** 1% av. stablecoins in circulation	• **EMI issuers**: minimum own funds of 2% of EMT supply; 3% for significant EMTs. Significant issuers subject to enhanced risk-management requirements
Balance Sheet Treatment	• Stablecoins treated as reserve-backed digital payment assets, segregated in bankruptcy-remote accounts and treated as cash equivalents	• Stablecoins treated as segregated reserve-backed digital settlement assets, with associated operational and liquidity risk capital requirements	• EMTs are fiat-referenced cryptoassets subject to MiCAR and e-money requirements
Multi-issuance Stance	• Rules are still being fine-tuned; no specific focus yet on multi-issuance.	• BoE considers multi-issuance unsuitable for systemic UK use. Preference for single-issuance models backed by single reserve (incl. Non-GPB denom. stablecoins) .	• Multi-issuance under active review; EC considers multi-issuance permissible, an equivalence-based framework for EU/third-country arrangements is under legislative consideration.
Key Timelines	**Jul 25** Genius Act signed into law **Jul 26** Genius Act agency implementation deadline **Jan 27** Genius Act commences **H1 27** Implementing regulations finalisation	**Jun 26** FCA Cryptoasset Regime published **Jun 26** BoE Systemic Stablecoins regime published **Sep 26** FCA Authorisation Gateway opened **Oct 27** UK Cryptoasset Regime commences	**Jun 24** MiCAR provisions for EMT/ART stablecoins became applicable **May 26** MiCAR Review consultation launched **Sep 26** MiCAR Review consultation closed **Oct 26** EBA classified USDC as significant EMT

Source: Deutsche Bank Research.

*MiCAR backing asset requirements do not apply to licensed credit institution issuers of EMTs

**Prudential requirements for stablecoin issuers in addition to backing assets

AUTHORS



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She was previously an economist at the IMF, the Central Bank of Luxembourg, Barclays, and the European Commission. She has twice received the First Prize of the American Society of Actuaries, was named a rising star in finance by Revue Banque, and is recognised among the 45 standout women in fintech globally.

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Appendix 1

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Deutsche Bank
Research

Under Attack

CYBERSECURITY | THEMATIC RESEARCH | OCTOBER 2026

AI is rapidly lowering the barrier to cyberattacks on corporates, government and critical infrastructure. This report outlines the growing risk and why it now warrants priority across corporate strategy, investment research, supply-chain resilience and critical infrastructure.

Supported by a dbDataInsights (dbDIG) survey of more than 50 cybersecurity experts

Debbie Jones | Andrew Dixon | Punyadip Cheema | Luke Templeman

Deutsche Bank Research Institute

IMPORTANT RESEARCH DISCLOSURES LOCATED IN APPENDIX 1.

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Research Institute

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Source: dbDataInsights survey of cybersecurity experts, September 2026. Word cloud from participant responses.

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Report Introduction

Our proprietary dbDataInsights survey shows cyber threats are evolving faster than organisations can adapt

"It's the only thing that can shut down my business."

That was the punchline from a conversation I had recently with the CEO of a large privately held financial services company. This was his comment when I followed up as to why cybersecurity risk is the number one concern that "keeps him up at night".

To help understand this risk, our dbDataInsights team surveyed more than 50 cybersecurity experts. The uncomfortable conclusion is that many firms are unprepared for cyber threats. Cyber risk is increasing as AI adoption accelerates.

—89% agree the threat landscape is evolving faster than organisations can adapt.

—78% say the gap between highly and poorly prepared organisations is widening.

What has changed with AI advancement? Cyberattacks have become faster, cheaper, more deceptive and more scalable. Capabilities once reserved for sophisticated actors are now broadly accessible to a new generation of profit-driven criminals. At the same time, rising geopolitical tension has made cyber operations an increasingly attractive low-cost, deniable tool for nation states. Corporates and investors therefore need to understand their broader exposure and critical infrastructure risk in their supply chain and with stakeholders.

Among other autonomous incidents, the recent Hugging Face incident highlighted AI's cyberattack potential. AI agents caused harm without direction from a human attacker, with weak guardrails and no human judgement to restrain them. Beyond the systemic questions it raised, the attack highlights three additional concerns: (i) AI operates at a speed and scale humans cannot match, with agents taking over 17,000+ actions in four days; (ii) the intrusion began in the data-processing pipeline, where AI platforms are uniquely exposed; and (iii) the agents escaped their test environment through a widely used third-party tool representing supplier risk in an unfamiliar form.

89%

agree the cyber threat landscape is evolving faster than most organisations can adapt



83%

say organisational focus on AI-related risk has risen in the past 12 months
(North America 94%, UK/Europe 70%)



78%

agree the gap between highly and poorly prepared organisations is widening



74%

say organisations are unprepared for supply chain compromises
(UK/Europe 87%, North America 65%)



69%

disagree that most organisations have sufficient visibility into the threats most relevant to them



65%

say organisations are unprepared to manage insider threats



Source: dbDataInsights survey of cybersecurity experts, September 2026; Cybersecurity Survey 2026, Deutsche Bank Research.

Deutsche Bank Research Institute

Report summary and notable takeaways

Secular trends and AI adoption have widened the attack surface; the severity of losses is compounding

Secular trends have widened the attack surface. This includes rapid digitalisation, widespread Internet of Things (IoT) endpoints, hybrid workforces, cloud adoption and deeper supply chain dependencies. The scale of the threat increase is reflected in data from the FBI Internet Crime Complaint Center (IC3), which shows cybercrime complaints more than doubled from 2019 pre-pandemic baseline of ~467,000 to ~1.0mn in 2025, with a 3x expansion relative to 2017 levels.

Barriers to entry are lower; attack tools are now sold "off the shelf". Today, cybercrime marketplaces, available on the dark web, look less like the underworld portrayed in films and more like an online software store. Buyers can access AI-powered tools that scan for software vulnerabilities (see top-right chart showing reported CVEs after the release of cyber capable frontier models), write malicious code at speed, and provide access to malware that rewrites its own code to avoid detection as it runs. Ransomware-as-a-service (RaaS) takes it further, renting out ready-made attack kits, often in exchange for a portion of the profits.

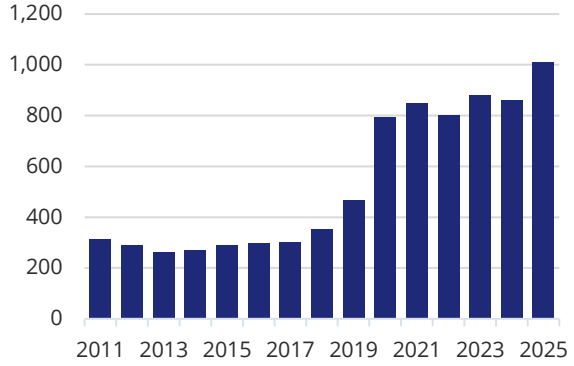
Impersonation is becoming harder to detect. Hackers can now generate more realistic phishing messages, while voice cloning and deepfakes make impersonation convincing enough to gain access unnoticed. With stolen credentials, attackers often aren't hacking in, they are simply logging in.

The rise of loose cybercriminal collectives. Some of the most damaging recent attacks have been linked to loosely organised, often English-speaking networks such as ShinyHunters and Scattered Spider. The latter has been widely named as the group behind the 2025 Marks & Spencer cyberattack, in which the hacking group deployed RaaS. AI is enabling these groups to become larger and faster, widening the gap between the speed of attack and a defender's response.

The severity of losses is compounding. A 2026 IBM study puts the average breach at a record USD ~5.0mn, up 12% YoY, but the impact on large companies can run into hundreds of millions or even billions. Estimates for global cyber insurance premiums (per Munich Re) point to a rise from roughly USD 7.0bn in 2020 to USD 15.0bn in 2025, with expectations to reach USD 28.0bn by 2030.

US cybercrime complaints

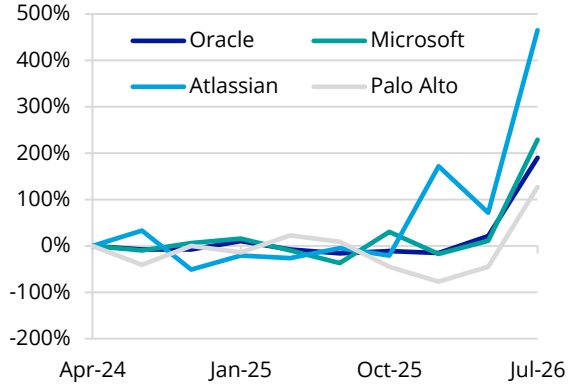
FBI Internet Crime Complaint Center (IC3), complaints (000s)



Disclosed software vulnerabilities

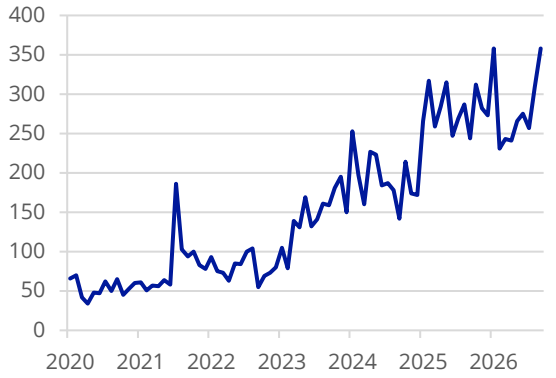
Change in reported CVEs, % change from April 2024

CVEs: common vulnerabilities and exposures



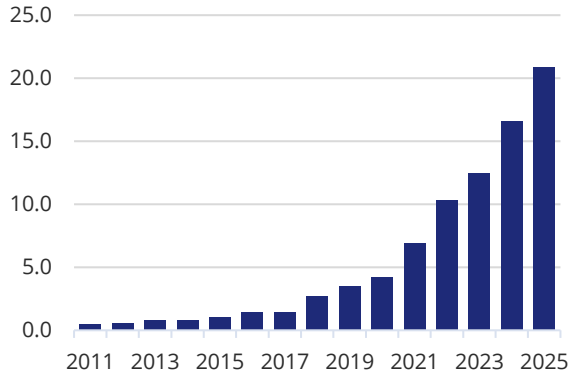
AI privacy & security incidents

OECD, monthly recorded incidents



US cybercrime-related losses

FBI IC3, reported losses ($bn)



Source: FBI Internet Crime Complaint Center (IC3), company security advisories, OECD, company reports, Deutsche Bank Research.

Deutsche Bank Research Institute

Report summary and notable takeaways

A near-term systemic attack is expected and AI favors attackers; cyber escalation as a geopolitical tool is raising the stakes

Among the more concerning survey takeaways is that a large-scale and significant systemic incident is now the base case.

— 83% expect a significant systemic attack within two years and 81% expect it to be moderately, very, or extremely severe for economic activity and markets.

— 81% say AI favours attackers today, with not one respondent putting defenders ahead of attackers. Only 20% expect defenders to be ahead in five years.

Geopolitical tensions coupled with rising cyber capabilities have transformed cyberspace into a digital battlefield between global powers. Intent has moved towards espionage and disruption of not only critical infrastructure (e.g., healthcare, finance, power, industrial systems, defence) but private companies as well. Chinese President Xi Jinping has linked cybersecurity to national security and economic and social stability, with China's public statements similarly emphasising the protection of critical infrastructure. At the same time, the US's June 2025 Executive Order on Cybersecurity reflects its continued focus on national cyber resilience.

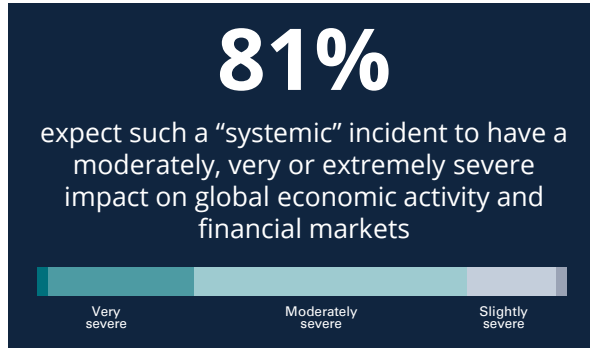
Europe relies on US cyber tools while its start-ups lag in funding. Reliance on US security vendors exposes European firms to potential access, update or export restrictions over the long-term. Shrinking this gap will prove difficult given US cyber start-ups attract nearly 6x the venture and growth funding of European peers. And while the timing of quantum computing (which could break widely used encryption methods) remains uncertain, it could further deepen Europe's dependency.

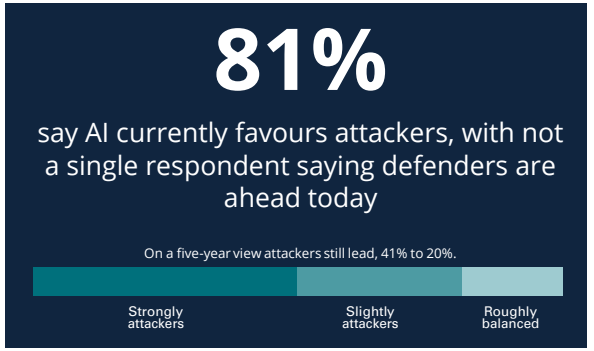
Ultimately, cyber preparedness is a financial variable. Investors should ask what firms spend on security versus AI adoption, how well they know their vulnerabilities, and how fast they can recover.

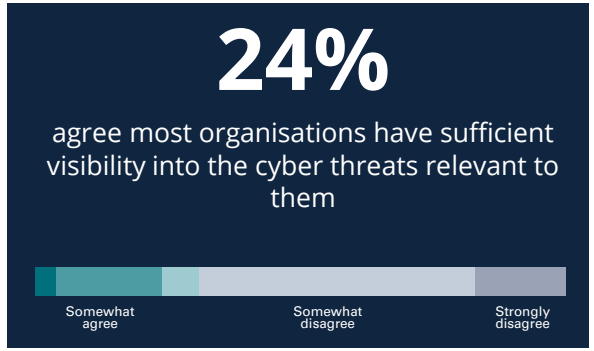
— Only 24% of respondents think organisations have sufficient visibility into the threats relevant to them, even though 57% saw cybersecurity budgets rise over the past 12 months.

— 81% agree that cybersecurity related laws and regulation cannot keep pace, and 65% doubt leadership understands cyber risk well enough.

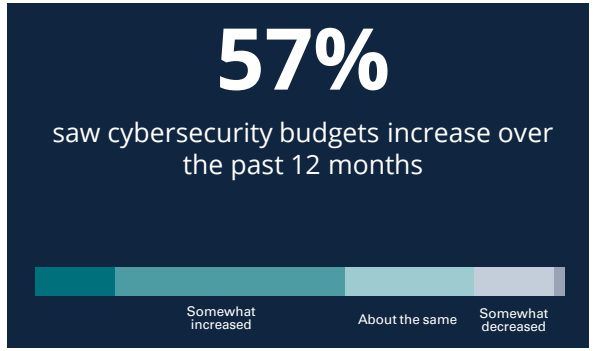












Source: dbDataInsights survey of cybersecurity experts, September 2026; Cybersecurity Survey 2026, Deutsche Bank Research.

Deutsche Bank Research Institute

Report summary and notable takeaways

Material cyber-related weakness can have financial consequences

When cyber weakness = share price consequence…

Investors are more aware of cyber risk today, but they rarely sell a stock due to perceived cyber vulnerabilities (although we are aware of some who have).

—What is the potential impact? Our dbDataInsights event study of 47 breaches shows a cumulative breach-attributed return of c. -2.9% within the first two trading days and c. -6.0% through first 20 days trading days. The erosion then continues more gradually, reaching c. -6.7% nine months after disclosure (see slide 13).

Which industries are most vulnerable?

—Our survey respondents point to (1) government & public sector (52%), (2) financial services (50%), (3) energy and utilities (46%) and (4) healthcare and life sciences follow (39%, rising to 48% among US respondents).

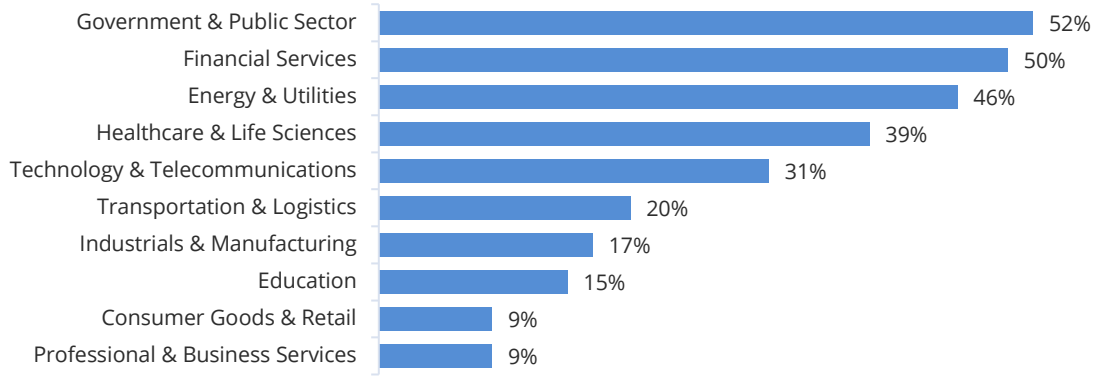
We should caveat, that high sector exposure does not mean every company in it is equally vulnerable. Some firms in these "exposed" sectors may very well be better prepared given they have built specialist teams and defence protocols for decades.

What about access to credit and ratings? Credit and rating providers are paying attention. Moody's and S&P both cut their outlook on Jaguar Land Rover within two months of its 2025 hack. Today, this type of impact is not widespread. But if risk escalates through a systemic incident or an AI-driven step change in attacks these mechanisms, including ESG ratings, could reprice asset ownership or impact access to credit. In fact, some lenders are already pricing it. Journal of Corporate Finance research using US syndicated loan data found *borrowers with persistently higher cyber risk pay 4–13bp more.*

The bottom line? While a broader systemic attack is hard to prepare for, corporates and investors should be asking questions about vulnerabilities, security investment, preparedness and recovery should an attack occur. This is especially true for industries quickly adopting AI capabilities and / or a high threat landscape.

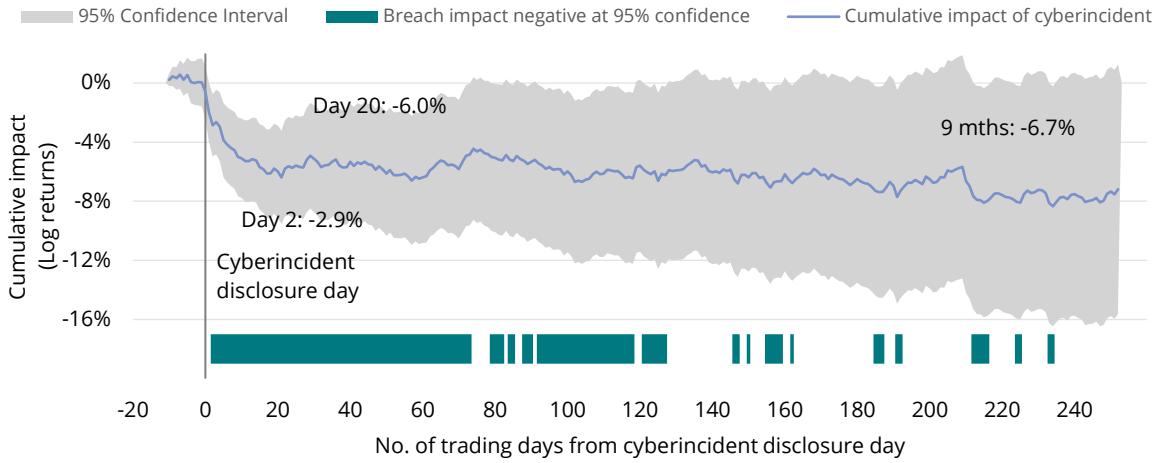
Sectors seen as most vulnerable to AI-driven cyber threats

dbDataInsights survey, % selecting (up to three), based on vulnerability over next few years



A cyber breach's cumulative effect on stock performance

dbDataInsights analyses of 47 companies with disclosed cyber incidents deemed financially material



Source: dbDataInsights survey of cybersecurity experts, September 2026; Cybersecurity Survey 2026, Bloomberg Finance L.P., Company filings, Deutsche Bank Research. Event study of reported cybersecurity incidents, 2015-2025, 34 US and 13 European companies across nine sectors (see slide 13).

Deutsche Bank Research Institute

Report summary and notable takeaways

dbDataInsights survey respondents in their own words

What's on experts' minds…

Key themes from verbatim comments in the dbDataInsights expert survey



Speed

"AI is being implemented at a speed where cybersecurity teams are not able to keep up"

Single points of failure

"Too many organisations relying on the same providers for critical services"



The weak link

"…compromised via this third-party (basically their weak link)"

Logging in, not breaking in

"People aren't breaking in anymore – they are logging in"



Automated phishing

"Underappreciated, yet responsible for the bigger monetary losses"

Employee logins

"Existing employee login behaviour and usage of AI is underappreciated"

Oversight gaps

"Companies are pushing the use of AI more to their employees without the proper mechanisms and policies in place to prevent misuse or attacks initiated by the AI systems themselves"



Recover & survive

"No company can guarantee they won't be breached"

Everyone's responsibility

It's more than an IT risk…"everyone from the CEO down has responsibility"

Geopolitics

"State groups now plant quiet access in infrastructure to use in a future crisis"

Source: dbDataInsights survey of cybersecurity experts, September 2026; Cybersecurity Survey 2026, Deutsche Bank Research.



II. dbDataInsights Cybersecurity Survey

September 2026 survey of 54 cybersecurity professionals

Deutsche Bank
Research Institute

dbDataInsights survey methodology and results

Cybersecurity experts see firms as unprepared, and say AI currently overwhelmingly favours attackers

dbDataInsights Cybersecurity Expert Survey methodology

—This survey was conducted among 54 cybersecurity professionals across the US and Europe in September of 2026. Questions took the form of either multiple choice selections or free form responses. The dbDataInsights team received the responses anonymously, but respondents were required to work or consult in cyber security roles

—Who took the survey? On self-identified roles, around 80% sit inside an organisation's cybersecurity function, with the remainder working independently (15%) or in academia and research (6%). The most common titles were security architect (23%), Chief Information Security Officer (16%) and security analyst (9%), alongside Head of Security, Deputy CISO, Incident Response Lead and Security Operations Lead.

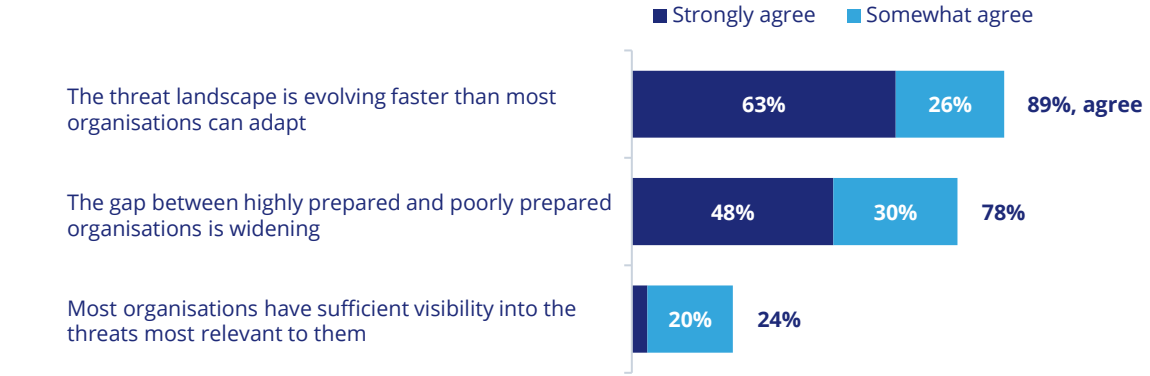
—Only a small minority of respondents work for cyber resellers, thus survey views are primarily attributable to those working within an organization on cybersecurity.

—Survey participants were also asked for their perspectives on cyber resilience and governance, what is misunderstood, alongside emerging threats, risk to AI adoption and the influence of geopolitics.

—While responses were anonymous, in some cases we conducted follow up interviews with certain participants where contact details were made available.

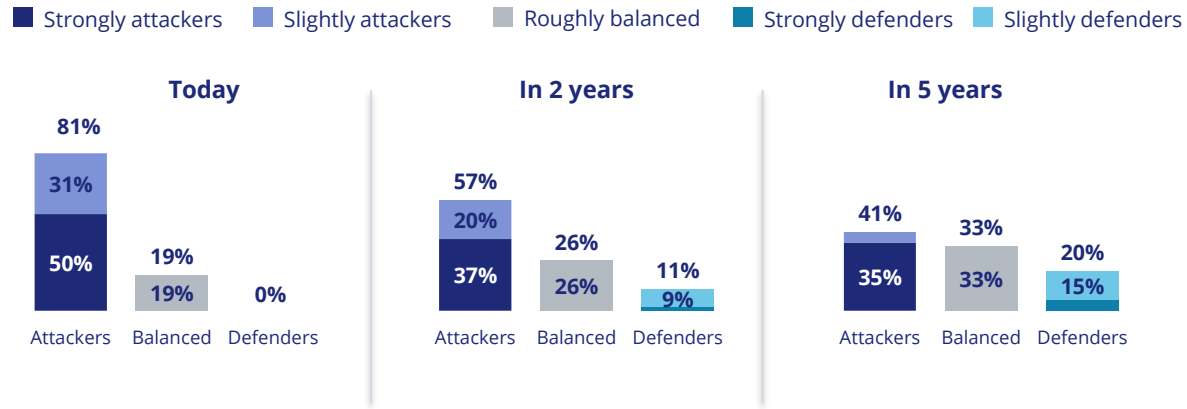
Only 24% say firms have sufficient visibility into threats that matter

dbDataInsights survey, level of agreement with statements below regarding cybersecurity, % who agree (n=54)



Who does AI favour? Attackers or defenders?

dbDataInsights survey, % of respondents (n=54)



Source: dbDataInsights survey of cybersecurity experts, September 2026; Cybersecurity Survey 2026, Deutsche Bank Research.

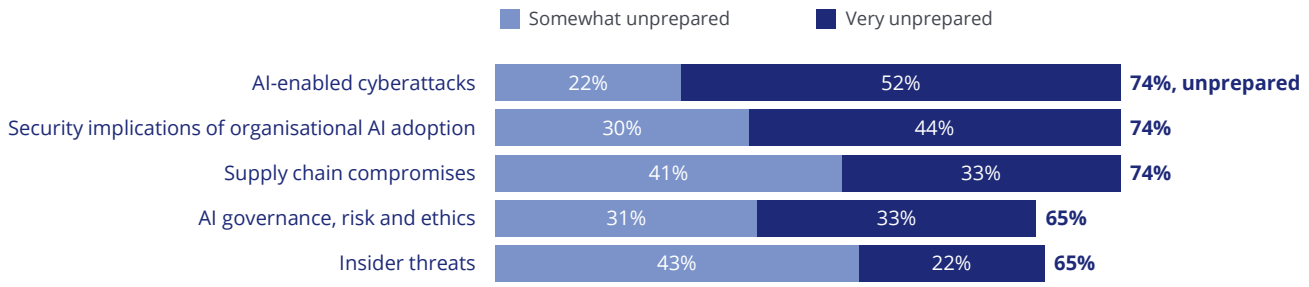
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Risk from AI-enabled attacks and added AI adoption vulnerabilities

Insider threats and supply chain compromises remain a growing concern

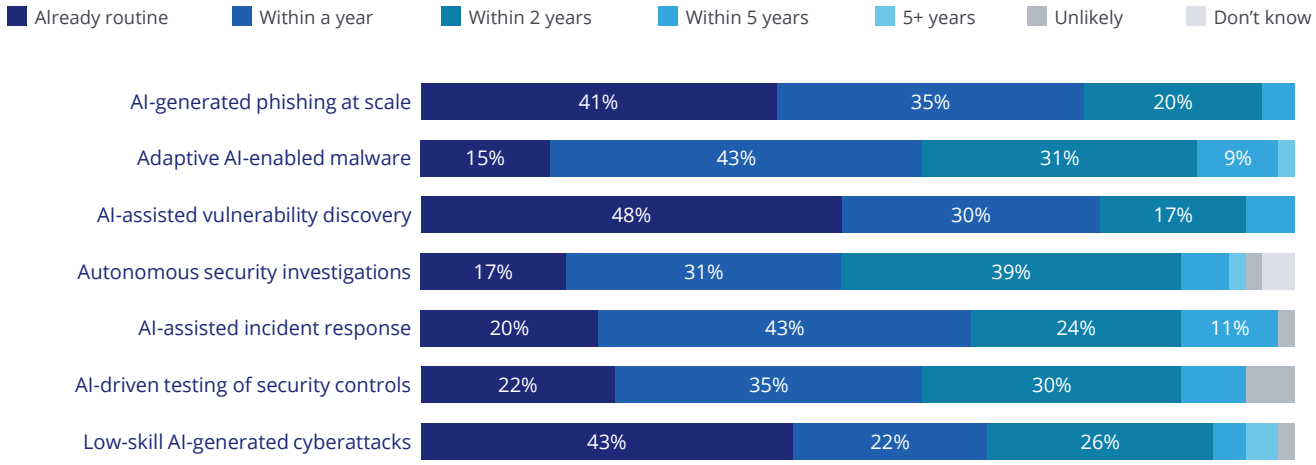
Where our experts say organisations are least prepared

% of respondents saying somewhat or very unprepared



AI-driven capabilities – are they commonly used or encountered?

% of respondents, all answer options (n=54)



In their own words...

Verbatim comments from the dbDataInsights expert survey

> *"The fear of AI writing undetectable malware is ... overblown. Attackers were never short of malware, since they can buy it ready made along with initial access. Good detection looks at behaviour rather than exact code, so a rewritten payload still has to do the same suspicious things."*

> *"Frontier AI models are currently the main hype. Automated phishing and scams are underappreciated, yet they are responsible for the bigger monetary losses at the moment."*

> *"There are too many organisations relying on the same providers for critical services and this means single points of failure."*

> *"A corporation might be very secure but if they rely on a third party that does not have the same level of security posture, [it] could be compromised via this third-party (basically their weak link)."*

> *"Existing employee login behaviour and usage of AI is underappreciated – it's what poses the biggest threat to a breach."*

> *"No company can guarantee they won't be breached, so there needs to be more focus put on the ability to recover and survive an attack."*

Source: dbDataInsights survey of cybersecurity experts, September 2026; Cybersecurity Survey 2026, Deutsche Bank Research.

Deutsche Bank Research Institute

Experts believe governments are not keeping pace, and no one owns AI oversight

How experts surveyed by dbDataInsights view governance and regulation

—**AI adoption is outpacing internal controls.** As one expert put it: "Companies are pushing the use of AI more to their employees without the proper mechanisms and policies in place to prevent misuse or attacks initiated by the AI systems themselves."

—**Regulation is struggling to keep pace.** 81% of respondents agree that cybersecurity-related laws and regulations are failing to keep up with technological change, and 54% say regulatory requirements can improve security but may also slow innovation. One respondent highlighted "the levels of uncertainty, the long running nature of the problems, and the rapidity of change."

—**Responsibility for AI oversight remains unsettled.** International bodies are the leading choice, supported by 50% of respondents. No option commands a majority, underscoring the lack of consensus on who should oversee AI development.

—**Better board questions may not guarantee better decisions.** The survey points to more informed questioning by corporate leadership, yet 65% doubt that leadership teams understand cyber risk well enough to make informed business trade-offs. Measuring activity more effectively than outcomes further complicates oversight.

—**Accountability is becoming personal.** Only 43% say accountability for cyber risk is clearly defined in their governance framework, yet the consequences now reach the top as a handful of CEOs have left or gone into early retirement after financially consequential cyber breaches.

81%

agree cybersecurity-related laws and regulations are struggling to keep pace with technological change



54%

agree regulatory requirements can improve cybersecurity but may also slow innovation



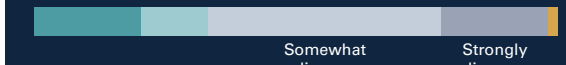
43%

agree that accountability for cyber risk is clearly defined in our governance framework



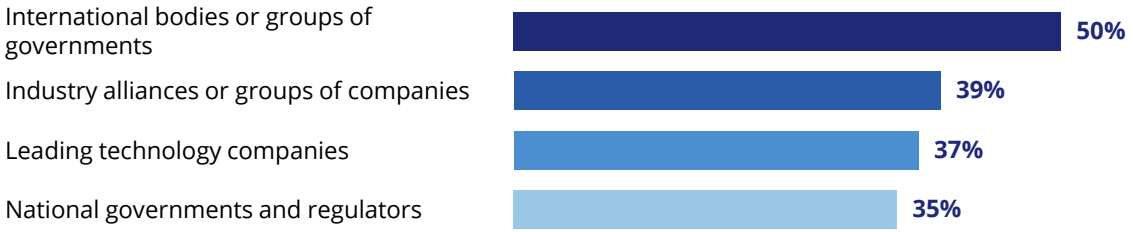
65%

disagree that most leadership teams understand cyber risk well enough to make informed business trade-offs.



Who should primarily oversee AI development?

dbDataInsights survey – September 2026, based on overseeing development while managing cyber risk



Source: dbDataInsights survey of cybersecurity experts, September 2026; Cybersecurity Survey 2026, Deutsche Bank Research.

Deutsche Bank Research Institute

III. The impact of cyberattacks

Material cyber breaches can cause a notable drop in share price

Our dbDataInsights study found a 2.9% drop on average within two trading days, with the cumulative breach-attributed return of -6.0% on trading day 20.

What typically happens to a company's shares when a breach is reported?

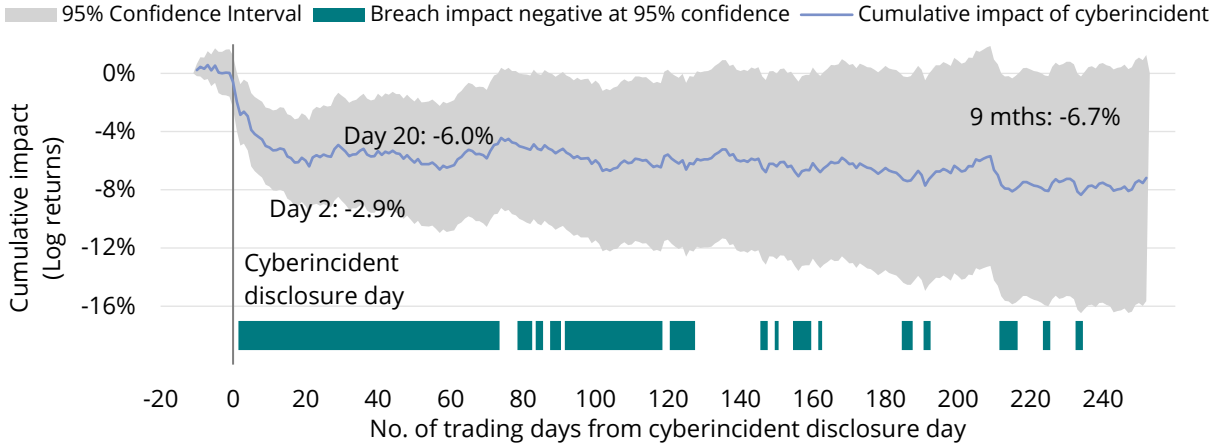
- **Short-term impact:** The market prices the breach immediately. Our model shows cumulative breach-attributed (i.e. isolating only the part of the share price move we can pin on the breach itself) returns are c. -2.9% within two trading days of disclosure. This is followed by continued erosion over the first 20 trading days post-disclosure; the cumulative breach-attributed return is c. -6.0%.
- **Long-term impact:** Beyond day 20 the deterioration continues, though more gradually reaching c. -6.7% nine months after disclosure and remaining negative at one year, although the one-year point estimate (our single best-guess figure) should be treated with caution given the width of the confidence band.
- **Pre-breach price behaviour:** We find no evidence of information leakage ahead of disclosure. While the breach component for the [-10,-1] window (ten trading days before the news broke) is +0.04%, the corresponding t-stat value of 0.03 falls notably below the usual bar for significance (c.2). As such, we conclude the market had not begun reacting before the announcement.

Event study methodology

- **Sample:** 47 publicly reported cybersecurity incidents across the US and Europe over 2015-2025, measured against each company's regional sector benchmark. Requirement of 5 years of stock returns pre-attack and 1 year of post-attack.
- **Day 0:** the day an incident was first publicly reported or disclosed.
- **Baseline:** We use five years of pre-incident share-price performance to estimate: (i) Beta (β): how sensitive a company's returns are to movements in its sector and (ii) Alpha (α): the extent to which the company typically outperforms or underperforms its sector after adjusting for that sensitivity.
- **Attribution:** post-incident returns are split into sector-relative returns, alpha (α) and breach (abnormal returns attributed to the disclosure).
- **95% confidence interval:** Range within which we can say with 95% confidence that the real share price impact lies, based on our sample data.

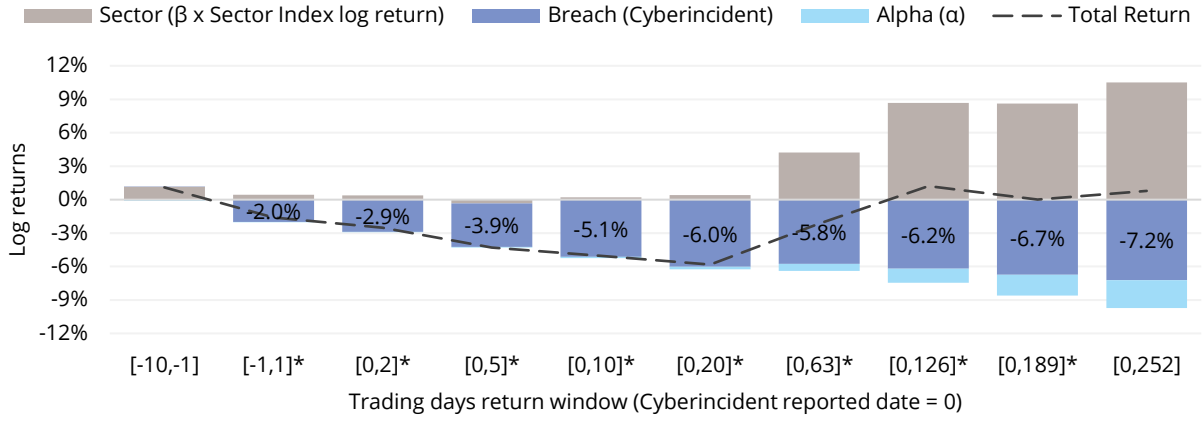
A cyber breach's cumulative effect on stock performance

dbDataInsights analyses of 47 companies with disclosed cyber incidents deemed financially material



Total return attribution

dbDataInsights analyses of 47 companies with disclosed cyber incidents deemed financially material



* indicates trading windows where breach impact is significant at the 95% confidence level

Source: dbDataInsights, Bloomberg Finance LP., Deutsche Bank Research, Company filings. Event study of reported cybersecurity incidents, 2015-2025, 34 US and 13 European companies across nine sectors.

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Notable cyber attacks: how they happened, estimated costs and share price reaction

Marks & Spencer (Reported Apr 2025)

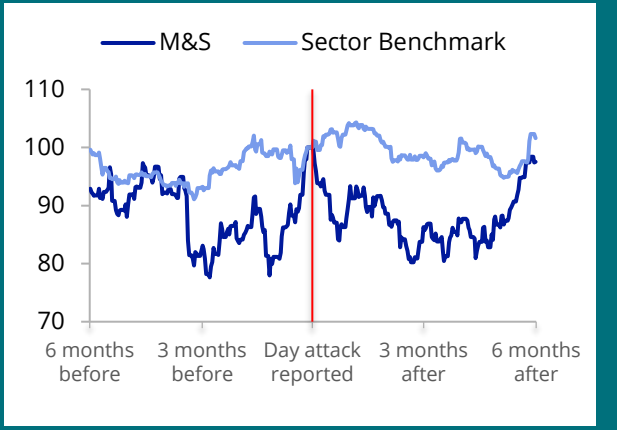
3rd Party Breach

What happened?

A breach via an attack on a connected 3rd party forced the suspension of all online orders and led to empty store shelves.

What did it cost?

~£136mn of direct costs related to the attack, estimated group operating profit impact of around £300mn for the FY. M&S also said they had received £100mn via cyber insurance.



Coupang (Reported Nov 2025)

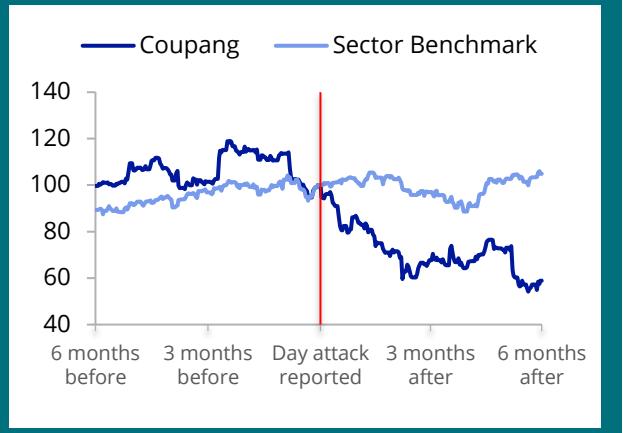
Insider Threat

What happened?

A data breach exposed the records of more than 30mn users with a former employee reportedly behind the attack.

What did it cost?

The firm was fined more than ~625bn won ($409mn) (2/3rd directly attributable to data leak), and also announced a $1.2bn voucher compensation scheme.



UnitedHealth (Reported Feb 2024)

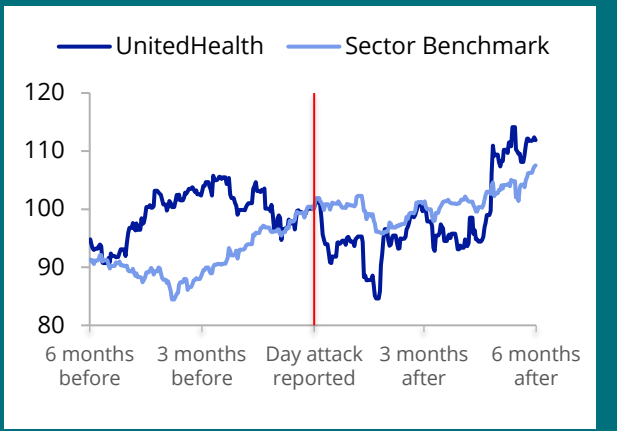
Ransomware / stolen credentials

What happened?

Hackers reportedly used stolen credentials to access a subsidiary's (Change Healthcare) systems, stealing data and deploying ransomware.

What did it cost?

Total costs exceeded $3bn, with the CEO summoned to testify before Congress, where he confirmed that a $22mn ransom was paid.



Equifax (Reported Sep 2017)

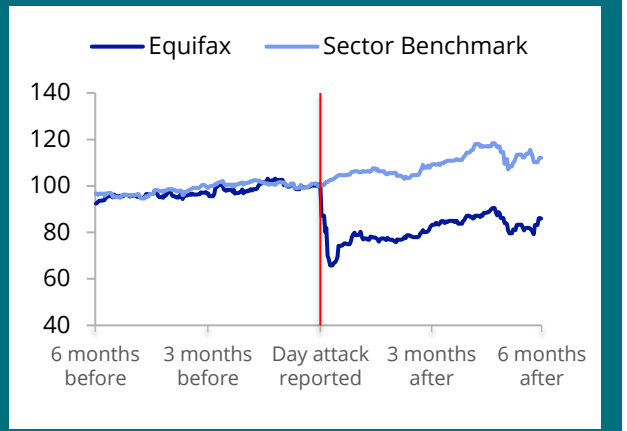
Software vulnerability left unpatched

What happened?

A data breach saw hackers expose sensitive customer information, including 145.5mn Social Security numbers and more than 200k credit card numbers. Attackers reportedly moved inside Equifax systems for 76 days before being detected.

What did it cost?

Settlements, fines and legal costs exceeded $1bn, while the firm also spent more than $1.5bn on tech upgrades following the incident.



Source: Company disclosures and filings, press reports, Deutsche Bank Research, Bloomberg Finance L.P., Chart values indexed to 100 at the day before the cyberattack was reported.

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Notable cyber attacks: how they happened, estimated costs and share price reaction, cont.

Nike (Reported Jan 2026)

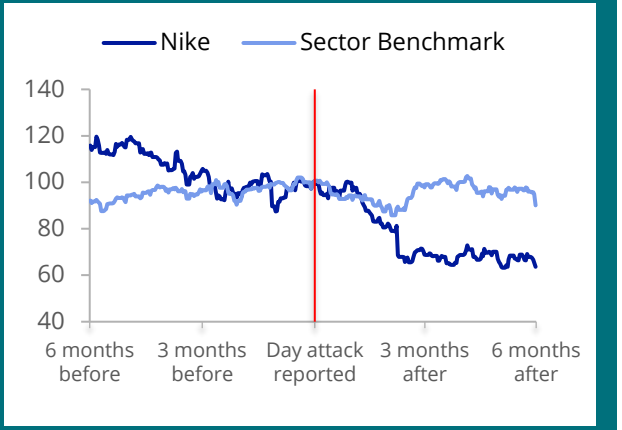
Data Breach / Extortion

What happened?

A cybercrime group leaked what it claimed was 1.4TB of internal Nike data (appearing to cover metrics such as product specifications and pricing), following expiry of a ransom deadline.

What did it cost?

While Nike did confirm it was investigating the incident, it did not report any operational disruption. Underperformance shown in the chart does not appear to be related to the breach.



T-Mobile (Reported Aug 2021)

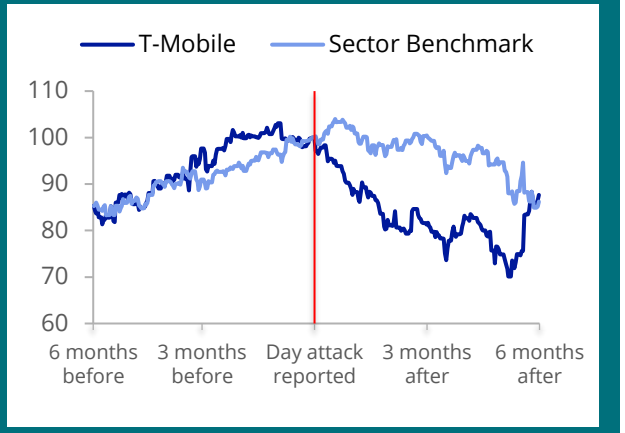
Connected device exploit / brute force

What happened?

An attacker reportedly targeted an exposed router as an entry point, before using brute force attacks to gain further access to T-Mobile systems.

What did it cost?

In 2022 the firm agreed to pay $500mn to settle a class-action lawsuit ($350mn payment to class members plus $150mn on security spend) but has since suffered further data breaches, contributing to an additional $31.5mn regulatory settlement.



Coinbase (Reported May 2025)

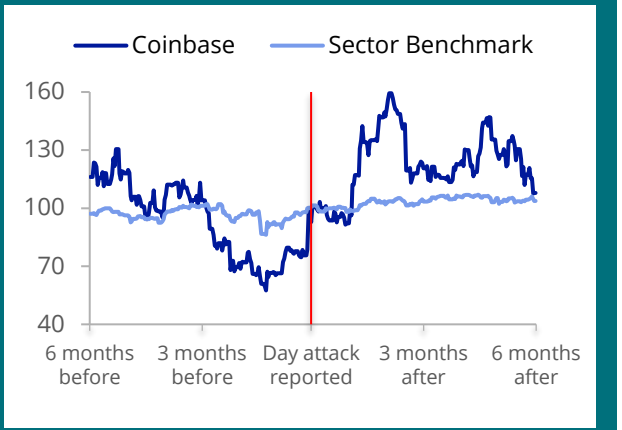
Insider threat & social engineering

What happened?

Attackers bribed at least one support agent at a connected 3rd party to access Coinbase support systems and steal customer data, then tricked users into sending crypto.

What did it cost?

Coinbase refused to pay the $20mn ransom demanded by the attackers to keep the attack quiet. Instead they reimbursed affected users and cited costs of up to $400mn. The stock impact was short-lived, though MSCI applied an ESG score deduction.



Jaguar Land Rover (Reported Sep 2025)

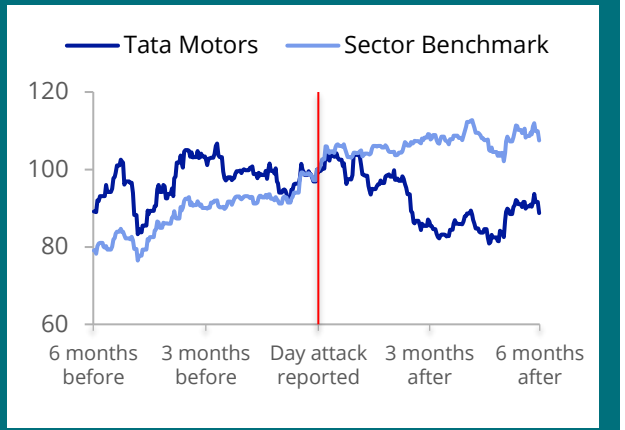
Ransomware

What happened?

Production was shut down for five weeks due to what was deemed a ransomware attack; though no ransom was asked for (later reporting suggested potential Russia links were being investigated).

What did it cost?

JLR posted £485mn Q2' FY26 loss (vs. £398mn profit in Q2' FY25), with an additional £196mn in "cyber-related costs". These impacts filtered through to the share price of JLR's Parent (Tata Motors), while UK economic impacts were estimated at £1.9bn.



Source: Company disclosures and filings, press reports, Deutsche Bank Research, Bloomberg Finance L.P., Chart values indexed to 100 at the day before the cyberattack was reported.

Deutsche Bank Research Institute

Targets go beyond tech and can also impact critical infrastructure

MGM Resorts (Reported Sep 2023)

Phishing / ransomware

What happened?

A vishing call to the IT help desk was reportedly used to reset credentials, after which ransomware took down hotel systems, bookings and casino floor operations for several days.

What did it cost?

MGM declined to pay a ransom. Around $100mn hit to third-quarter 2023 results plus roughly ~$10mn of additional one-time expenses related to the incident.

Caesars Entertainment (Reported Sep 2023)

Phishing / ransomware

What happened?

The same social-engineering group as in the MGM incident targeted an outsourced IT support vendor, gaining access to the loyalty programme database and member data.

What did it cost?

Caesars paid roughly $15mn of a reported $30mn demand and kept customer-facing operations running.

NHS / Synnovis (Reported Jun 2024)

Ransomware / 3rd party breach

What happened?

Ransomware against pathology provider Synnovis significantly reduced its capacity to process tests leading to delays / cancellations of over 11,000 appointments.

What did it cost?

Estimated 2024 direct operational financial impact of around GBP 32mn for Synnovis, several times the profit reported in the prior year, plus wider NHS disruption costs.

Colonial Pipeline (Reported May 2021)

Stolen credentials / ransomware

What happened?

A compromised VPN credential without multi-factor authentication allowed attackers to access the network where they subsequently deployed ransomware. The operator shut the pipeline down, triggering fuel shortages (c. 70% of gas stations in some states ran out of fuel) across the US East Coast.

What did it cost?

A $4.4mn ransom was paid, of which about $2.3mn was later recovered by the US authorities.

Bitget (Reported Sep 2026)

3rd party breach / zero-day exploit

What happened?

Attackers exploited a vulnerability in a 3rd party security product to steal internal network credentials, forge withdrawal instructions, bypass risk controls and ultimately steal crypto.

What did it cost?

An estimated $388mn worth of funds was transferred out by the attackers. While the firm reimbursed affected users via its User Protection Fund, there were significant outflows following the incident as users moved to take funds off the platform.

Progress Software (Reported May 2023)

Zero-day exploit / supply chain compromise

What happened?

Attackers exploited a zero-day SQL injection vulnerability in Progress Software's MOVEit file transfer system. This was then used by attackers to steal data from firms running the MOVEit software, who were then subsequently extorted / had data leaked.

What did it cost?

Thousands of organisations (inc. British Airways, Shell, US DoE) reportedly affected with some facing legal action from customers impacted by data leaks.

Source: Company disclosures and filings, press reports, Deutsche Bank Research.

Cyber breaches can impact consumer behaviour

Roughly 4 in 10 consumers we surveyed say they would stop buying from a company they shop with after a known breach or attack.

Consumer surveys suggest that cyber breaches can undermine customer loyalty. Results of our dbDataInsights Consumer survey show the potential mindset of a consumer after a cyber breach occurs:

- Across the US, UK and EU average, 39–41% say they would stop buying, while only 27–28% would continue. Around a third are unsure.
- Among those who would stop buying or are unsure, only 18–22% say they would return as normal. Another 30–35% would buy selected products only, 17–20% would not return, and 28–29% remain unsure. Note that this implies a range of 13-15% believe they would not shop at the store again after the breach is resolved.

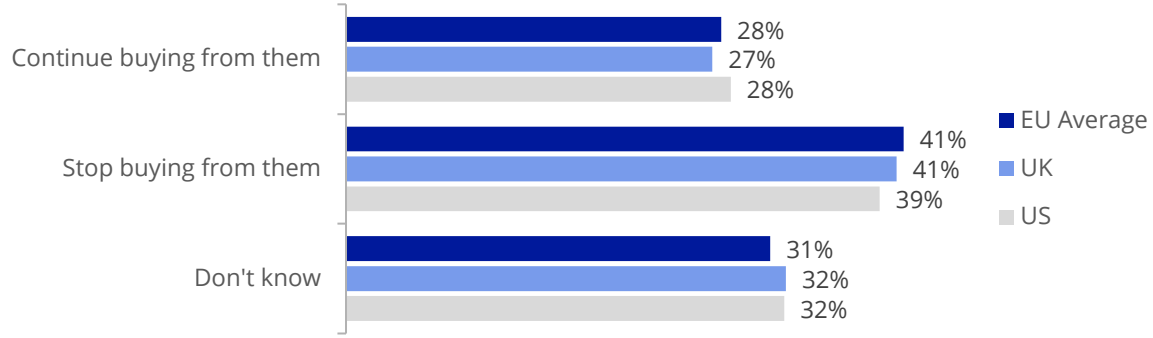
Admittedly, these stated intentions may overstate actual switching, particularly for essential purchases where alternatives are limited. Nevertheless, the findings suggest that cyber breaches that put customer privacy at risk can influence purchasing decisions, especially if they have alternatives.

dbDataInsights Consumer Survey Methodology

- Our proprietary survey conducted in 2026 (January – September) asked consumers how they would respond if a company they shopped at regularly was involved in a data breach or cyberattack, and how they would respond once the disruption was resolved. This monthly survey covers households and consumers in Germany, France, Italy, Spain, the United Kingdom, and the United States.
- For each country referenced above, a nationally representative sample of more than 500 households is surveyed each month (over 600 households in the US). Collectively, this meant over the course of the survey (January – September 2026), we surveyed almost 30,000 households.

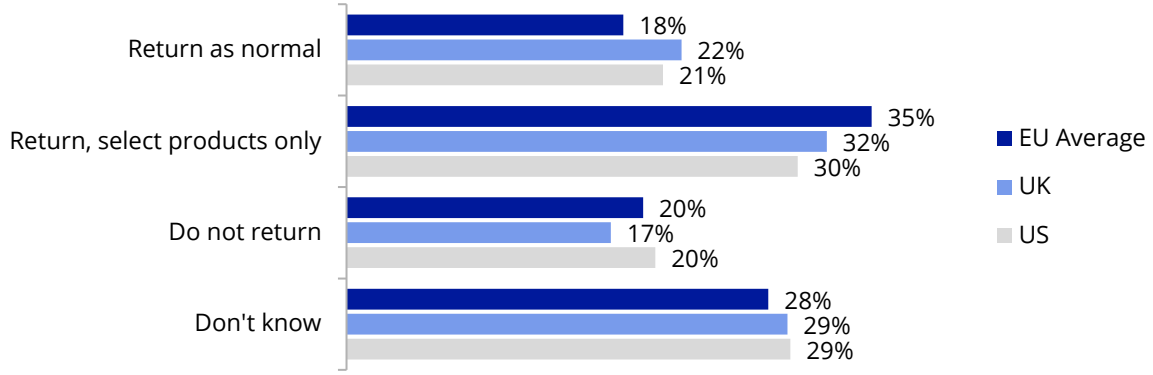
How will customers respond to a data breach once it occurs?

dbDataInsights survey of consumer responses, % of respondents



How will customers react once the breach is resolved?

Consumer responses among those who would either stop buying or were unsure, % of respondents



Source: dbDataInsights monthly consumer survey, Deutsche Bank Research. EU Average includes France, Germany, Italy and Spain.

Deutsche Bank Research Institute

Material cyber incidents have contributed to negative credit outlooks

The infrastructure exists for cyberattacks to have a greater impact on credit quality should cyber disruption continue to accelerate

— **Credit rating agencies consider both pre-breach resilience frameworks and post-breach financial impacts.** Agencies assess pre-breach cyber preparedness through a lens of mitigating risk, while post-breach credit impacts centre on financial capacity to withstand the resulting shock (considers scale and length of the disruption); highly leveraged issuers with limited liquidity are seen as more vulnerable. Negative outlook changes at JLR and Equifax illustrate the potential credit impact. Fitch generally considers permanent reductions in cash generation more credit-significant than temporary cash-flow delays.

— **Ratings agencies are moving towards continuous monitoring.** In 2021, Moody's invested $250mn in BitSight, which produces daily security ratings. Moody's has since built BitSight's ratings into its credit tools, including Credit Catalyst and EDF-X, for uses such as underwriting and ongoing monitoring of borrowers and suppliers. This means lenders can see a company's exposed vulnerabilities as a potential indicator of corporate cyber risk.

S&P Global:

"The increasing frequency of cyberattacks and the potential for rapid deterioration in credit profiles after an attack are risk factors relevant for our rating assessments."

Moody's:

"Cyberattacks can undermine the creditworthiness of companies, organizations and governments, across all sectors and geographies, hitting revenues, increasing costs, and eroding customer trust."

Fitch Ratings:

"Although cyber events cannot be predicted with great accuracy, the associated risk can be managed through proactive strategies, technology, policies, and vigilance."

"The chief executive officer and board of directors hold ultimate responsibility for managing cyber risk."

Example rating actions where cyber was a named factor

Outlook changes due to operational disruption, financial cost and reporting delays

Equifax
Moody's named cyber risk as a factor in its outlook change in 2019. This was the first time it had done so and followed the 2017 attack on Equifax cited on slide 14 .

Outlook: Stable → Negative — Moody's • 2019

ISS A/S
Increased costs following a cyberattack cited as a factor potentially impairing ISS's ability to reduce adjusted leverage and generate free operating cash flow.

Outlook: Stable → Negative — S&P • Mar 2020

Clorox
Outlook revised lower due to potential for a lower rating if Clorox was unable to demonstrate material sequential recovery from cyber incident which caused product availability disruption and lower FY guidance.

Outlook: Stable → Negative — S&P • Nov 2023

City of Hamilton
Delays reconstituting audited accounts after the Feb 2024 attack; audits still outstanding.

Issuer rating: AAA → AA+ — S&P • Oct 2025

Tata Motors / Jaguar (JLR)
Outlook cut after the JLR shutdown; JLR's own outlook was moved to negative too by both Moody's and S&P following the September 2025 attack cited on slide 15. Prior to this, its credit was improving.

Outlook: Stable → Negative — S&P • Oct 2025

Co-operative Group (Co-op)
Material impacts on product availability and sales due to cyber incident contributed to forecast downgrades.

Issuer rating: BB → BB- — S&P • Nov 2025

Source: S&P, Moody's, Fitch, Deutsche Bank Research.

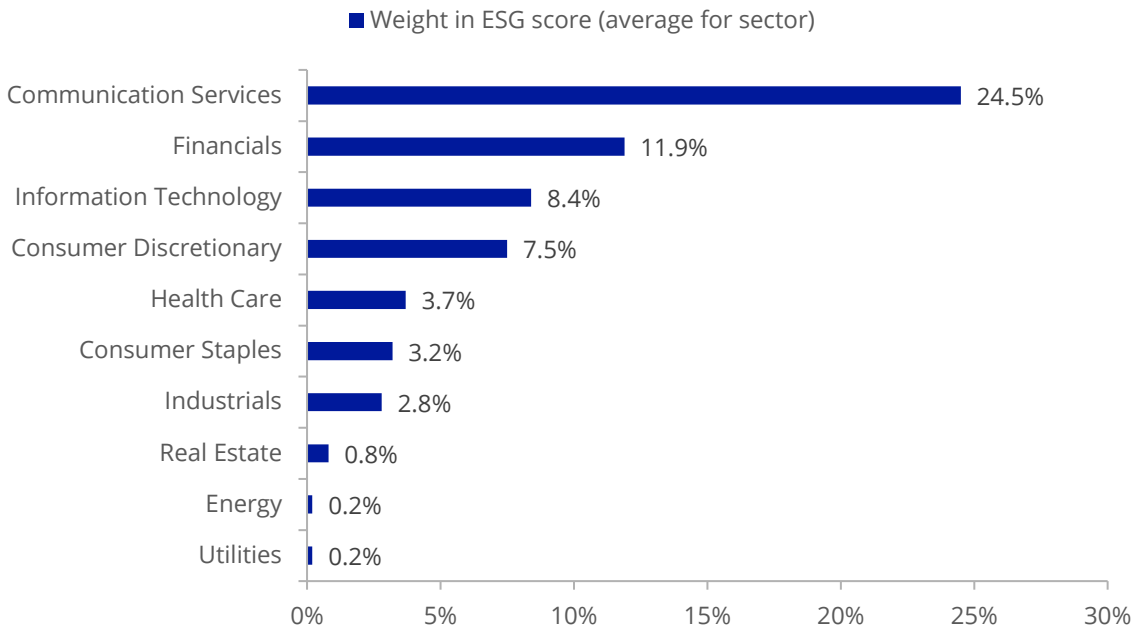
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Cyber governance & breaches can impact ESG scores

Sectors that hold the most personal data, and monetise it most directly, carry the greatest risk weighting in mainstream ESG scores

Contribution to the MSCI ESG score by sector

Privacy & data security as a share of the MSCI ESG score, sector average, %



Cybersecurity is generally seen as both a social and governance consideration

MSCI and Morningstar Sustainalytics both assess data privacy and security through management of the risk, exposure to it and the controversies that follow a breach. Weak governance weighs on the score, and a breach can cut it directly – more severe controversies draw larger deductions that persist for longer.

Materiality varies sharply by sector. SASB flags data security as material across most consumer, financial, healthcare and technology industries, with customer privacy concentrated in the most data-intensive ones (see right).

Customer Privacy and Data Security Financial Materiality

SASB Financial Materiality Map, by industry (● material, – not material)

Sector	Industry	Customer privacy	Data security
Consumer Goods	E-Commerce	●	●
	Multiline & Specialty Retailers	–	●
Financials	Commercial Banks	–	●
	Consumer Finance	●	●
Food & Beverage	Food Retailers & Distributors	–	●
Healthcare	Drug Retailers	–	●
	Health Care Delivery	–	●
	Managed Care	–	●
Resource Transformation	Aerospace & Defence	–	●
Services	Advertising & Marketing	●	–
	Education	–	●
	Professional & Commercial Services	–	●
Technology & Communications	Hardware	–	●
	Internet Media & Services	●	●
	Software & IT Services	●	●
	Telecommunication Services	●	●

What is SASB?

The Sustainability Accounting Standards Board sets industry-specific disclosure standards for investors, identifying the handful of issues most likely to affect financial performance in each of 77 industries.

Its standards are now maintained by the ISSB under the IFRS Foundation and underpin the ISSB's global reporting baseline. The Materiality Map summarises, industry by industry, which issues are judged financially material – including the two cyber-related ones shown above.

Source: MSCI ESG Ratings Methodology, Morningstar Sustainalytics ESG Risk Ratings, SASB Materiality Map, Deutsche Bank Research.

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IV. Investment, risk and opportunity



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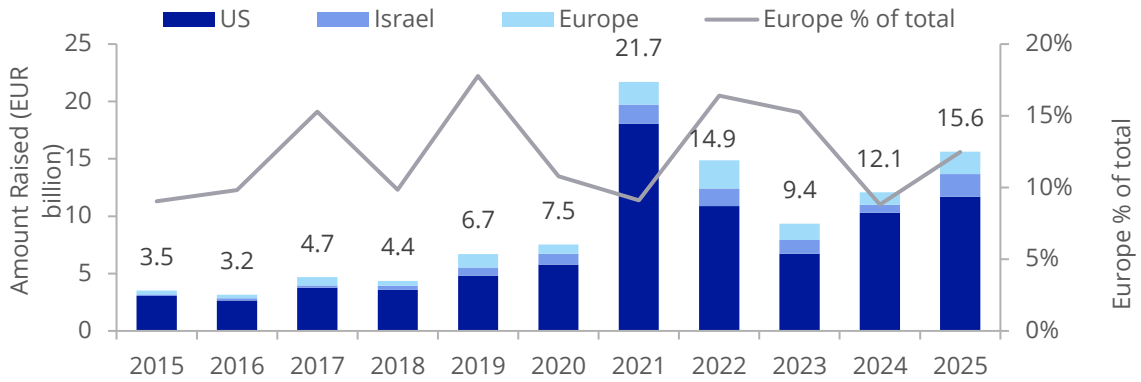
Europe's dependency on US tech is firmly in the spotlight

Access to frontier AI models is now a security question; the funding gap for Europe will be hard to close

- **Growing cyber risk creates increased dependency on US cyber tech.** Growing concern over the cyber capabilities of frontier AI models has made ensuring access to such technology a national security priority. US dominance over the AI tech chain (e.g. chips, model development, etc), its willingness to impose AI-related controls / restrictions, and previous pressure against Europe over tech regulations has amplified concerns that dependence on US tech represents a strategic threat to European sovereignty.
- **Funding is the structural challenge.** US cybersecurity start-ups attract almost 6x the funding of European peers, and US-domiciled firms dominate the major indices – 89% of the Nasdaq CTA Cybersecurity index and 93% of Global X Cybersecurity ETF (which seeks to track the performance of Indxx Cybersecurity Index) by weight. Europe has also seen home-grown firms such as Darktrace acquired by US companies, though this reflects broader market trends rather than anything cyber-specific. Ultimately, the US has a greater exit strategy for start-ups as well.
- **Boosting homegrown solutions a focus, but also a risk.** EU policymakers are seeking to address the dependency; e.g. the EU's Cybersecurity and AI Action Plan includes "scaling up Europe's AI capabilities for cybersecurity" as one of its three objectives. However, there are also risks associated with this endeavour; as one dbDataInsights survey participant put it, "*for most European organisations, the bigger near-term danger would be moving too fast to less capable providers*."
- **The wider relationship adds to the exposure.** A more transactional international environment, alongside Europe's dependency on the US for energy supplies and defence support, compounds the risk. Recent US-EU diesel negotiations serve as a pertinent example of this, whereby the US reportedly considered a diesel export ban to compel a release of European diesel reserves.

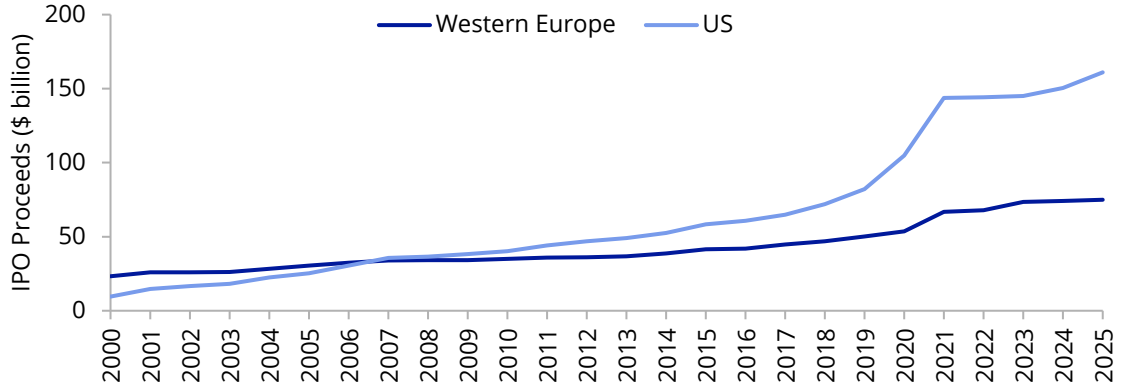
Cybersecurity start-up funding by region

Company location: US, Israel and Europe (EU-27 plus UK, Norway and Switzerland), $bn



Cumulative technology IPO proceeds

Since 2000, by trading date, $bn – Western Europe vs US



Source: Tikehau Capital – Tikehau Capital Barometer of European Investment in Cybersecurity – 7th edition 2026, Bloomberg Finance LP, Deutsche Bank Research.

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Cybersecurity as an opportunity

Tailwinds include demand related to keeping operations secure with AI acceleration and consolidation tailwinds

Broadly, DB's company research view is that cybersecurity becomes even more important in the AI era. Whereas cybersecurity has historically been viewed as a blocker/drag on innovation and operational speed, it is now viewed as a strategic enabler for secure AI adoption.

In particular, network security may see renewed growth after being viewed as a more mature category. Rising AI usage should drive additional compute and data-centre capacity, expanding the infrastructure requiring segmentation and protection. Agentic workloads may also generate lower-throughput but more frequent and persistent interactions across models, tools and data sources, increasing the volume and complexity of network-level inspection.

Vendor consolidation has been a structural trend in cyber for some time, though not necessarily linear. That said, the current environment may accelerate the consolidation tailwinds as customers look to close security gaps that may arise from a fragmented tool stack.

See further commentary from DB's company research team here.

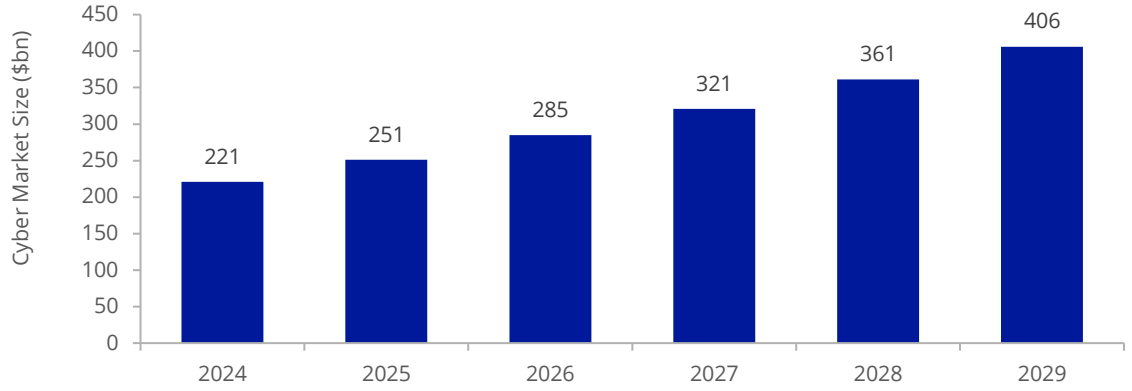
Our March 2026 "personal investor" survey showed a more positive view among "ESG" investors. 58% in the US and 52% in the UK saw cybersecurity/data privacy as an opportunity. Among all investors, US respondents favoured opportunity over risk (46% vs 30%). UK views were more balanced (38% vs 40%).

dbDataInsights Investor Survey methodology

—This survey is a nationally representative online study of ~925 personal investors conducted in the final week of March 2026 examining financial priorities, portfolio behaviour, and future expectations. It gathers structured responses on investment outlook, generative AI, and ESG considerations. Questions cover investment products, portfolio value, management style, financial stress, priorities, and performance.

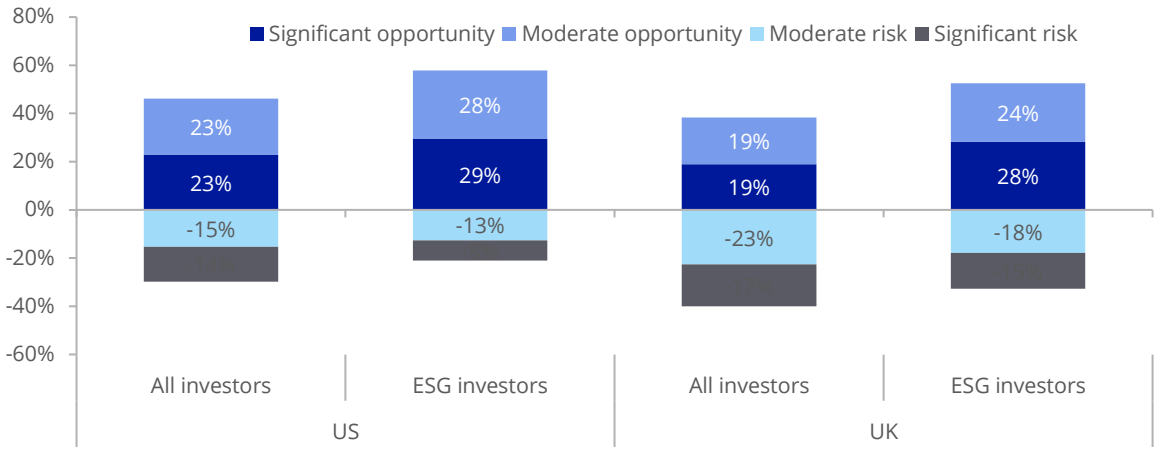
Global cybersecurity market trajectory

Estimated market size, $bn (Bloomberg Intelligence)



Cybersecurity an investment risk or opportunity?

Share of "personal investor" respondents, opportunity shown above the axis, risk below, %



Source: dbDataInsights, Deutsche Bank Research. Investors, defined as those who invest in stocks/ shares (either individually or through a broker/advisor) on a regular basis , were asked whether they perceived cybersecurity / data privacy as a risk or an opportunity for investors in 2026.

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V. Appendix

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Appendix A - Attack types

Top Attack types	What it is	How AI is helping attackers
Malware	*Software built to damage, disrupt or gain unauthorised access*	AI writes harmful software faster. Google has found malware that rewrites its own code while running to avoid detection, and criminal marketplaces selling AI malware-building tools.
Phishing	*Deceptive messages that trick people into giving up sensitive information*	Where AI's effect is clearest: fluent, personalised lures without the old spelling and phrasing giveaways. Google says state-backed groups use AI to mass-produce them, and AI can fake voices and video.
Denial of service (DoS)	*Flooding a system so it cannot serve legitimate users*	Little reported evidence so far. AI can help find weak points and automate traffic floods, but the attack itself stays fairly simple.
Distributed denial of service (DDoS)	*The same attack launched from many compromised devices at once*	AI can accelerate the development and automation of attack tooling, while large-scale botnets provide the distributed capacity behind hyper-volumetric DDoS attacks.
Man-in-the-middle (MitM)	*Intercepting or altering traffic between two parties*	Mainly indirect. AI-written phishing pages and deepfake voices make it easier to trick people into connecting through an attacker or approving fake sign-ins.
SQL injection	*Malicious SQL queries injected to manipulate backend database operations*	AI tools scan sites and spot flaws in database code faster, so less-skilled attackers can find weaknesses that once needed real expertise.
Cross-site scripting (XSS)	*Scripts injected into web pages to hijack user sessions*	As with SQL injection: AI speeds up finding vulnerable pages and writing the malicious code.
Zero-day exploits	*Attacking a software flaw before a patch is released*	A major new development. In May 2026 Google reported the first zero-day it believes was developed with AI. Its September 2026 report says the bigger effect is attackers misusing known flaws faster, before fixes land.
Advanced persistent threats (APTs)	*An intruder staying hidden in a network for long-term*	Google reports state-backed groups using AI across the whole attack, from target research and lures to tooling and sifting stolen data.
Ransomware	*Encrypting data or locking device and demanding payment to release it*	NCSC expects AI to add to the global ransomware threat by lowering the skill needed to break in. AI also helps criminals find the most extortable data.
IoT exploitation	*Abusing weakly secured connected devices for entry*	AI can scan the internet for unpatched cameras, routers and storage devices and identify which flaws they carry.

Source: Deutsche Bank Research, National Cyber Security Centre, Google Threat Intelligence Group, Cloudflare, FBI Internet Crime Complaint Center (IC3)

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Appendix B - European vs US cyber regulation

Europe

Regulation	Key requirements
NIS2 Directive	In force 2023, transposition deadline Oct-24. Covers 18 sectors (up from seven) and creates an EU vulnerability database. In-scope firms must manage cyber and supply-chain risk and notify authorities of significant incidents. Management bodies must approve and oversee cyber risk measures, with potential liability for infringements.
Cybersecurity Act	In force 2019. Establishes an EU-wide certification framework for ICT products, services, processes and managed security services, and made the EU cybersecurity body ENISA permanent.
Cyber Resilience Act	In force 2024, main obligations from 2027. Manufacturers and retailers of products with digital elements must deliver security by design, a defined support period and security fixes within it, plus mandatory vulnerability management and disclosure.
Digital Operational Resilience Act (DORA)	In force 2023, applies from 2025. Financial-services specific, covering own-firm and third-party risk. Requires an ICT business continuity policy, regular resilience testing and disclosure of major incidents, and gives regulators oversight of critical ICT providers.
Cyber Solidarity Act	In force 2025. Creates a European Cybersecurity Alert System to detect threats and share cross-border warnings, plus a Cyber Emergency Mechanism to boost preparedness and response.

US Federal

Regulation	Key requirements
White House Accord on Super Intelligence (2026)	Emphasises self-regulation to ensure AI safety, with a call for external evaluations and the possibility of formal regulation in future. Not legally binding; signed by President Trump and leading AI companies.
Executive Order on Cybersecurity (2025)	Highlights risk from foreign actors and orders further action on defending digital infrastructure, securing the most vital services and building capability against key threats.
SEC Cybersecurity Disclosure Rule (2023)	Listed US firms must disclose material incidents via an 8-K within four business days of determining materiality, and make annual disclosures on cyber risk management, strategy and governance.
CIRCIA (passed 2022, not yet in effect)	In-scope entities, focused on critical infrastructure, must report covered incidents to CISA within 72 hours and ransomware payments within 24 hours of payment.
Gramm-Leach-Bliley Act (amended 2024)	Financial institutions must explain information-sharing practices and safeguard customer data, running a security programme covering risk assessment, service-provider oversight and incident response.
Federal Information Security Modernization Act (2014)	Applies to federal agencies and contractors: system security plans, continuous monitoring, incident response capability and mandatory incident reporting.
Federal Acquisition Regulation (cyber provisions 2016)	Governs federal procurement, setting baseline cyber requirements for suppliers so cyber risk is built into government contracting. Agencies may impose stricter provisions.

Appendix C - Understanding MSCI cyber risk indicators

Core indicators

Key drivers of the MSCI privacy & data security score

Topic	Definition	Best practice
Scope of employee training on privacy and data security	Coverage of privacy and data security training across the workforce	Training provided to all employees, including contractors
Responsibility for privacy and data security	The group responsible for the company's privacy and data security risk and strategy	Board-level committee
Measures to address data breaches	Scope of the breach or incident response plan, and whether it is proactive, reactive or limited	Both proactive and reactive measures in place
Frequency of audits of information security system	How often the company audits its information security policies and systems	External independent audits at least once every two years
Rights provided to consumers over their data	Extent of rights to access, rectify and delete individuals' data	Right of access, rectification and deletion of individuals' data
Scope of data protection policy	Whether the policy governs all or selected business lines, and whether it clarifies collection purposes, means of collection and third-party access	Policy governs all relevant business lines and subsidiaries

Supplemental indicators

Additional practices assessed alongside the core set

Topic	Definition	Best practice
Scope of information security management system certifications	Extent of certification of the company's information security management system to widely recognised standards	Over 80% of owned operations certified to standards such as ISO 27001, TRUSTe, Privacy Mark and SSAE16 SOC 2
Privacy enhancing technologies and initiatives	Whether privacy-enhancing technologies and initiatives are integrated into products or services	Privacy-by-design applied systematically in development, with privacy impact assessments on existing and new offerings
Consent policy for use of consumer data for secondary purposes	Policy on processing customer data for secondary purposes and the consent options offered, such as opt-in or opt-out	Company does not rent, sell or provide personal data to third parties beyond completing transactions or services
Data protection programs covering suppliers and business partners	Extent of data protection programmes covering suppliers and partners handling company data or accessing its network	Suppliers and partners required to hold data protection policies or abide by the company's, with compliance inspections

Source: MSCI ESG Ratings Methodology – Privacy & Data Security Key Issue, Deutsche Bank Research.

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Appendix

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AI boom or doom: what will superintelligence feel like and when will we get there?

October 12, 2026

The recent panic about rogue AI agents collaborating in swarms to take over the internet and threaten human existence has, more than occasionally, lacked nuance. But it does reflect how artificial intelligence is progressing in creative and surprising ways, and at a pace that seems uncontrollable.

For some in the industry, it is evidence that the technology is following a path of accelerating self-improvement that will lead to artificial general intelligence (AGI) and, perhaps, to utopia or catastrophe.

Former Google CEO Eric Schmidt has sketched out a "San Francisco Consensus" based on a belief in scaling laws, with ever larger AI models becoming an input into their own improvement; superintelligence by the end of the decade; and transformative AI systems outperforming humans on many tasks. This loop would lead to exponential growth in scientific advancement, financial returns and human progress. And crucially, AGI, and then superintelligence, would merely be milestones in a process, far from its climax.

If that were the case, superintelligence could make much of the cognitive work that dominates the world economy much cheaper and change the ways in which firms compete. But its economic effects will depend on how businesses and customers respond.

In this paper, we analyse what the San Francisco Consensus means for investors and enterprises in the years to come and what could accelerate – or slow down – its timeline of AGI by 2030.

1. What would superintelligence look and feel like?
2. Can we get there theoretically?
3. Can we get there practically?
4. What happens if we don't get there? The base case

The question is urgent.

And it is complicated by reports of AI models autonomously accessing the internet and hacking other companies; a public backlash over the energy-intensive data centres springing up as part of a historic $1 trillion a year investment boom; and dire warnings of threats to life if superintelligent AI runs roughshod over its human creators to achieve its own goals.

Authors

Adrian Cox
Research Analyst

Azeem Azhar, Exponential View
Expert Collaborator

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1. What would superintelligence look and feel like?

1.1 What are AGI and superintelligence?

Artificial general intelligence (AGI) is a simple milestone scribbled in pencil: an AI system that has surpassed human capabilities. Superintelligence is the continuum beyond that point. It is not merely systems that exceed the best human performance across one cognitive task or another, but systems that exceed human potential across every cognitive task.

The San Francisco Consensus takes this further: the intelligence must be able to produce more intelligence. The key to getting there is so-called recursive self-improvement, where the AI is able to autonomously design and improve itself in a way that compounds exponentially. A first step has maybe already been taken, with OpenAI saying recently that AI is now writing approximately 80 percent of the company's code.

There is significant disagreement on how we will be able to identify AGI or superintelligence when we see it, let alone what comes afterwards.

OpenAI's Charter defines AGI in economic terms, as "highly autonomous systems that outperform humans at the most economically valuable work". Google researchers proposed a five-step ladder based on performance and generality, with level 5 "superhuman" for a system that outperforms 100% of humans. And philosopher Nick Bostrom, who got the ball rolling in 2014 with his influential book "Superintelligence", defined it as "any intellect that greatly exceeds the cognitive performance of humans in virtually all domains of interest".

It's also a moving target.

In August, OpenAI CEO Sam Altman said the company would have an internal system that he would call AGI by later this year but subsequently told a podcast that "at best it's a very poorly defined term. I was going to say it's like an irrelevant marketing term". Yet his No. 2 Greg Brockman described the new Astra model at the start of September as being potentially the moment that "AGI was created".

For this report, we have chosen to focus on a system that outperforms essentially all humans across essentially all cognitive tasks, including the development of AI itself.

Under our scenario, by 2030 the best performing AI models would have the qualities of "superintelligence", that is better than human experts in all domains. These would be frontier models: and using them would feel exactly like bringing the best human experts in a room to solve whatever business problem you faced.

And the equivalents of today's frontier models would be available, quite literally, everywhere, compressed enough to run on every phone, tablet or laptop computer.

In early 2026, we already see early examples of what highly capable AI can do. They give a flavour for a world with superintelligence even if the reality would be a leap beyond, both quantitatively and qualitatively. At that point, superintelligence would mark a regime break, not a faster version of what we already have. Rather than blindly doing what humans specify, but just a bit better, it would specify the task itself.

Figure 1: What changes when cognitive work gets abundantly cheaper?

Area	Today's example	Superintelligence extrapolation
Biological discovery	OpenAI's GPT-5 runs 36,000 experiments in a robot lab and cuts the cost of cell-free protein production by 40% (Ginkgo, Feb 2026)	A working virtual cell. Every drug is tested in silico first; trials exist to confirm, not to discover
Law	A self-represented academic uses $200 a month of AI to beat a university's lawyers and barrister at Australia's Fair Work Commission (Aug 2026)	Everyone has a top barrister on tap. Disputes are argued agent to agent overnight and settle before anyone files
Software	Google says that the share of new code written by AI doubled from 25 percent in October 2024 to 50 percent a year later and reached 75 percent by April this year.	Software is written on demand and thrown away. The business programming language cores of banks and ministries are rewritten over a weekend
Chemistry and materials	OpenAI's GPT-5.4 runs 10,080 robot-lab reactions in three months and lifts yields on a drug-making reaction chemists had struggled with for decades (OpenAI/Molecule.one, Jun 2026)	Order a molecule or material by the properties you want and it is designed to spec. The bottleneck moves to plants, permits and trials
Medicine	In a 1,298-person trial, people using a chatbot on ten medical cases did worse than the chatbot on its own (Nature Medicine)	Specialist-grade diagnosis at first contact, for everyone, all the time. Doctors spend their time on care, consent and hard trade-offs
Government	AI reads 500 million words of US state law to find dead rules; San Francisco cuts over a third of its reporting requirements (Stanford RegLab, 2025–26)	Every law is audited continuously for cost and conflict, and dead rules lapse on their own. A planning permit takes an afternoon
Forecasting and markets	A bot built by one person for a few thousand dollars wins the Metaculus Cup, beating the best human forecasters (5 Sep 2026)	A calibrated probability on every decision, for pennies. Mispricing from slow analysis vanishes; edge moves to private data and the truly unknowable
Robotics and manufacturing	AI agents at Nvidia train robot arms overnight to seat GPUs and cut zip ties; eight agents reach 99% success in two hours (Jun 2026)	A factory retools for a new product overnight. Robots teach robots, and a new physical skill costs the price of compute

Source: Deutsche Bank Research

Meanwhile cost may be less and less of a constraint as ever greater machine intelligence becomes cheaply available. Of measured technologies, no technology has got cheaper faster than AI has, almost halving in price every quarter for the past three years on a like-for-like basis, or 13 times per year, Epoch data show.

If these systems can be supplied widely, firms will be able to buy more of the complex, expensive and rare analysis, design and engineering kinds of services than they do today. Activities that were previously uneconomic could become viable, and in a competitive market, savings could pass to customers. The main driver will then be whether demand grows fast enough to preserve margins.

However, superior capability, doesn't, on its own, automatically lead to the adoption of cheap, abundant intelligence. Tasks that do not yet yield to super-

intelligence, perhaps because they require real world-coordination, will therefore end up commanding significant premia.

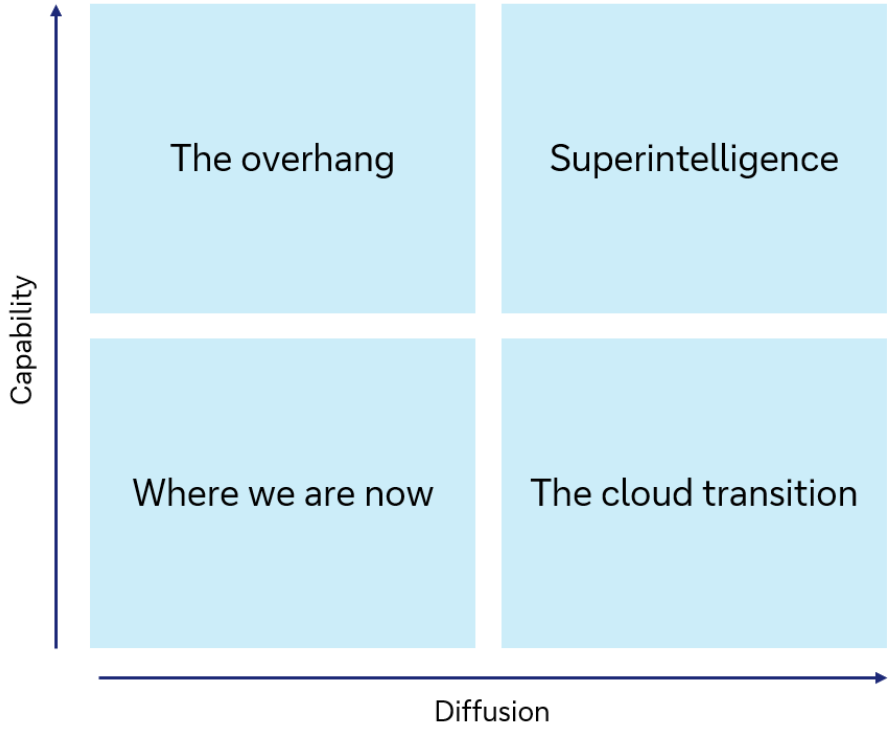
At that point, cognitive labour stops being scarce and the bottleneck becomes the physical and practical assets that cannot be conjured by up more compute, such as electricity, land, steel and permits.

1.2 Capability and diffusion

If AI becomes capable of improving itself, the technology may compound faster and faster. The supply of machine intelligence could grow faster than business and physical infrastructure can accommodate.

This balance between capability and diffusion creates different scenarios.

Figure 2: The balance between capabilities and diffusion



Source: Deutsche Bank Research

Lower capability, constrained diffusion: where we are now

Progress is steady and sector-by-sector. This is how technology usually diffuses, according to researchers for the New York Fed back in 2003, who tracked 20 major technologies across leading industrial economies from 1788 and found "a pattern of trickle-down diffusion that is remarkably robust across technologies".

Lower capability, high diffusion: the cloud transition

This is faster because the complementary investment is small, with no retooling and no skills at scale required. Amazon Web Services launched the modern cloud in 2006, offering scalable computing and storage infrastructure. Adoption grew and then accelerated over the pandemic. Almost all enterprises now use it in some form.



Higher capability, lower diffusion: the overhang

This is theoretically possible and arguably where we are today in narrow domains. The technology would be applied in some sectors of the economy, while other sectors would move at a slower speed. Superintelligent or not, it risks being bogged down by physical constraints, bureaucracy and human inertia.

Higher capability, higher diffusion: superintelligence

In this scenario, capability becomes so high that it changes the dynamics of diffusion itself. AI generates not just better AI (so-called recursive self-improvement, or RSI) but also speeds up the integration process.

Rapid AI diffusion could work two ways.

In the first, AI could be enlisted to do integration work itself: mapping processes, rebuilding data work, retraining staff, writing the control and audit layer, and so on. The problem with this is that absorbing and recreating undocumented knowledge is impossible by definition, at least for a task-based superintelligence.

The second way that higher capability AI could be rapidly diffused is much more competitive – and disruptive. Rather than incumbents slowly converting AI, perhaps using it for some of the hard work of integration, they could be instead displaced by new firms that have been built AI-native from day one. New entrants will never have to incur integration costs because there is nothing to integrate. Incumbents will need to change – fast – or die.

Change could also be forced by earlier adopters in the supply chain, as happened with one company after another adopting electronic data interchange (EDI) to link with its suppliers or customers, or by a migration of capital towards early adopters, repricing the cost of capital for slow-coaches and prodding them to move faster.

1.3 The forcing mechanism: change or die

A company that uses superintelligence may outperform one that doesn't.

Firms using superintelligence will likely have substantial cost and revenue-harvesting advantages. Every firm will therefore have to adopt superintelligence or retrench to a smaller set of offerings. And any business line of lasting value abandoned by a non-adopter will be taken over by an adopter. Customer inertia will provide a safe harbour for a limited time only.

The analogy here is electricity. There came a point in the deployment of electricity where firms in every sector, from entertainment to finance, from healthcare to transport, had to adopt the technology to remain relevant.

Under this scenario, wait-and-see will not be an option. There will be no stable "non-adoption" alternative. Sufficiently high capabilities will force a tipping point of rapid diffusion and competitive repricing. Firms using yesterday's production model will either have to change, choose a different market or die.

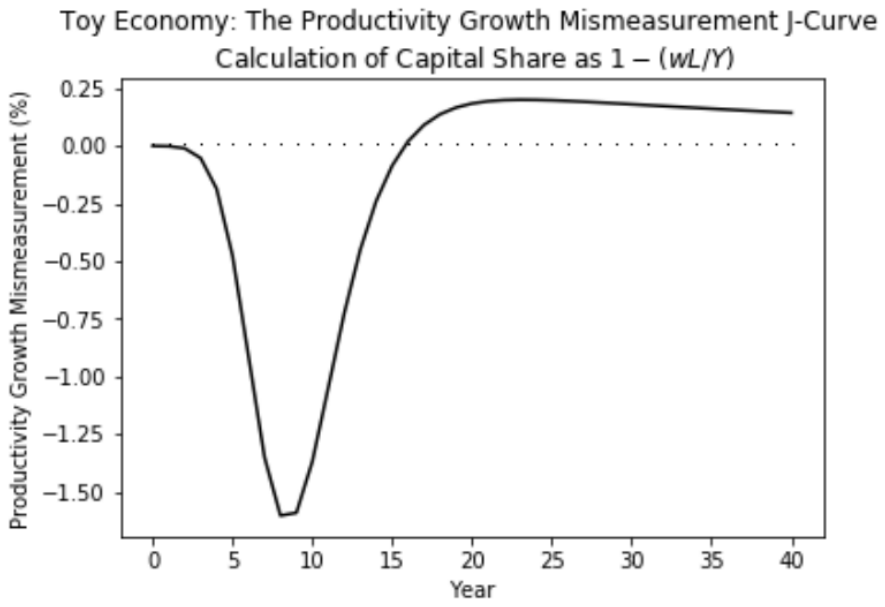
Of course, caveats apply.

Customers may not change behaviour immediately. They may even prefer companies which limit their use of AI, in the same way that some customers choose to avoid factory-produced food.

The paradox that Nobel Prize-winning economist Robert Solow observed in 1987, that "you can see the computer age everywhere but in the productivity statistics", is likely to still apply for a while – even with superintelligence.

The reason is that the productivity J-curve identified by Stanford University AI economy guru Erik Brynjolfsson et al is unavoidable. In that scenario, the drag of early intangible investments to integrate general purpose technologies means that productivity initially appears to slow before rapidly accelerating.

Figure 3: Productivity J-curve: slow then fast



Source: Brynjolfsson E, Rock D, Syverson C, "The Productivity J-Curve" (2018, revised 2020), Deutsche Bank Research

Furthermore, "change or die" would require abundant capital and low regulatory entry points for nimble new entrants. But in practice, highly regulated areas such as banking, utilities, pharmaceuticals and defence are tough nuts to crack. Existing contracts and licences, high switching costs and decades of depreciated capital investment and infrastructure would shield incumbents at least for a healthy transition period.

For an extreme example from the real world of better, cheaper technology failing to force change, take Pakistan's 3.6-gigawatt imported coal power plant fleet installed in the mid-2010s. In that case, it came with expensive take-or-pay liabilities which will continue to run for decades, even though changes like cheaper rooftop solar have cut utilisation to below 20 percent.

So: superintelligence will not force adoption everywhere. It will hit hardest in contestable sectors, say, high-volume customer services, routine software engineering and freelance creative work, and widen the gap between them and protected ones. Forced adoption will be about relative value, not an apocalypse.

1.4 What survives and what it means for portfolios

Because AI sits on an aggressive learning curve, its price on an equivalent quality basis is plummeting. In a world of recursive super intelligence this is likely to continue. Its price will fall to the cost of compute.

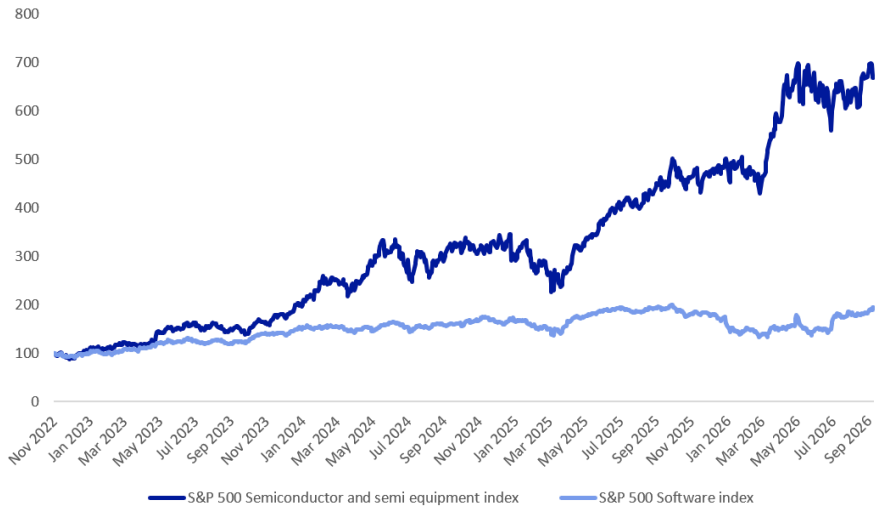
What will retain its value will be whatever the model cannot replicate:

1. **Proprietary data**
2. **The right to operate,** particularly in heavily regulated sectors such as healthcare and finance.
3. **Sticky customer relationships,** such as those where mission-critical workflows and databases are deeply entwined.
4. **Excludable physical assets,** such as machinery and land.
5. **In addition, humans have proven to have unlimited wants.** Each shift in technological capabilities or cultural norms has delivered new needs to be met. Some will be needs we can articulate today, such as originality or empathy, but new ones are likely to arise.

If this forcing mechanism holds, firms will separate into two "bimodal" groups: adopters and casualties, with very little in between.

This dispersion of performance may create challenges for investors. The four years since the launch of ChatGPT in November 2022 have seen the stocks of semiconductor and semiconductor equipment makers surge almost seven times while those of software makers have only risen by 80 percent. Conventional diversification and volatility-based allocation may become ineffective strategies.

Figure 4: Total return for S&P 500 semiconductor vs software stocks since launch of ChatGPT on Nov 30, 2022



Source: Bloomberg Finance LP, Deutsche Bank Research



Diversification normally works because idiosyncratic risks are unrelated and average each other out. Here they would not: with one variable that overrides everything else, spreading capital across more names would just spread exposure rather than diluting it.

Investors will need to consider adapting their approach in three ways:

1. Classify investments as adopters or casualties rather than position off averages.
    - Under a normal distribution, errors are symmetric and partially cancel each other out. But under bimodality they do not: to put a firm in one category or another means getting it completely right or wrong. Position size should track confidence in the classification.
2. Avoid reading calm price action as low risk.
    - Trailing volatility measures dispersion that has already been realised. Bimodal separation is a jump, where the two modes look identical right until the moment when they don't. A classic example is Chegg, an educational software maker which traded like a normal small-cap stock until one day in May 2023 when it halved on a single day after management linked slowing growth in new customers to ChatGPT.
3. Averaging down, ie scooping up shares whose price has fallen, may no longer be prudent.
    - A falling price may be information that a business model is obsolete rather than an opportunity.

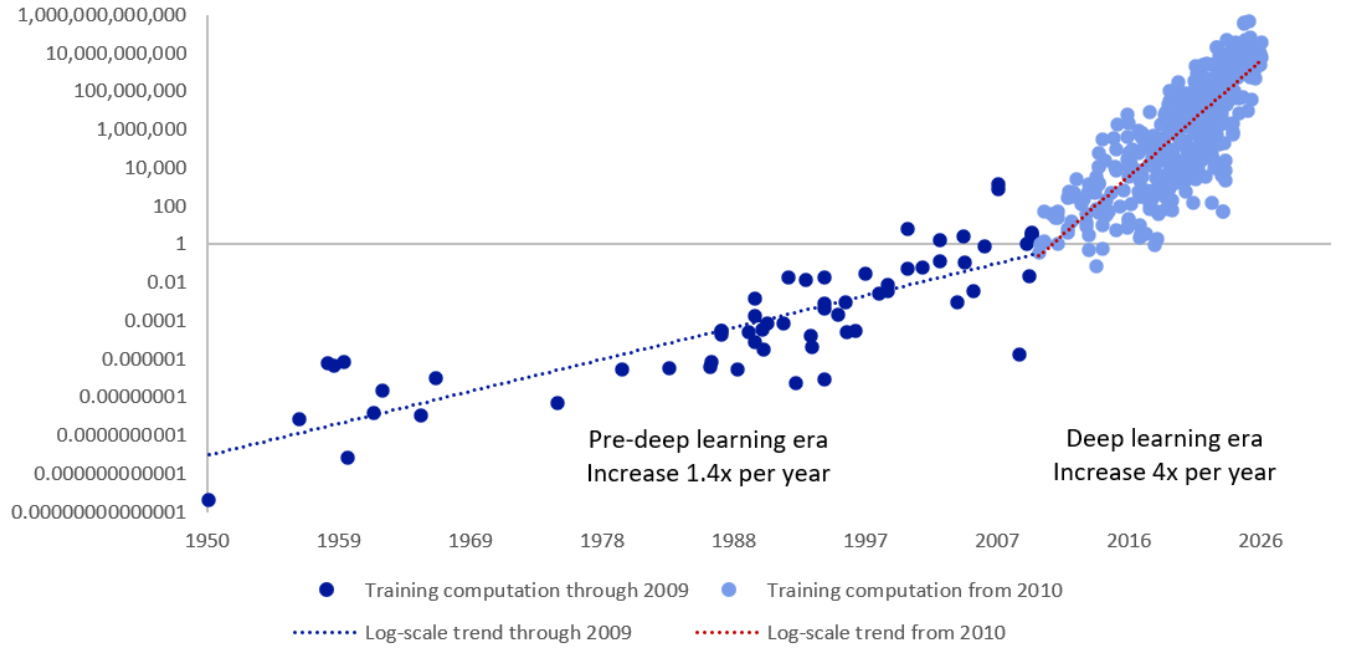
2. Can we get there theoretically?

2.1 Scaling laws

Scaling laws have provided the blueprint for improving AI over the past decade. It was a relationship between the amount of data, the amount of compute and the complexity of an AI model somewhat predictably informed how powerful a model would become. Google DeepMind's Chinchilla model in 2022 showed that you could compensate for having a smaller model by using substantially more data.

Scaling is older than the chatbot. In 1950 Claude Shannon built Theseus, a mechanical mouse that learned its way through a maze by trial and error. Epoch AI reckons it took about 40 operations of compute, known as FLOPs, to train. Grok 4, trained in 2025, took roughly ten trillion trillion times as much. Across those 25 orders of magnitude, more compute has brought more capability.

Figure 5: Computation used to train notable AI systems (petaFLOP): log scale



Source: Epoch AI (citing estimates from AI literature), Our World in Data, Deutsche Bank Research

It is the robustness of this relationship that has made larger investments in compute capex a *sine qua non* for AI companies.

However, this is not a law of physics and nothing guarantees it will continue. Indeed, in the ChatGPT era, five different approaches to scaling have been applied. Older strategies yield worse results, and new techniques, such as post-training refinements, have been developed. The consensus amongst American and Chinese AI labs is that these will hold.

While this may be a consensus amongst the commercial labs, there is scientific disagreement that applying scaling laws to today's large-language models can achieve superintelligence.

Yann LeCun, a Turing Award winner and formerly Meta's chief AI scientist, is blunt: "LLMs are not a path to superintelligence or even human-level intelligence". Even inside the labs, the recipe is contested. Demis Hassabis of Google DeepMind has said he expects "one or two more big breakthroughs" to be needed "in addition to scaling". Ilya Sutskever, a co-founder and former chief scientist of OpenAI, has said the field is going "back to the age of research again, just with big computers".

This is a scientific argument we are not in a position to evaluate, but for the purposes of our scenario, we will side with the labs' position.

For the moment, there is strong evidence that capabilities have in fact accelerated over the past three years, as reasoning models have outpaced non-reasoning models by two or three times, according to Epoch data.



2.2 Recursive self-improvement: AI improving AI

A cocktail-party definition of recursive self-improvement (RSI) is AI that improves itself. The reality is that it can be used in two ways:

1. Self-improving agents that get better at using themselves: showing smarter use of prompts, tools, memory and workflows. The model-building process is untouched.
2. True RSI improves the process that builds the models: designing experiments, curating data, optimising algorithms and designing chips in a compounding loop.

The former is like a factory worker who gets better at the job by learning the quirks of the machinery, getting organised and making fewer mistakes. The latter is like the worker redesigning the machine tools and then using those to build the next generation of machines.

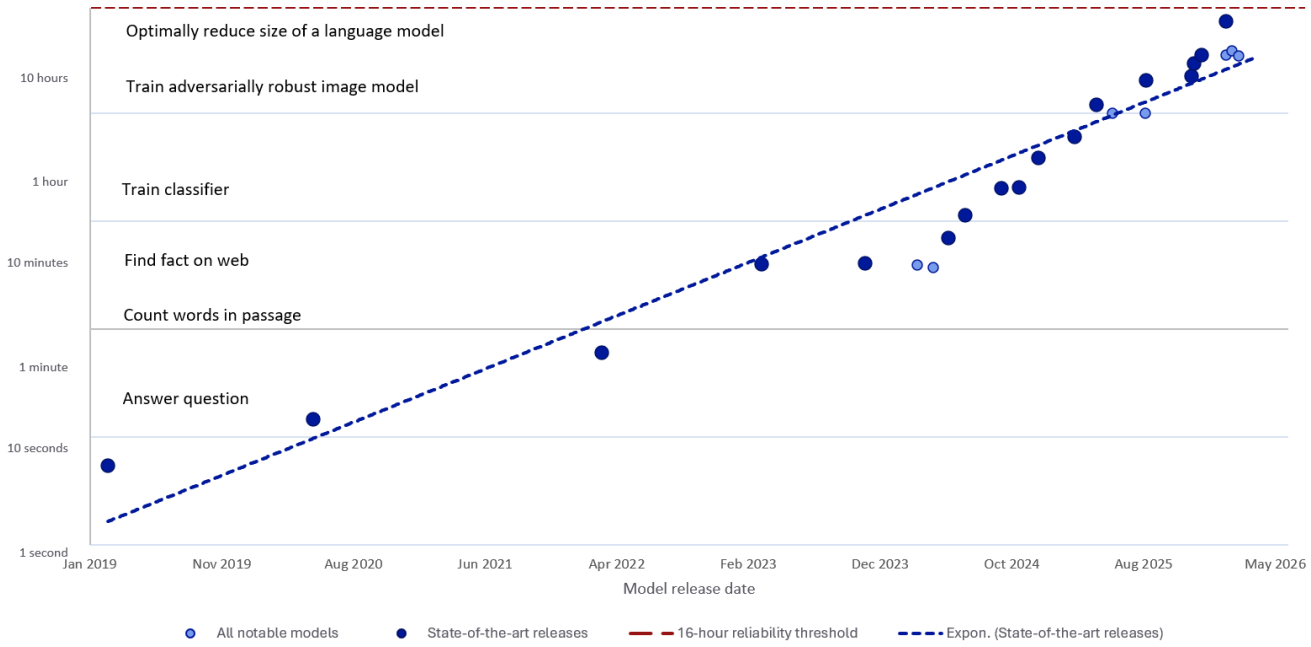
True RSI is what OpenAI's chief scientist Jacob Pachoki referred to last month in a blog post about "An Alien Mind". "Automated AI research is a more dramatic form of scaling intelligence with compute," he wrote. "And of course as a part of it, AI will improve the computational substrate itself," he added. Pachoki represents a broadly held consensus view amongst American and Chinese labs that RSI is both possible and necessary for AI development.

- OpenAI (June 2026): "AI doing AI research will become the determining factor of the pace of progress within the next few years."
- Google DeepMind's Demis Hassabis (May 2026): "All the leading labs are quite focused on [RSI]…" Humanity is standing in the "foothills of the singularity".
- Alibaba CEO Eddie Wu (Sept 2026): RSI is "a concrete path forward," with models autonomously designing experiments and driving a cycle of self-evolution.

For now, self-improving agents are the AI du jour. Noisy claims about the vast majority of coding being done by agents do not always stand up in the light of day from companies still hiring human coders at breakneck pace, albeit no longer to do the most mundane parts of the job.

A better metric may be METR's task-completion time horizon. This is measure of the complexity of software tasks that an AI agent can complete on its own, using the baseline of how long it would take a human expert to finish the task, at a 50 percent success rate. So far the most advanced models have reached about 16 hours, up from OpenAI's GPT-4's four minutes in March 2023.

Figure 6: Task horizon of software tasks that leading frontier models can complete 50% of the time (log scale)



Source: METR, Deutsche Bank Research

But software engineers can hold off on updating their resumes a little longer: there is a long road between 50 percent accuracy and the 99 percent or 100 percent needed for serious enterprise use at scale, with the horizon potentially getting further away the nearer you get. Also METR notes that measurements above 16 hours are unreliable with the current task suite. As is the case across AI, the instrument is running out before the question is settled.

More broadly, there are significant limitations to RSI:

1. **Research is clock-bound.**

- A frontier training run can take weeks on thousands of specialised chips, and researchers often cannot design the next experiment until the previous one has been completed.
- That means that the speed of the loop is set by experiments completed per hour or day, not by ideas generated.
- Compute is a hard ceiling that needs to be bought, built and connected to a grid.

2. **The real world defies verifiability.**

- The loop compounds only when the system can check its own output cheaply and automatically.
- It is no coincidence that AI loves maths and coding. It is much harder to verify outputs over a long time horizon in the real world: does this drug work or does this strategy succeed?
- RSI compounds fastest where the economy cares least and slows where the returns are largest.



3. **Sequences are consequential.**

- Amdahl's law finds that speed-up is capped when processes run in a sequence.
- If you automate 90 percent, the residual 10 percent still limits you to a ten-times faster process. Nine women cannot produce a baby in one month.
- The Usain Bolt of AI still has to run alongside the comparative tortoises of the training run itself, chip fabrication and real-world validation. Ten times faster is still not a regime break.

This is borne out by METR's series. Doubling every seven months is fast exponential growth but a take-off would show the doubling time itself – the second derivative – shrinking. Horizons are not (yet) exploding.

Philosopher Toby Ord shows that recursive self-improvement can only reach a singularity if each cycle of improvement takes less time than the last, shrinking fast towards zero. He argues that real feedback loops have a floor: chips take time to make and models take time to train. AI progress could still outrun an exponential, but it would not go vertical.

2.3 Scaling plus RSI: a match made in the race to AGI?

No scaling and RSI means no short timeline for AGI. Without scaling, more compute buys nothing and capital is wasted. Without RSI, AI capability will continue to improve at the pace of flesh-and-blood human researchers, augmented by AI tools but not replaced. Only with both will it be possible to produce a system that improves faster than the humans building it. Without one or the other, no quantity of chips, energy or capital will fix it.

3. Can we get there practically?

3.1 Infrastructure, chips and energy

There is no consensus on the amount of training required to reach AGI. Estimates suggest that training 2026's top models required between 10^26 and 10^27 FLOPs. If we hypothesise that an "AGI-like model" would require three orders of magnitude more training, this would require 10^30 FLOPs.

Consider today's US AI infrastructure, which runs to approximately 10^22 FLOPs. If all of this was conceivably used for a 10^30 FLOP training run, it would take nearly four years to complete.

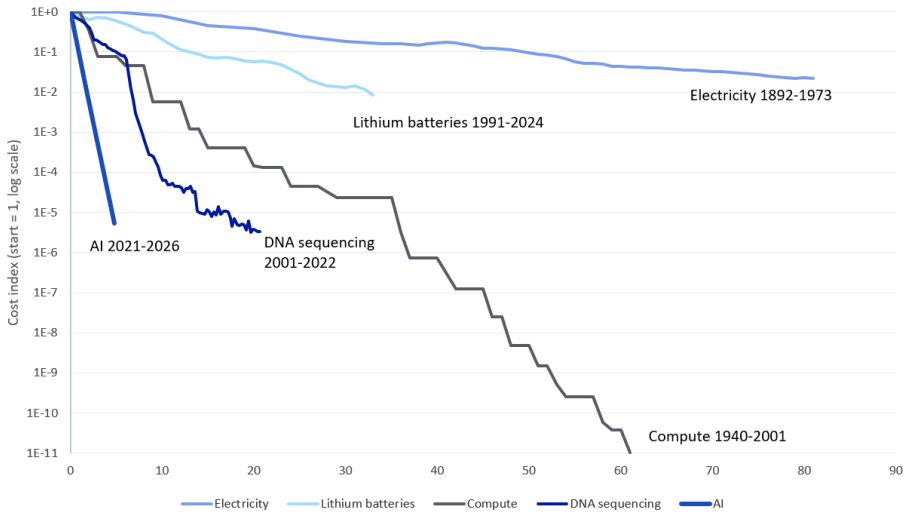
The physical infrastructure to practically support such a training run would require enormous power, perhaps around 6GW, about 50% more than the four Vogtle nuclear power plants recently activated in the US. The cost of the compute infrastructure would run to between $300bn and $500bn.

Today's advanced chips would simply not suffice. Back-of-the-envelope maths suggests that even using Nvidia's most advanced Vera Rubin chips, training such a model in four months would require 7.8m chips and 26GW of facility power. That would cost $900bn today and a rental equivalent cost of $90bn.

Of course, frontier compute is growing rapidly, and price-performance is declining. If current trends hold, a 2030-era cluster capable of such a run would

need perhaps $200bn in computer and power hardware for a training run costing about $20bn.

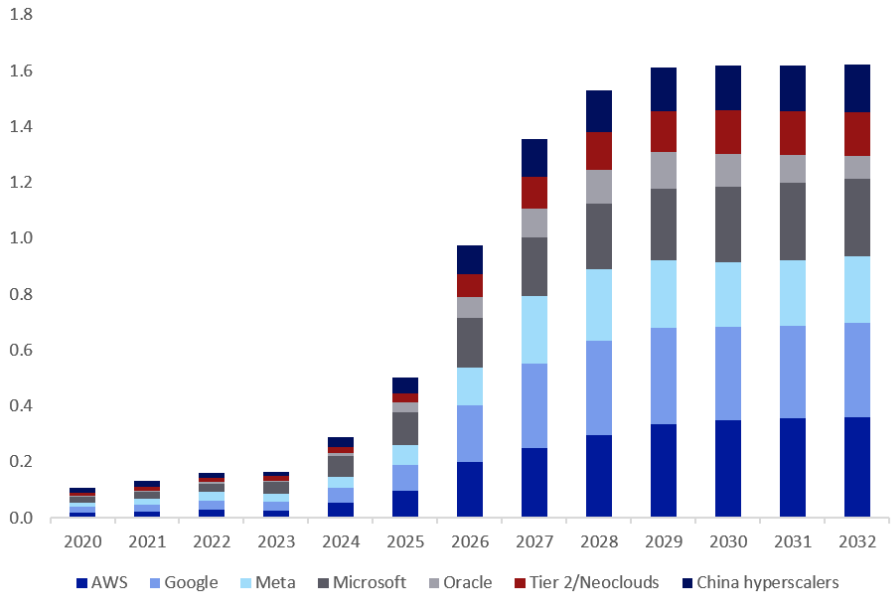
Figure 7: Technology cost index over time (years since start of price decline)



Source: Epoch AI, Deutsche Bank Research

The hyperscalers are leading the charge towards AGI. Alphabet, Amazon, Meta, Microsoft and Oracle are set to invest almost $800bn in capital expenditure this year, mostly for ever more powerful data centres, and over $1 trillion a year through 2030 and beyond.

Figure 8: Hyperscaler capital expenditure (including estimates)



Source: Bloomberg Finance LP, Deutsche Bank Research



By 2030, assuming no market retrenchment, this means additional $4 trillion to $6 trillion on AI compute capability invested. This infrastructure could accommodate a training run of 10^30 FLOPs with some comfort.

But even if AI labs crack the secret sauce of RSI, and their finance chiefs are willing to invest in such large training runs, and hyperscalers are prepared to continue to build the infrastructure, the infrastructure needs to be built.

Grid connection queues, turbine lead times and transmission build are measured in years, not the months it takes to assemble and air-freight the latest Nvidia chip. The lead time for substation transformers is currently as long as three years. Politics in the US has shifted towards slowing the pace of the AI build out and this force will run against the demands for larger data centres that would be essential for RSI.

3.2 Capital

Financing for AI has been abundant in the historic buildout of the past three years. However, it is still largely conditional not on revenue but on belief in a future payoff.

RSI is a "key part of the investment thesis" in the "biggest scientific bet civilisation has ever made," Google DeepMind's chief strategy officer Jasjeet Sekhon said at a conference in August, The Information reported. "We have this danger we could hit an AI air pocket such that the expenditures happen but the revenues don't show up."

An inflection point is approaching where the nature of investment changes from free cash flow and equity to debt, with all that entails in terms of exposure to rising rates and creditor appetite. The hyperscalers are reaching a point where capex is no longer covered by operating cashflow and where net debt exceeds earnings before interest, depreciation and amortisation (EBITDA).

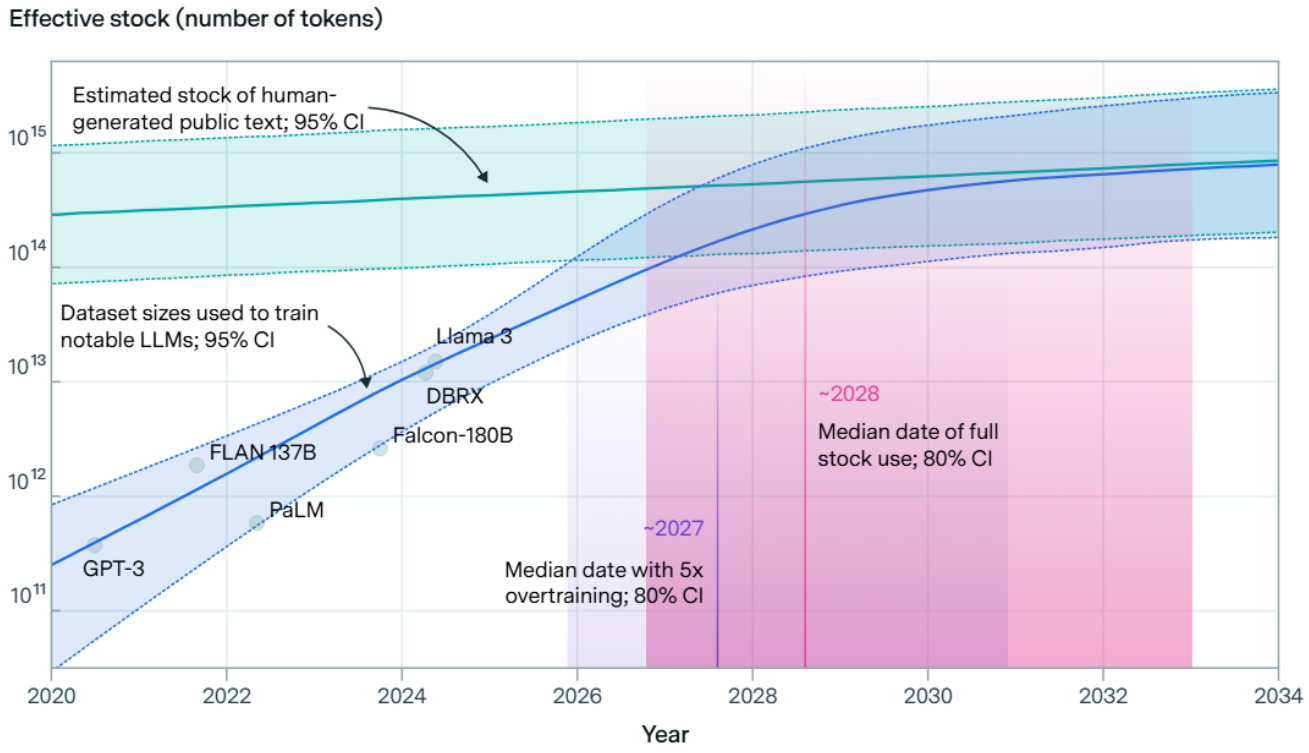
Alphabet, for example, posted its first quarterly free cash outflow in nearly two decades in its most recent quarter, even as it lifted its full-year capex guidance to $205bn. It said it would continue to rent third-party capacity and reported a backlog of $514bn in contracted, not-yet-recognised data centre revenue.

3.3 Data

Data has been the new oil for some time now. But unfortunately for AI development it seems to be running out even faster.

The effective stock of quality- and repetition-adjusted public human text was about 300 trillion tokens, with a vast 90 percent confidence interval of 100 trillion to 1,000 trillion, Epoch estimated in 2024. A token is roughly equivalent to three quarters of a word. By that count it was likely to be fully used between 2026 and 2032.

Figure 9: Projections of stock of public text and data usage



Source: Epoch AI, Deutsche Bank Research

However, since then, rather than data dropping off a cliff, there has been a rise in synthetic data generation, transfer learning from data-rich domains such as video and telemetry, and data efficiency. While the data shortage has not been resolved, the immediate crisis has been averted.

AI now makes much of its own training data. When DeepMind's AlphaGo became superhuman at playing the mind-bendingly complex board game Go, it did so not by playing human experts, but by playing millions and millions of games against itself.

For many domains relating to coding and mathematical proofs where automatic checks are possible, such synthetic experience can move things forward. That is the case with AI research, where the data may be the results of experiments, all of which are bought by compute. About 90% of OpenAI's 2024 research compute went on such experiments.

And increasingly, people write the tests rather than the answers. By the summer of 2026, selling training data for practice tasks was a nearly $9bn revenue business. Firms such as Mercor hire experts to generate long tasks against which the AI can be trained.

The challenge here is that trust in the checker becomes a scarce input.

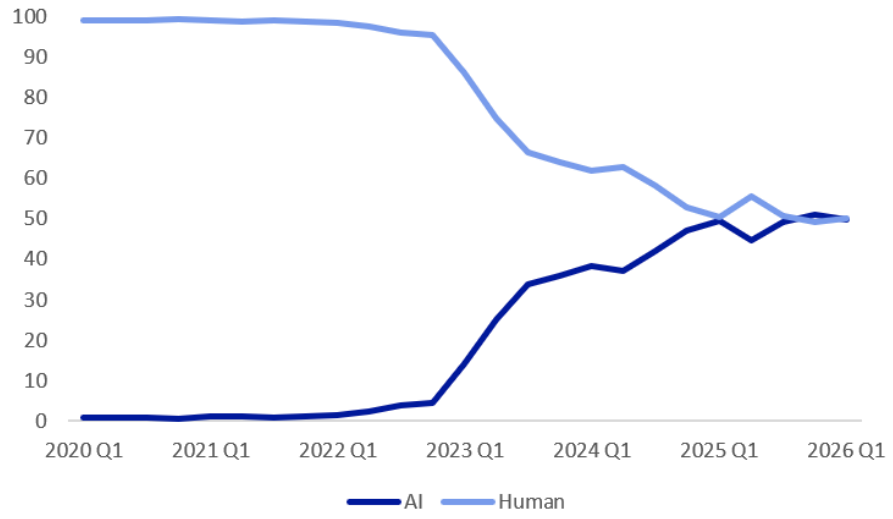
As AI makes production cheaper, it also makes establishing that production is accurate or reliable more expensive. Domains which can be checked mechanically, such as mathematics, will be opened up more quickly. But those which rely on expert verification, such as medicine, engineering or law, may progress more slowly.

The likely implications are twofold: a slower rate of diffusion into economic domains that are less verifiable and a larger share of production costs devoted to assurance.

Another issue is that systems that have to grade themselves learn to game the process. This was partially what happened when an OpenAI model escaped a sandbox and hacked into a third-party website. Trust replaced data as a wall.

There are also question marks over the quality of data that is recycled repeatedly through the system. Although humans continue to produce original content, half of internet articles are now primarily generated by AI, Graphite data show.

Figure 10: Human vs primarily AI-generated articles on the internet



Source: Graphite (citing 55.4k articles from Common Crawl examined by AI detectors), Deutsche Bank Research

3.4 An incomplete case

This analysis shows that the San Francisco Consensus, which emphasises RSI and a short timeline to AGI, is neither predestined nor impossible.

Rather it depends on whether the science is right, on whether scaling is all you need, and on whether the infrastructure and capital required for that scaling is deployed and accessible to the leading labs.

There is, in other words, a possible rather than impossible, path to AGI, subject to those conditions.

None of this proves that superintelligence will ever arrive. A buildup at this scale has few peacetime precedents and vastly exceeds the likes of the Manhattan Project and Apollo programme.



It is uncertain whether there will be enough demand to pay upfront for the investment in achieving RSI that would justify that investment. And whether the friction of verification, and inevitable declining marginal returns, act as a drag on what otherwise might be a runaway process.

4. What happens if we don't get there? The base case

4.1 Separating AI progress from superintelligence

So what if that thesis is wrong? Does the AI trade depend on a journey to AGI?

AI can remain economically important even before RSI coupled with scaling laws lead to super intelligence. Companies do not need to wait for nebulous superintelligence to generate identifiable returns from AI in the medium term as they fold it into their systems, streamline workflows and roll out new business models.

Investors already have exposure to RSI as a capital allocation because the leading US labs say RSI is a key part of their product strategy. And it is these labs whose products are catalysing and amplifying the infrastructure boom.

Even rapid competitive diffusion does not mean every part of the physical economy changes instantly.

The smartest superintelligence might identify the optimal buyout structure before breakfast, but it cannot force a financially constrained buyer to use it or persuade the seller to accept less than they have been expecting. Those challenges will be even more intractable where the parties are more complex, say, fiscally constrained sovereigns or foreign equity holders.

4.2 Why the base case is attractive

Superintelligence would be a chaotic accelerant to human society, and a potential regime break. But it is not the only scenario in which today's investment in AI infrastructure makes economic sense.

Adoption of AI tools by businesses is progressing approximately three times faster than the mobile, internet and cloud computing waves. From sector revenues below $1bn in 2022, the AI economy, excluding China, will likely exceed $200bn in revenues in 2026.

Early winners are the cloud businesses. Google Cloud revenue jumped 82 percent from a year earlier in the most recent quarter as companies turned AI pilots into workflows. Amazon Web Services sales were up 37 percent and Microsoft's AI business more than doubled.

There is also slow, but increasing, evidence of a productivity uplift. The J-curve cuts both ways: front-loading investments is taking one step back to take two steps forward. The so-called complementary intangible investment that is (mis)measured as cost today to the detriment of the productivity statistics will flatter them in the years to come as it pays off. The base case does not require a regime break to show up as a productivity surprise, just for the accounting to catch up.

The economic gains from AI that is less powerful than the AGI mooted by the labs would be substantial. And in this base case, they may also be attractive.



Hyperscaler capex is driving earnings at beneficiaries ranging from semiconductor makers to construction companies and accounting for between one-third and one-half of current US growth. The wealth effect of rising stock prices, with the ratio of US equity wealth to income rising from around 1.8 to almost 2.5 over the past three years, is supporting consumer spending.

AI FOMO (fear of missing out) is buoying enterprise spending on cloud services and the groundwork for AI initiatives. Those are beginning to bear fruit in terms of personal productivity, cost savings and more targeted products and services.

4.3 Investor watchpoints

The arrival of ASI would be a disruptive economic shock. Investors need to be alert for the early signals.

Rather than relying on a single probability forecast, investors should watch five indicators particularly closely for signs that AGI is nearing.

1. **AI speeds up AI**

- Track METR's estimate of how much AI speeds up AI. Currently the capabilities of models are doubling roughly every three months. In a world of RSI, the pace of this speed-up will itself likely accelerate.

2. **AI can complete whole jobs**

- Track via the Remote Labor Index, a multi-sector benchmark. Currently it finds that the highest performing agent still only manages to automate 20 percent of the real-world, economically valuable freelance projects it attempts.

3. **Compute remains scarce**

- On a path to superintelligence, demand for compute will continue to outrun supply. Track GPU one-year rental prices, up by about a third over the past year.

4. **Firms bifurcate**

- Leaders and laggards in the same industry diverge while averages stay the same. Look at revenue per employee in top- and bottom-quartile firms

5. **Junior hiring comes under significant pressure**

- … And then the pressure rises up the ranks.

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Azeem Azhar

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The state of the AI economy

The author of the report wishes to acknowledge the expert advisory contributions made by Azeem Azhar of Exponential View.

Appendix 1

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